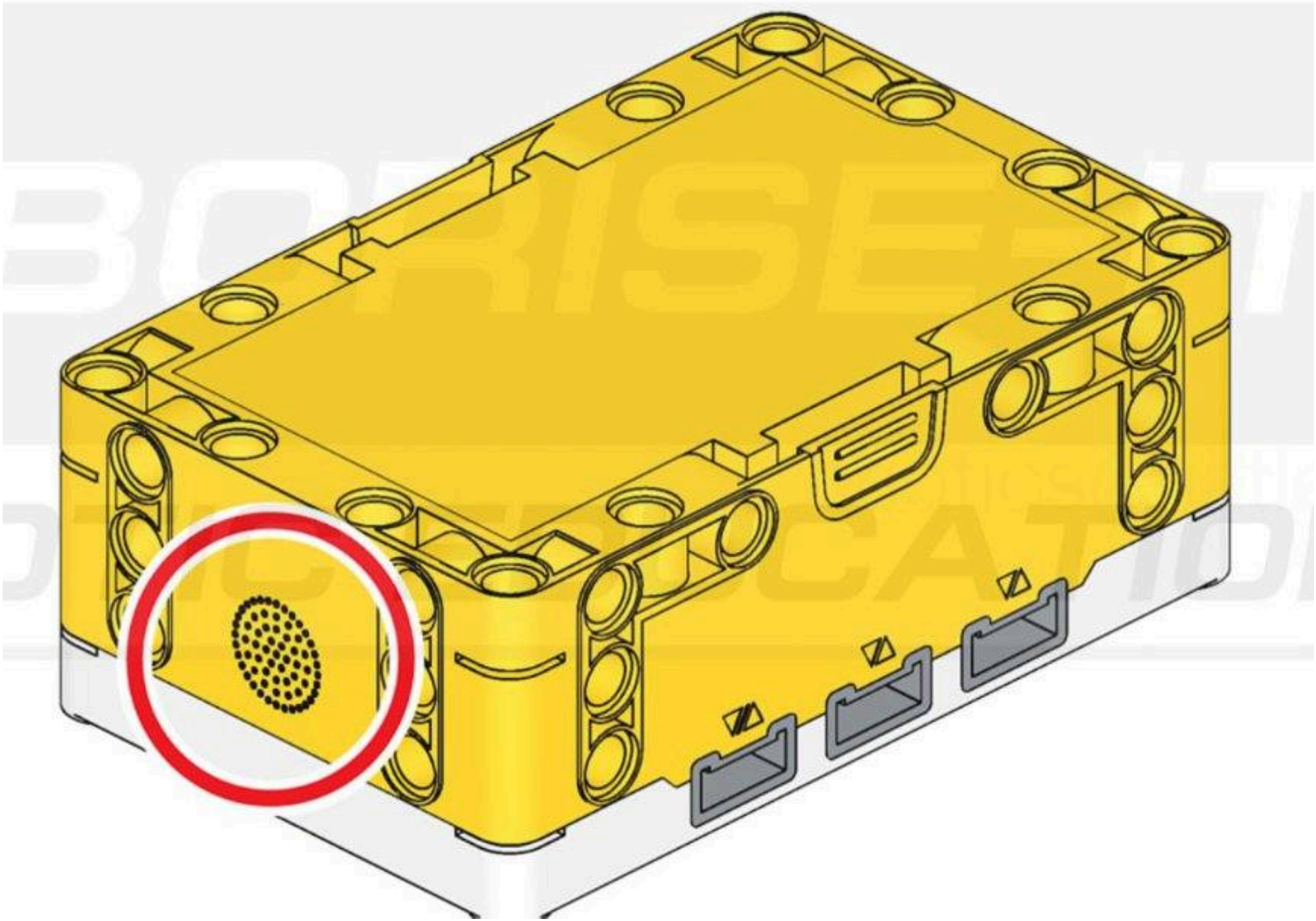
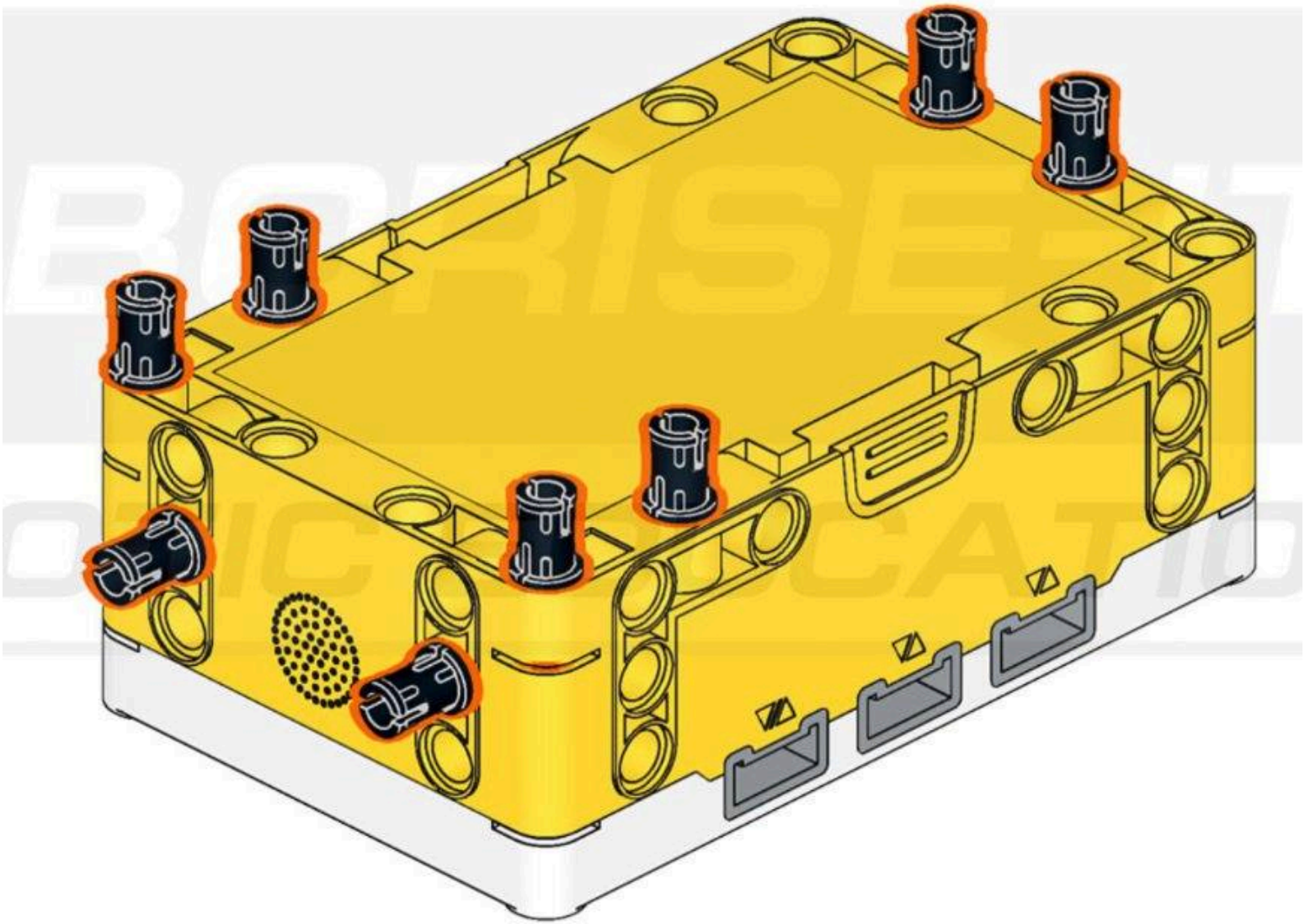

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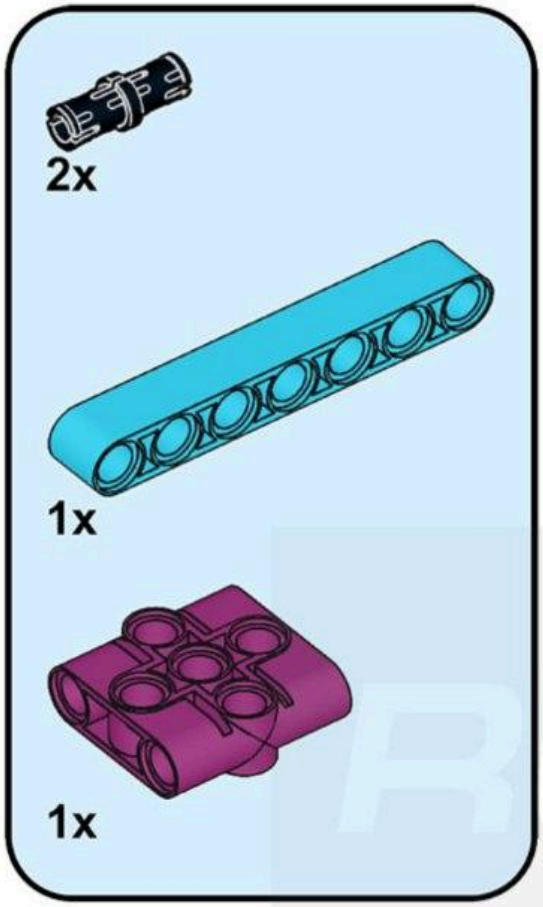




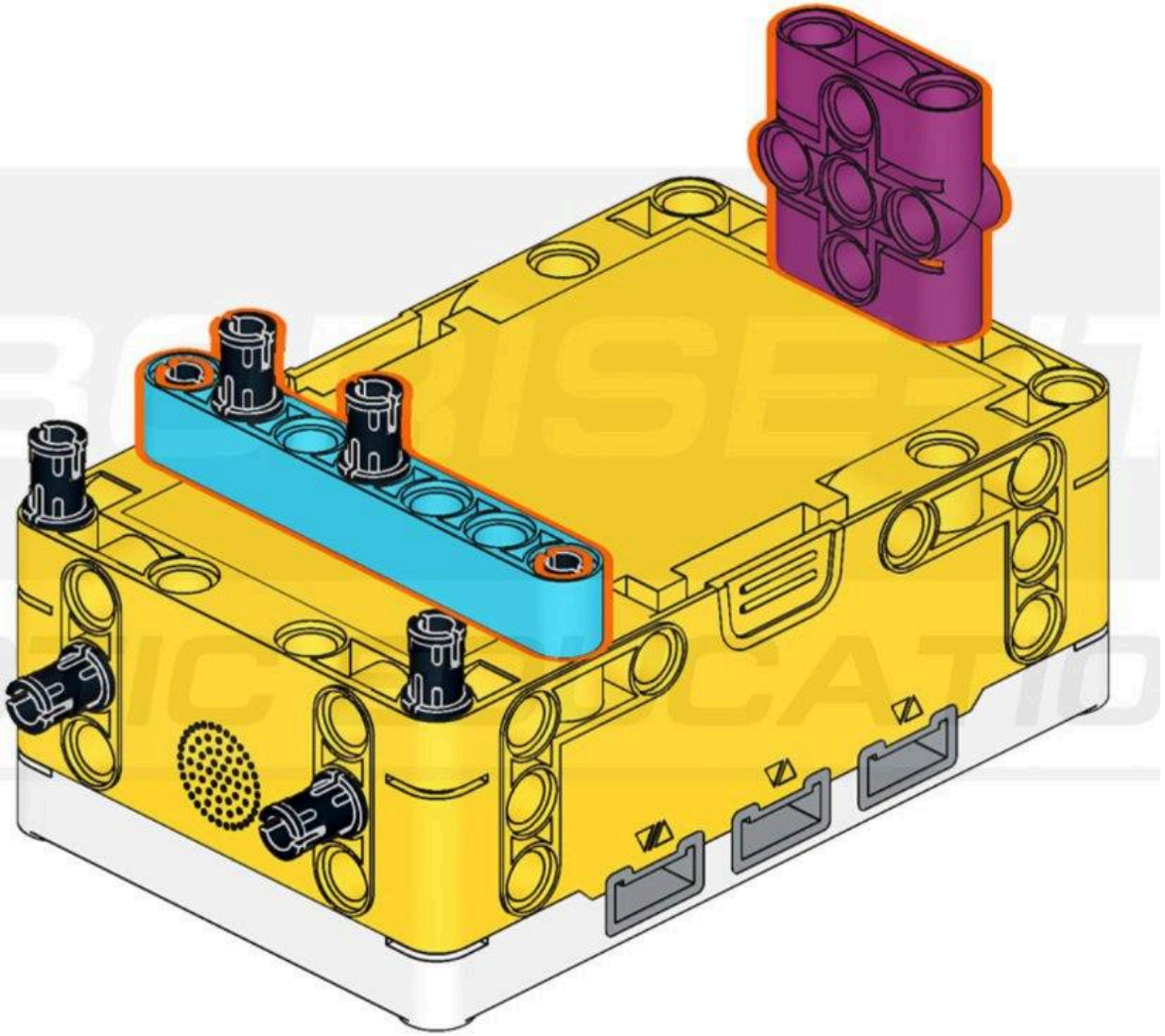
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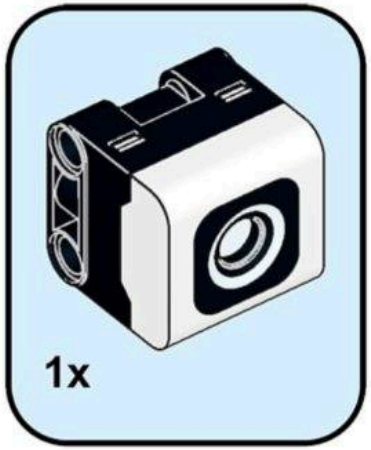
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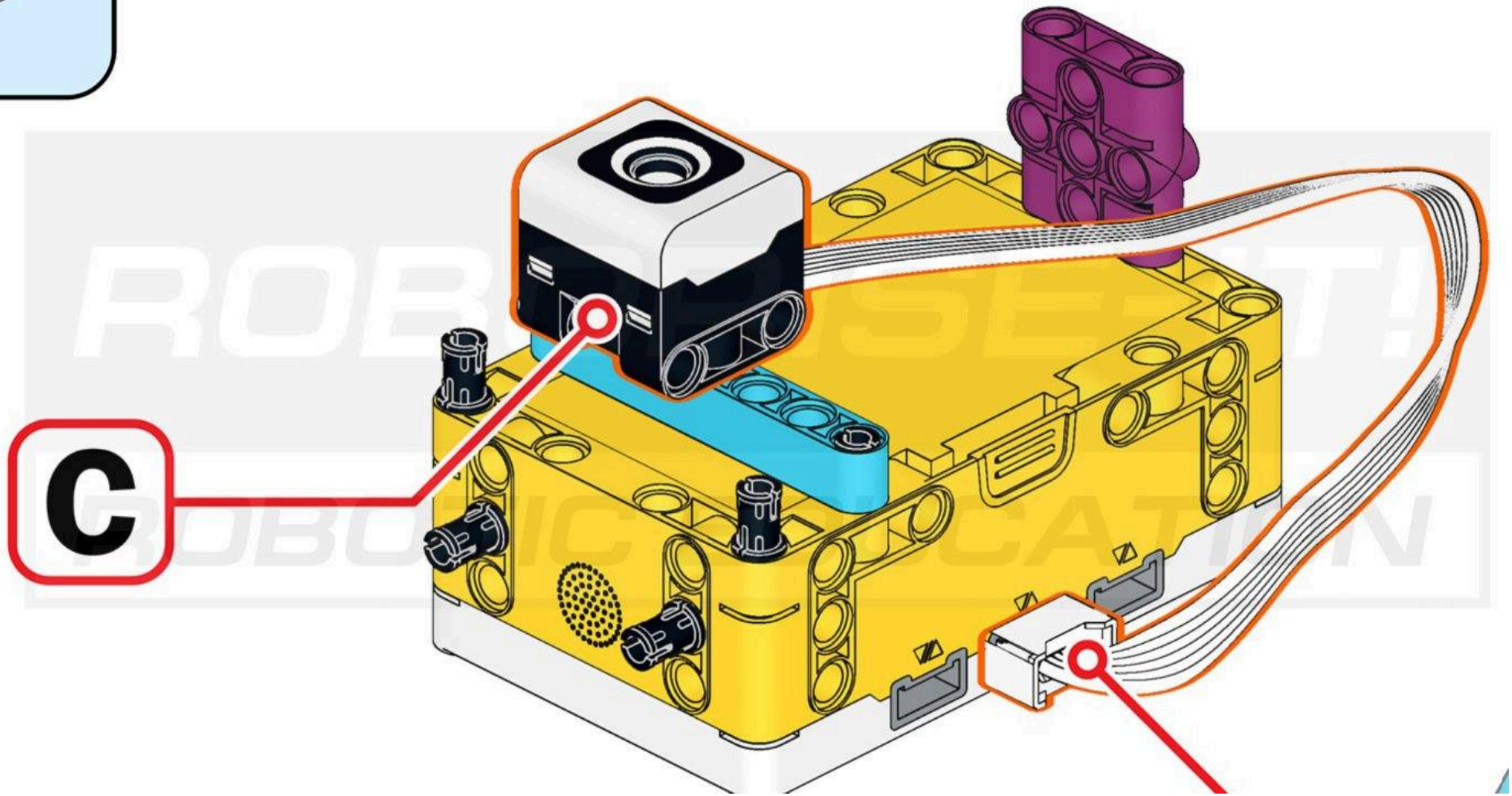
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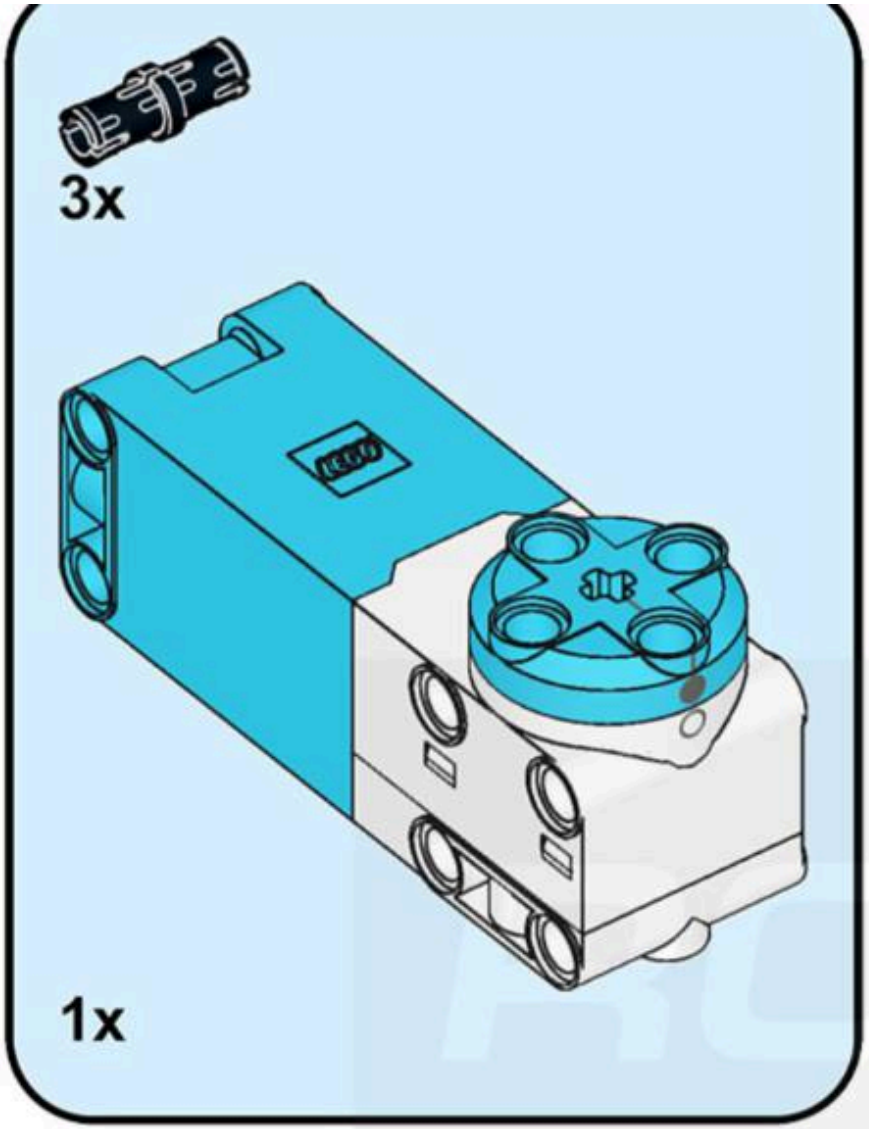


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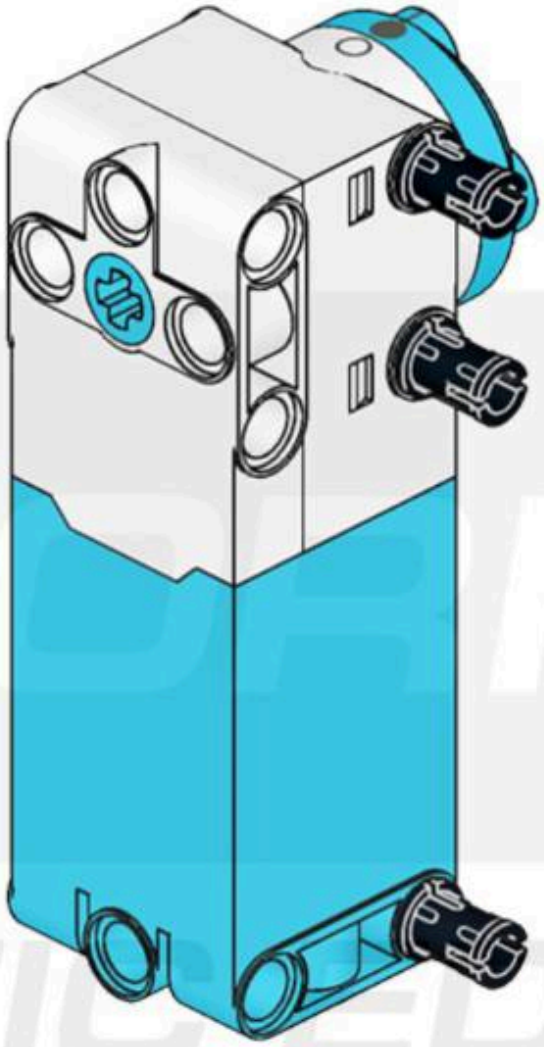


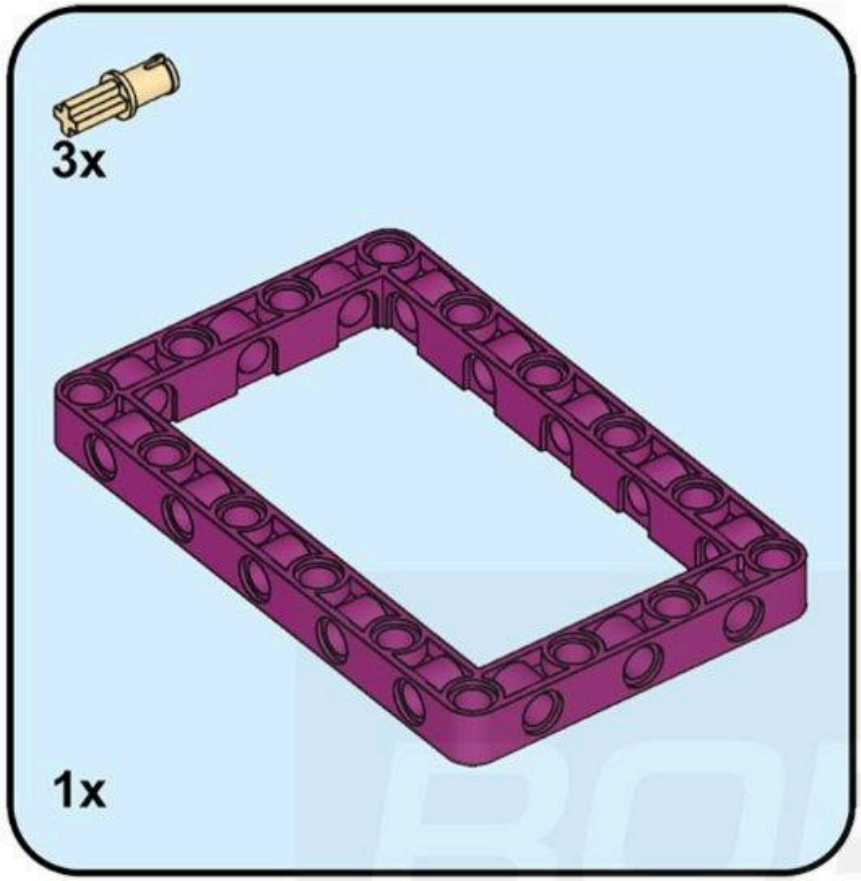
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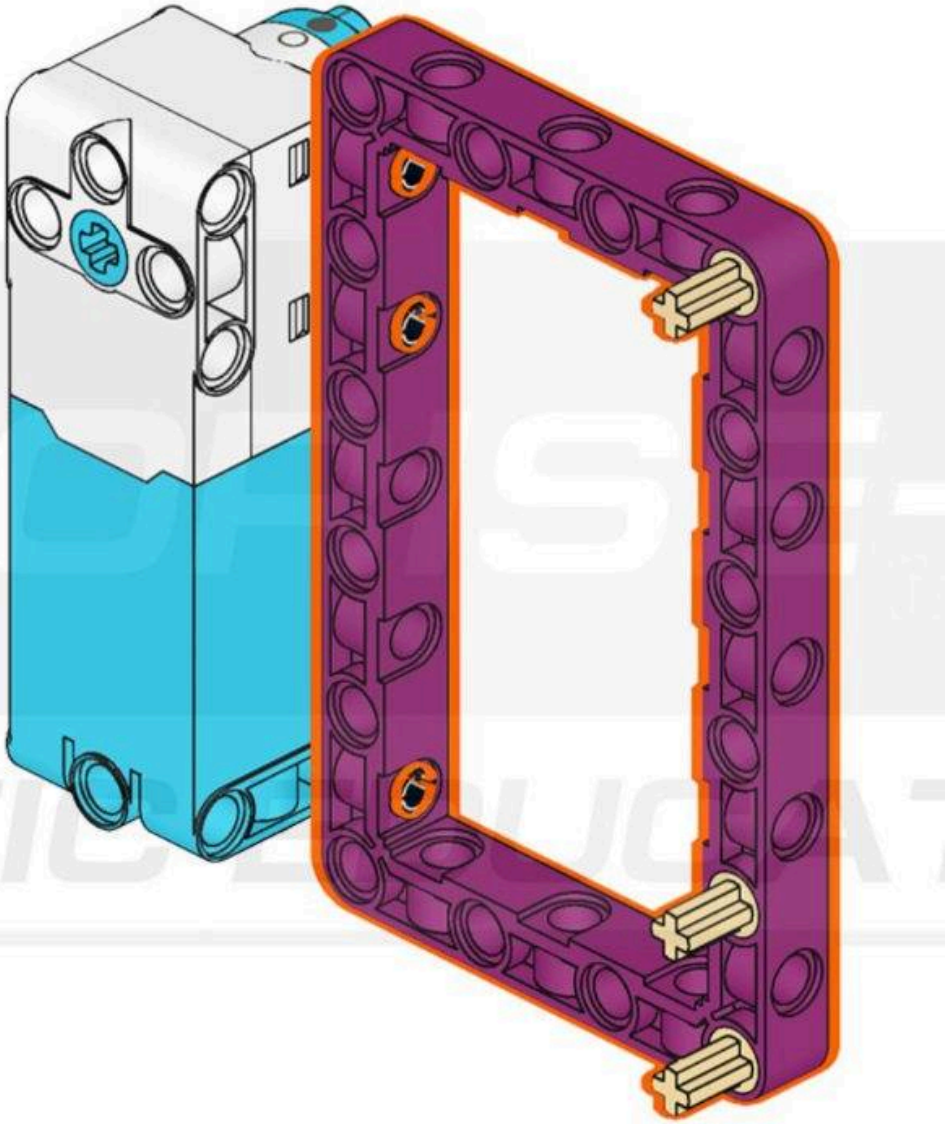


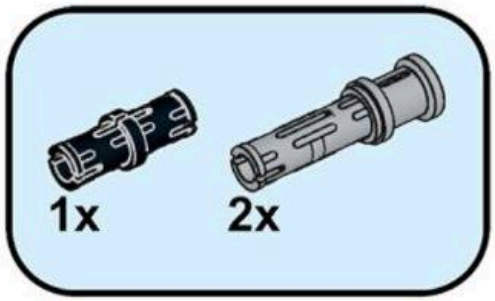
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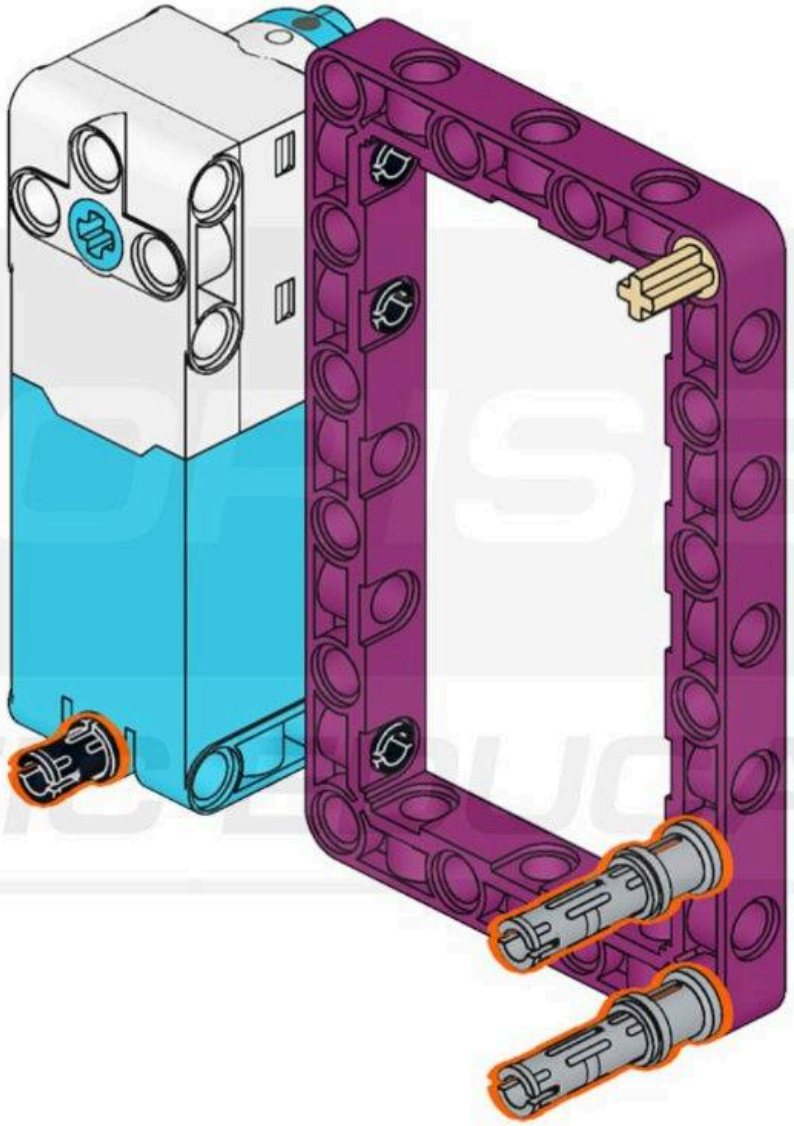


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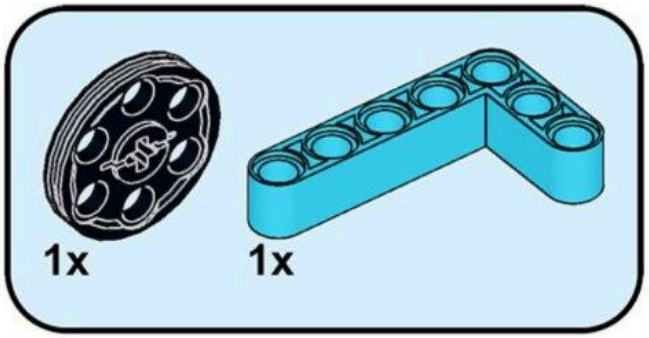




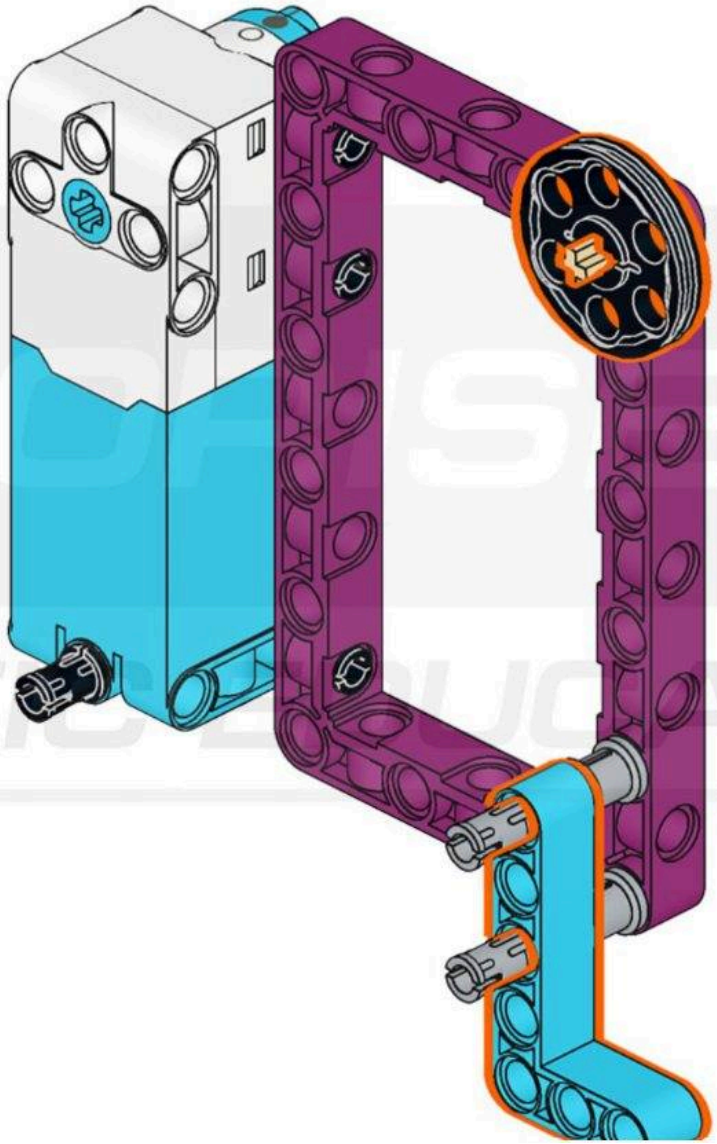
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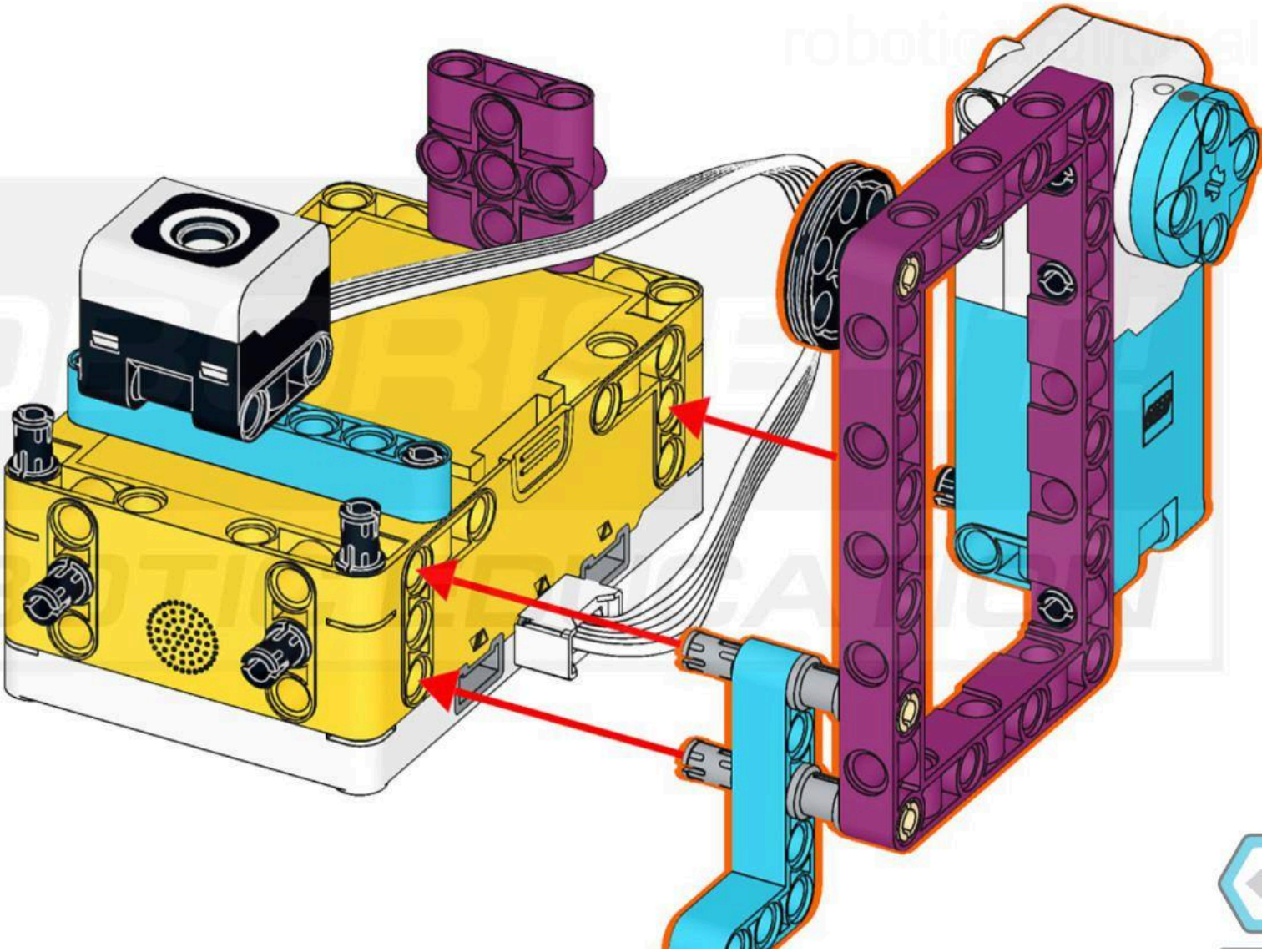


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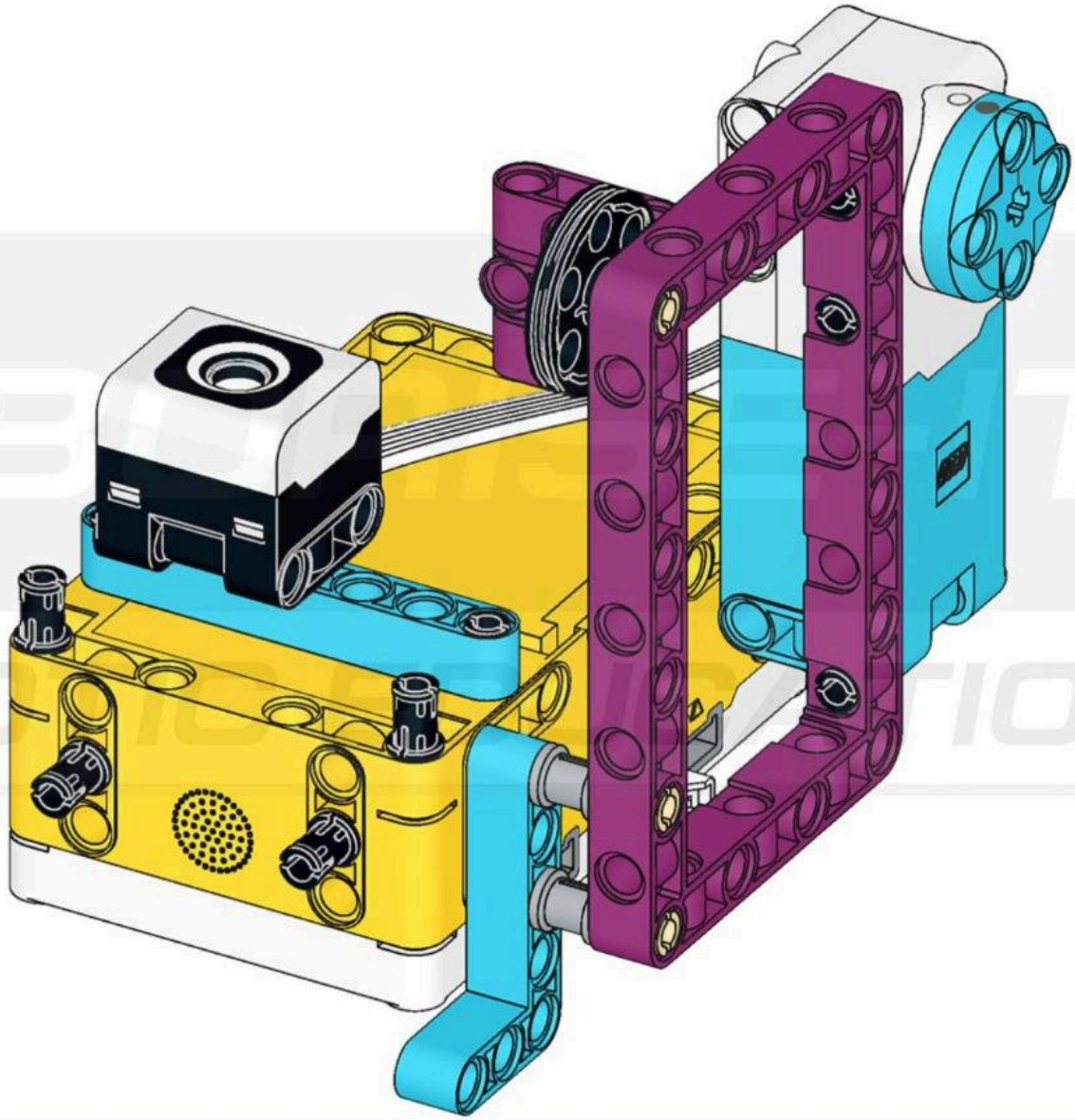


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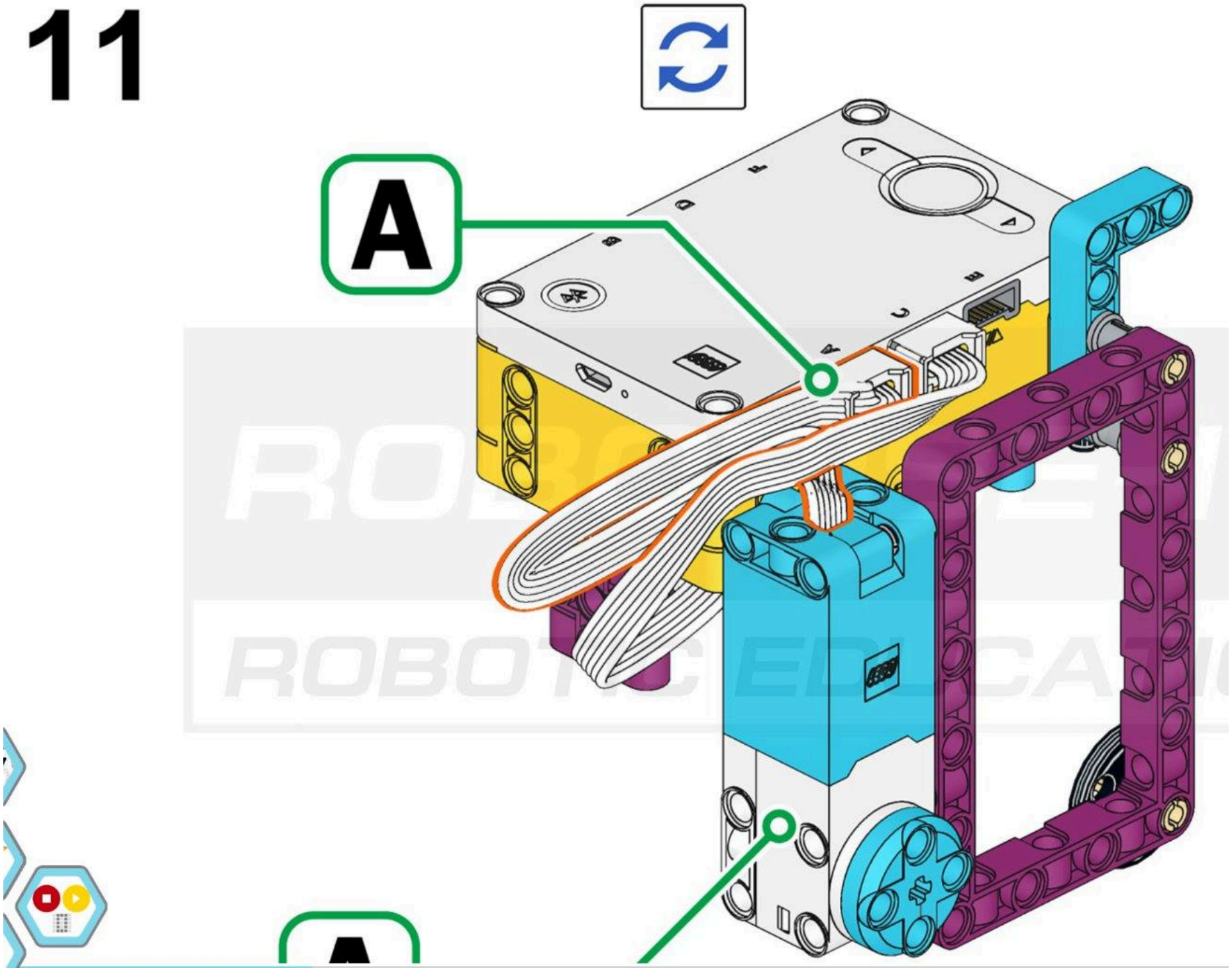
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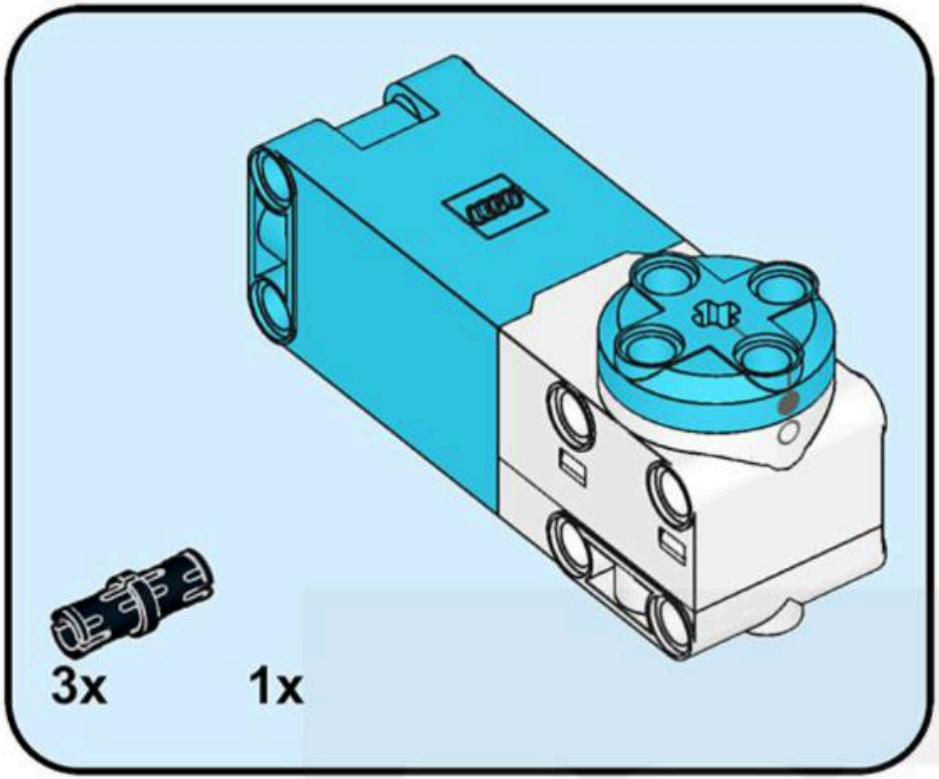


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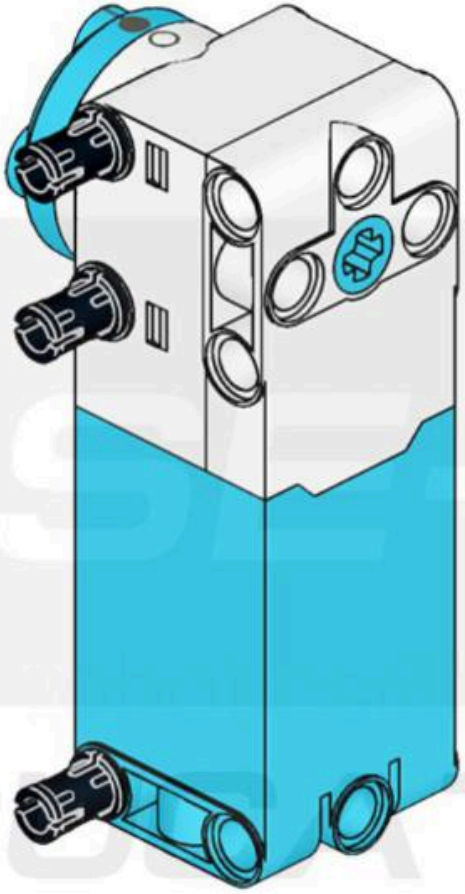


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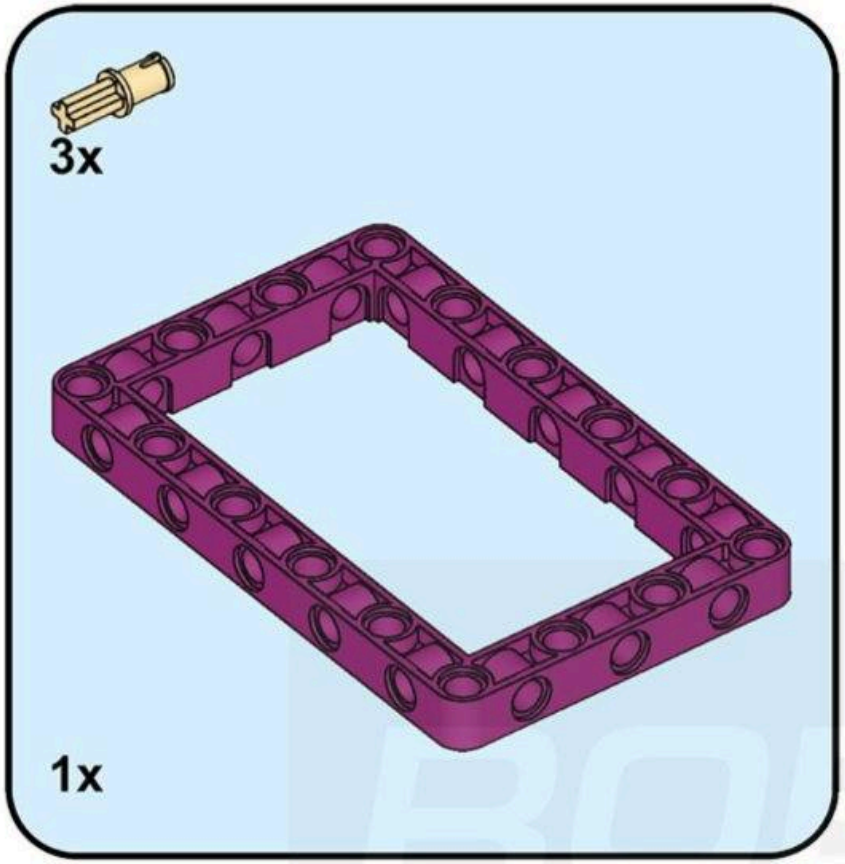




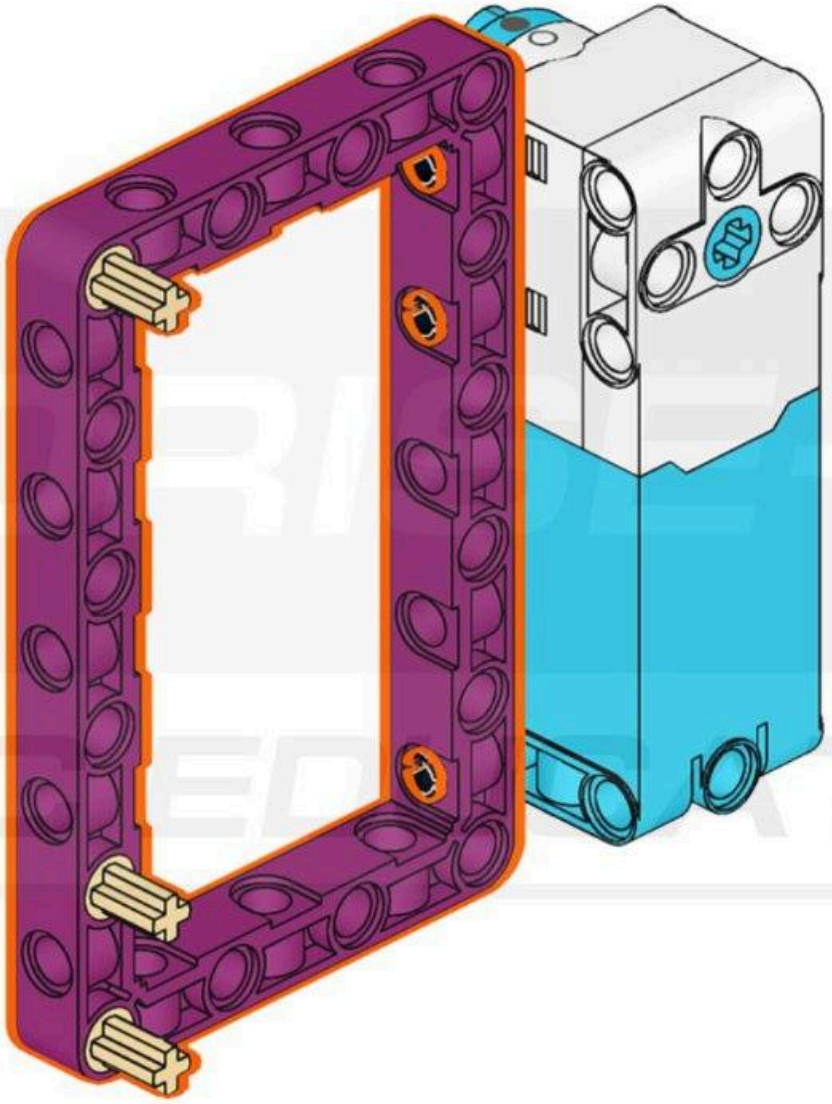
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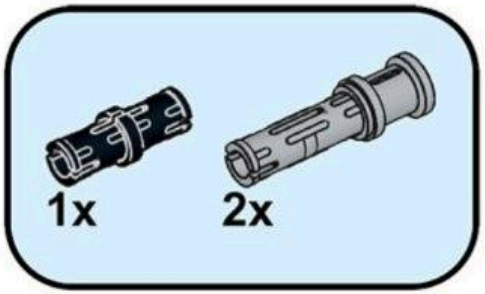


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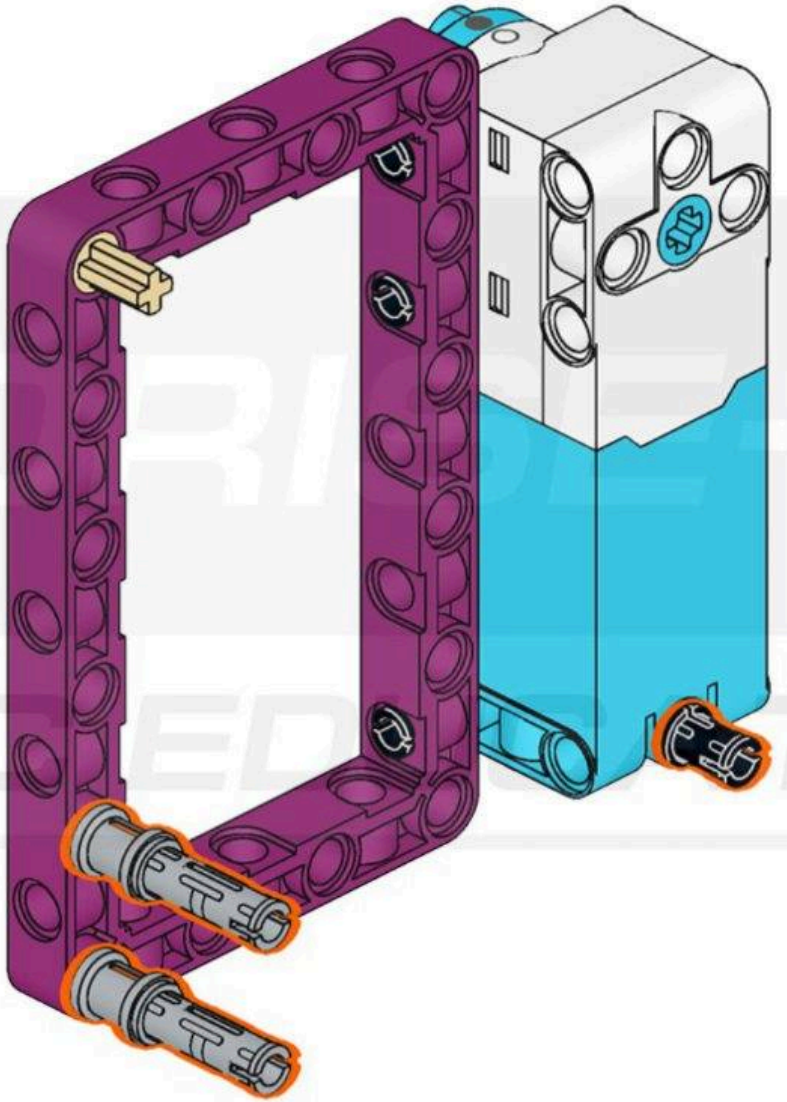


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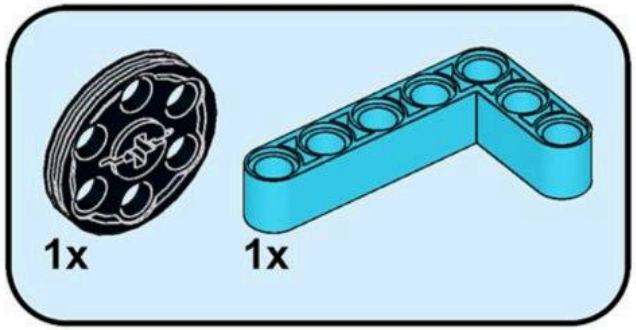




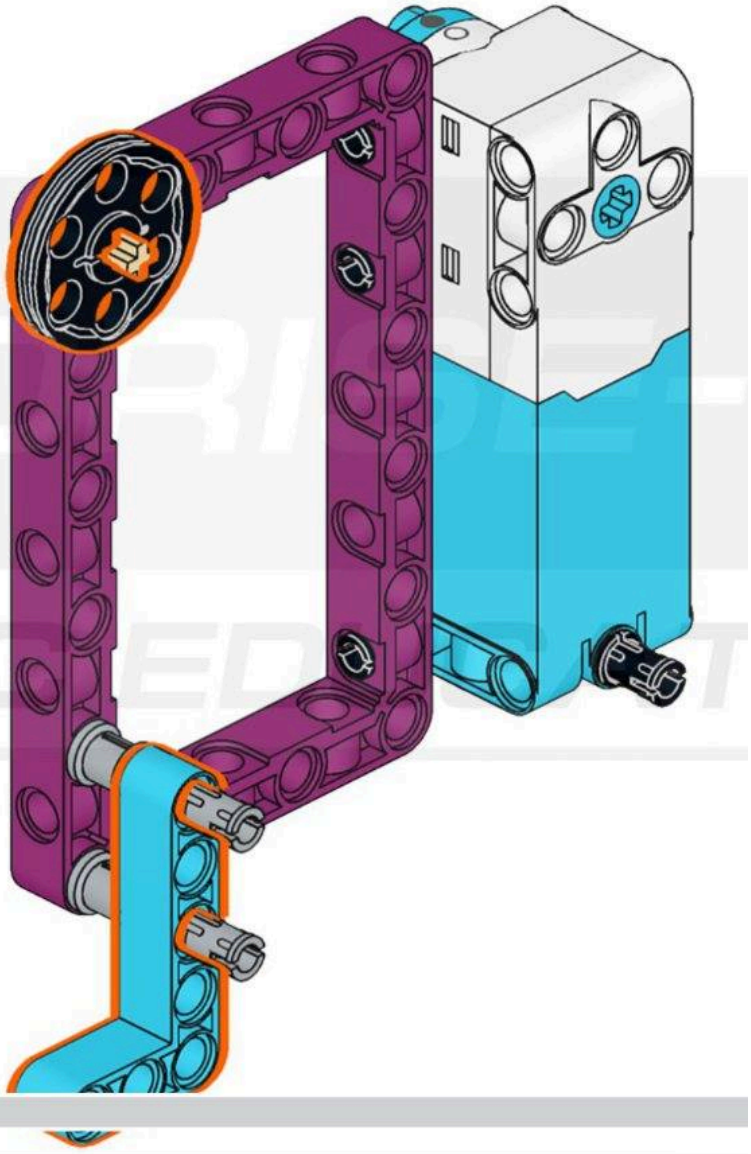
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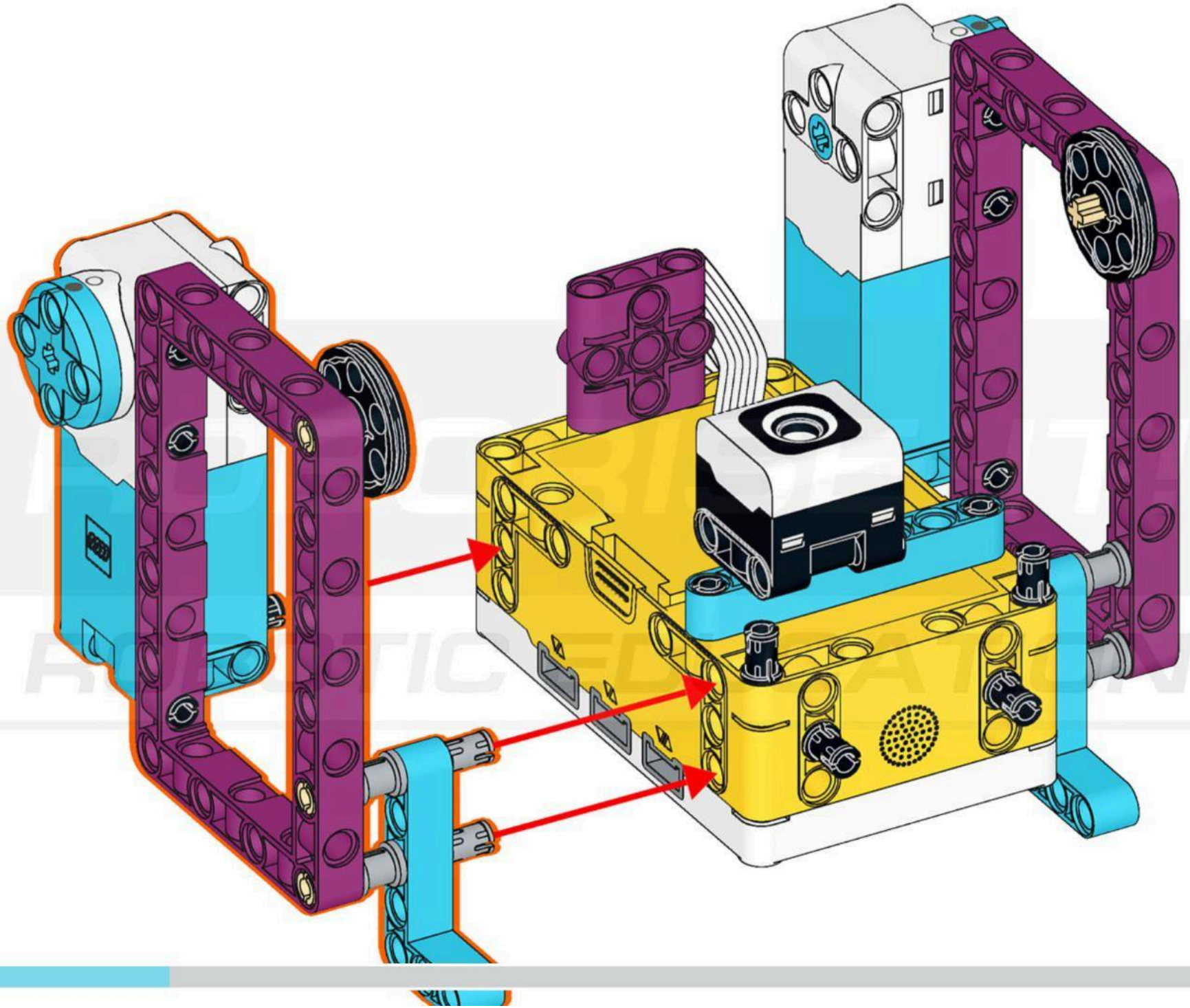


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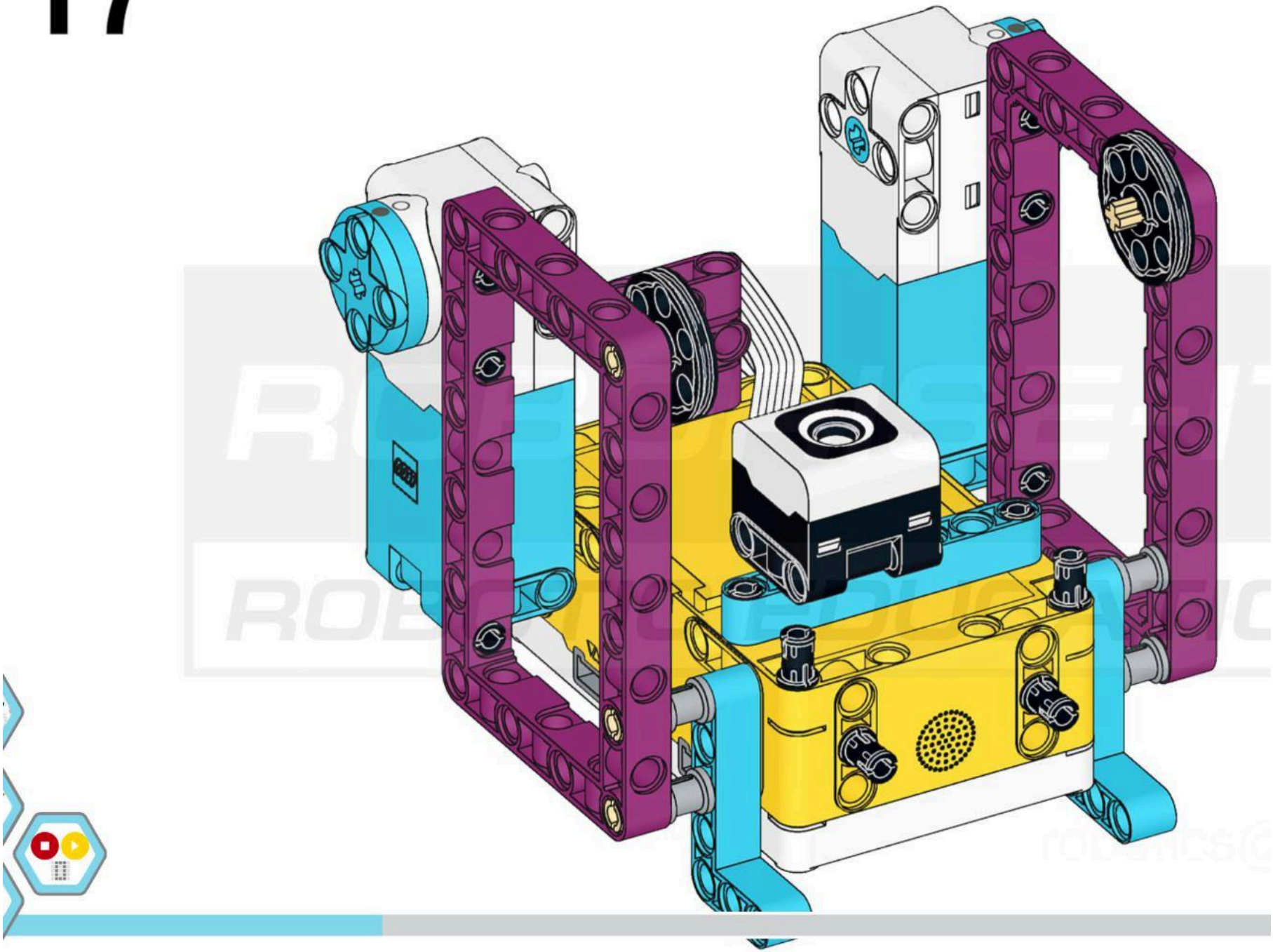


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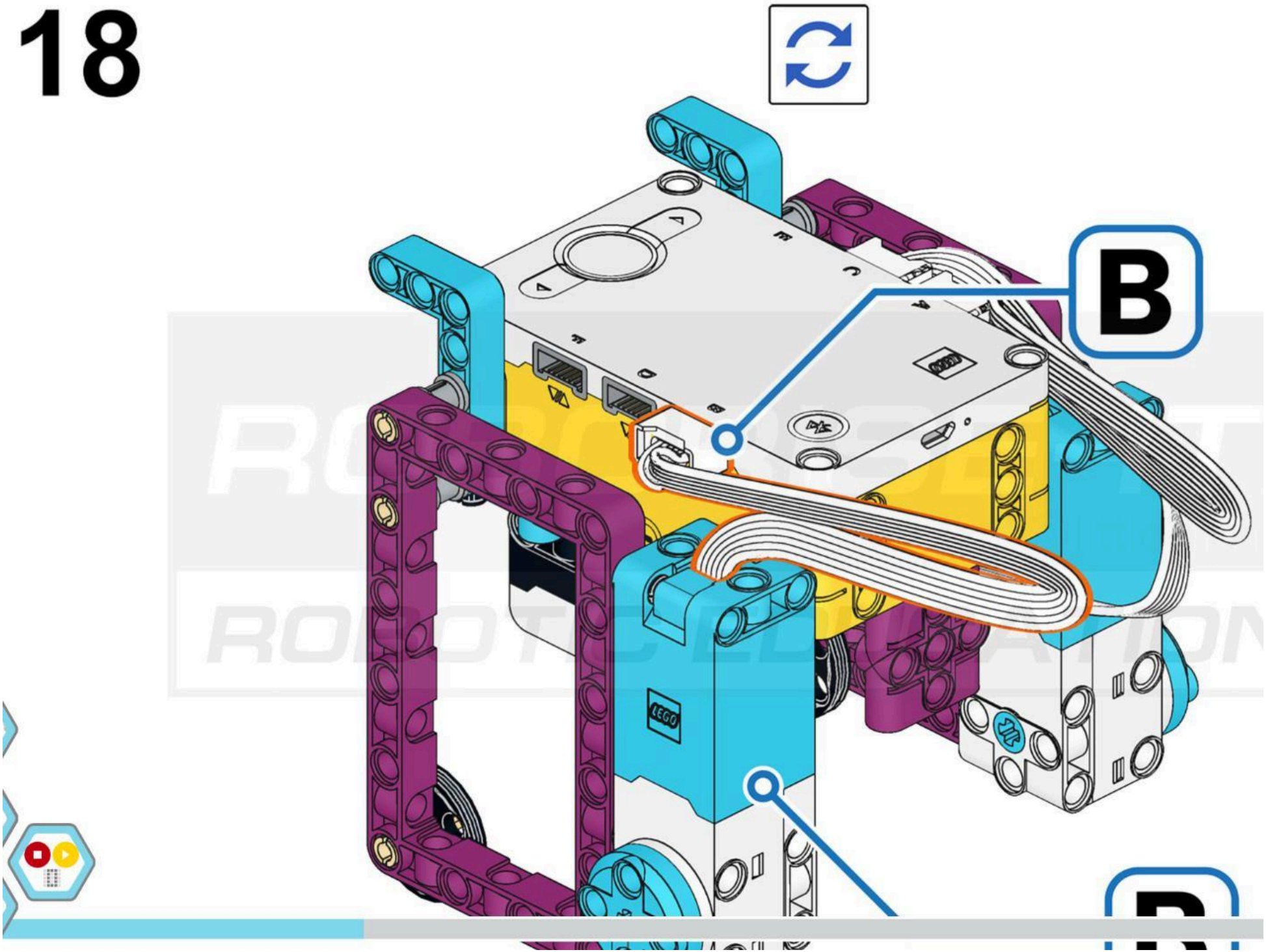
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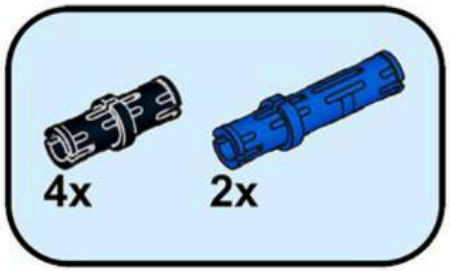


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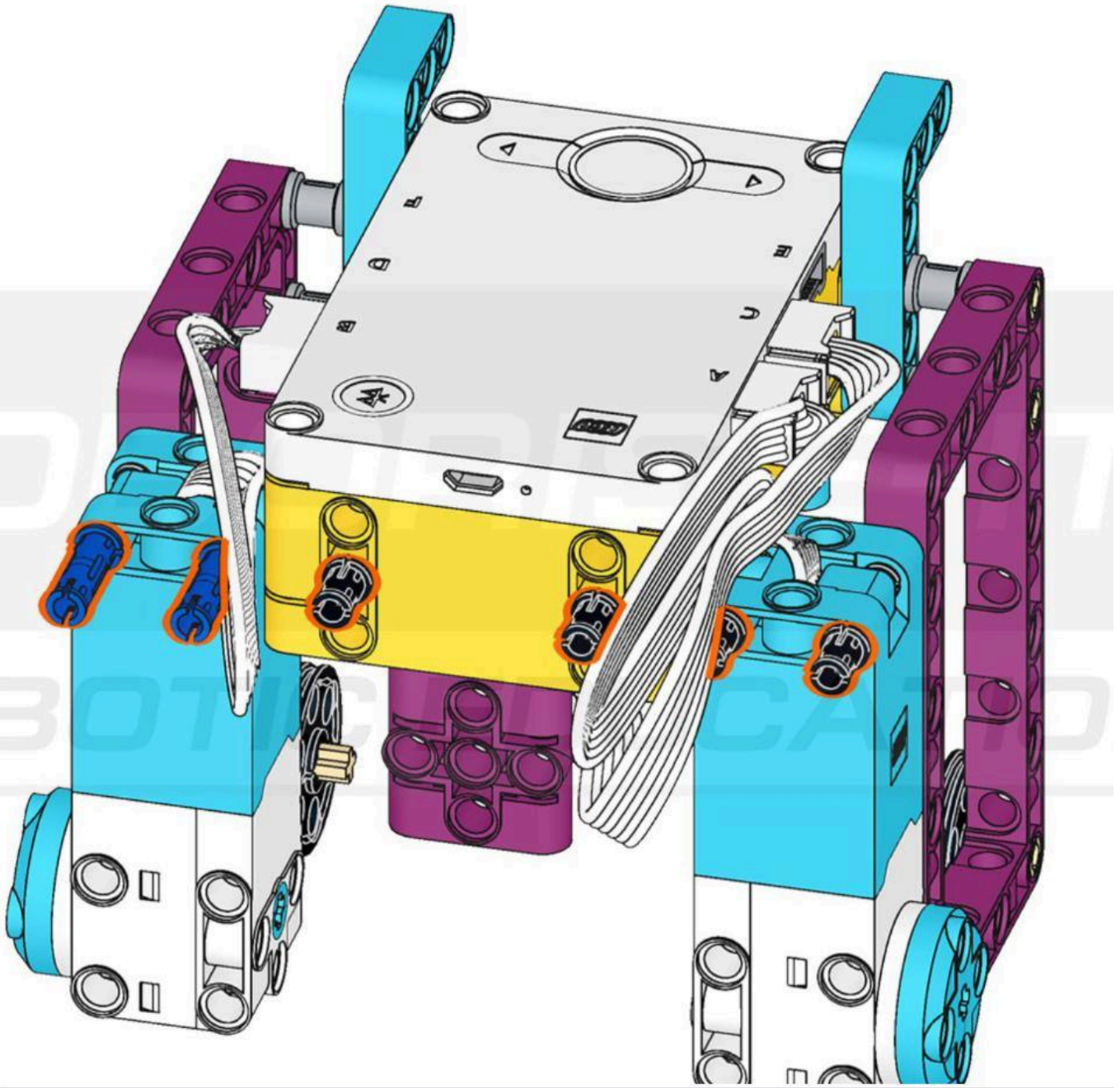


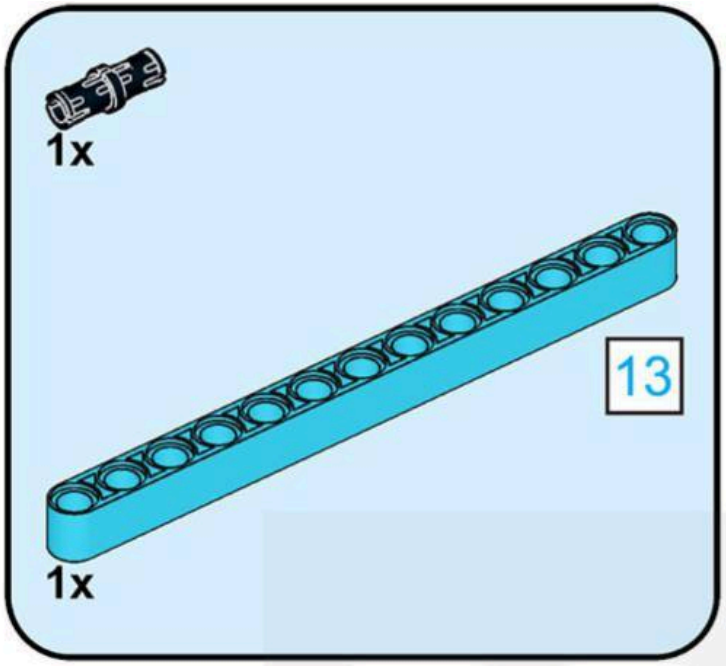
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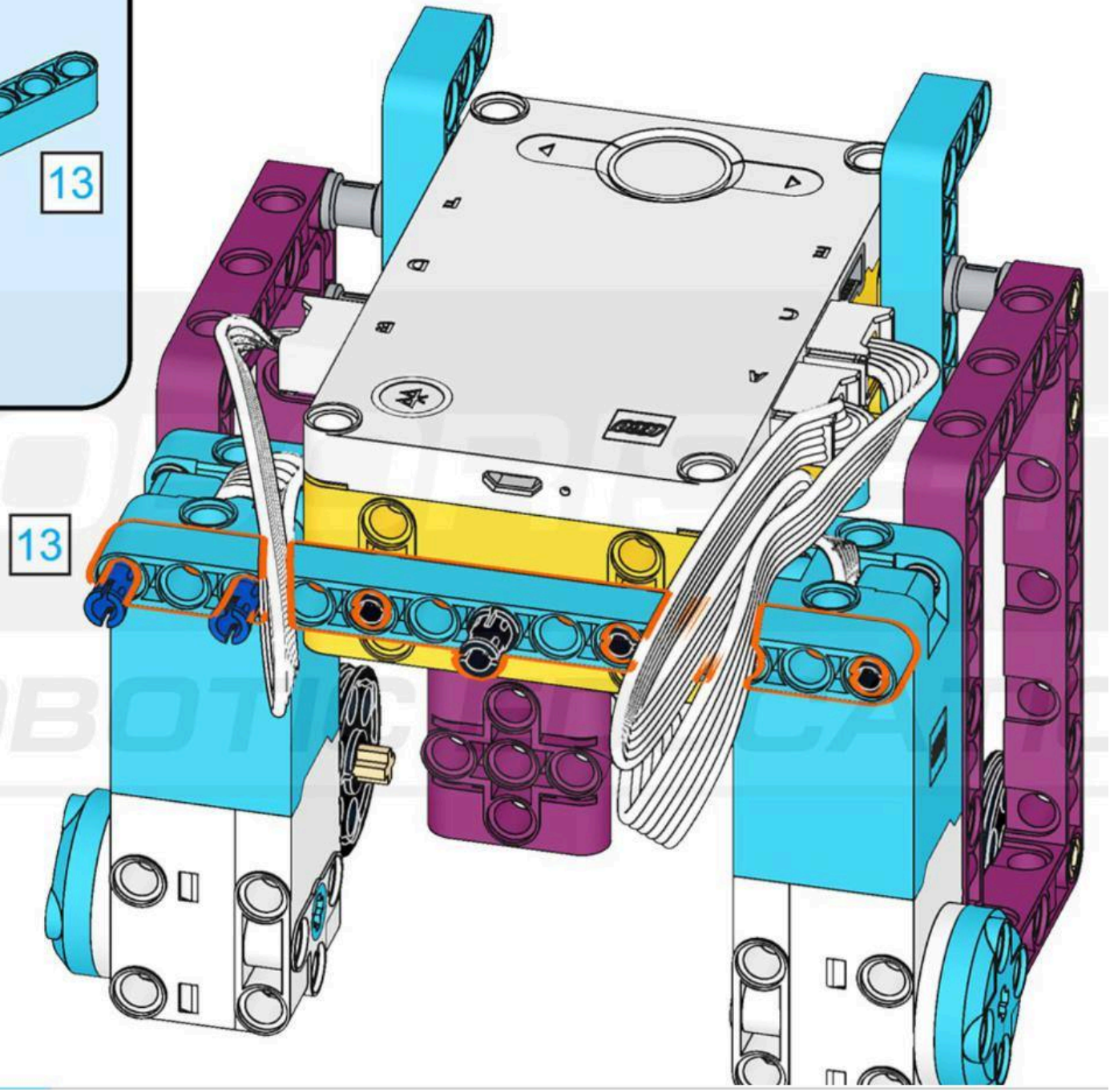


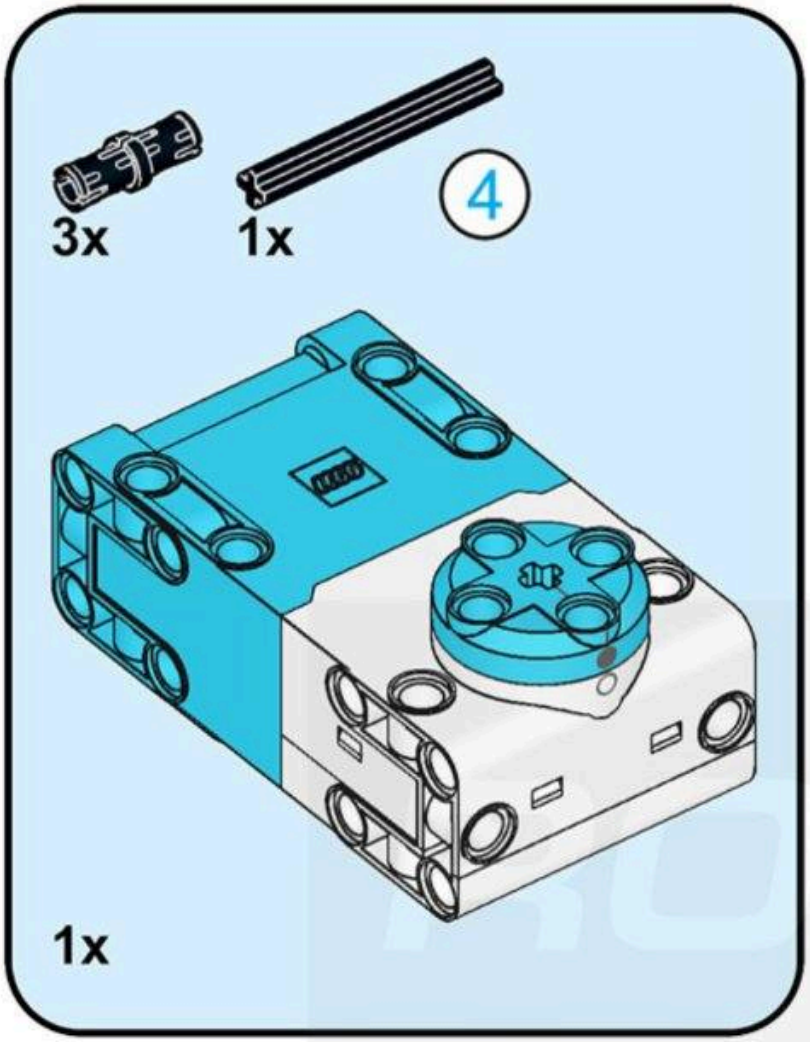
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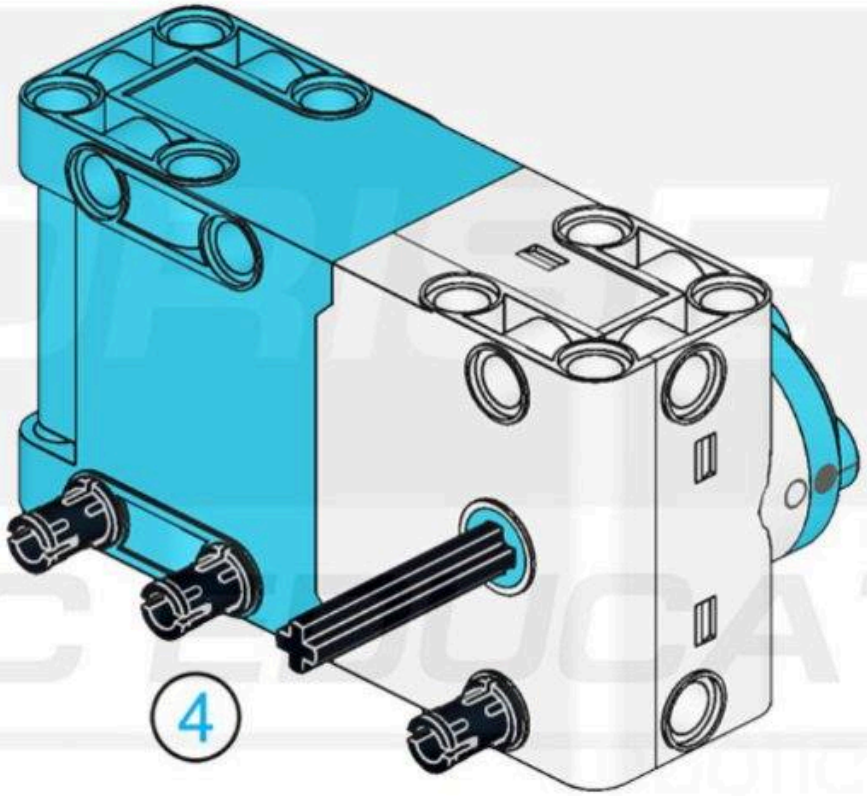


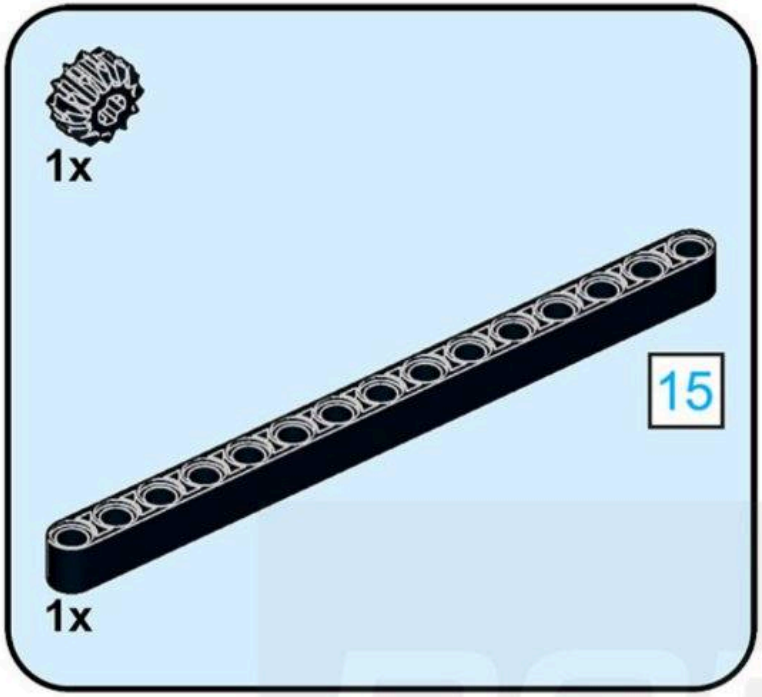
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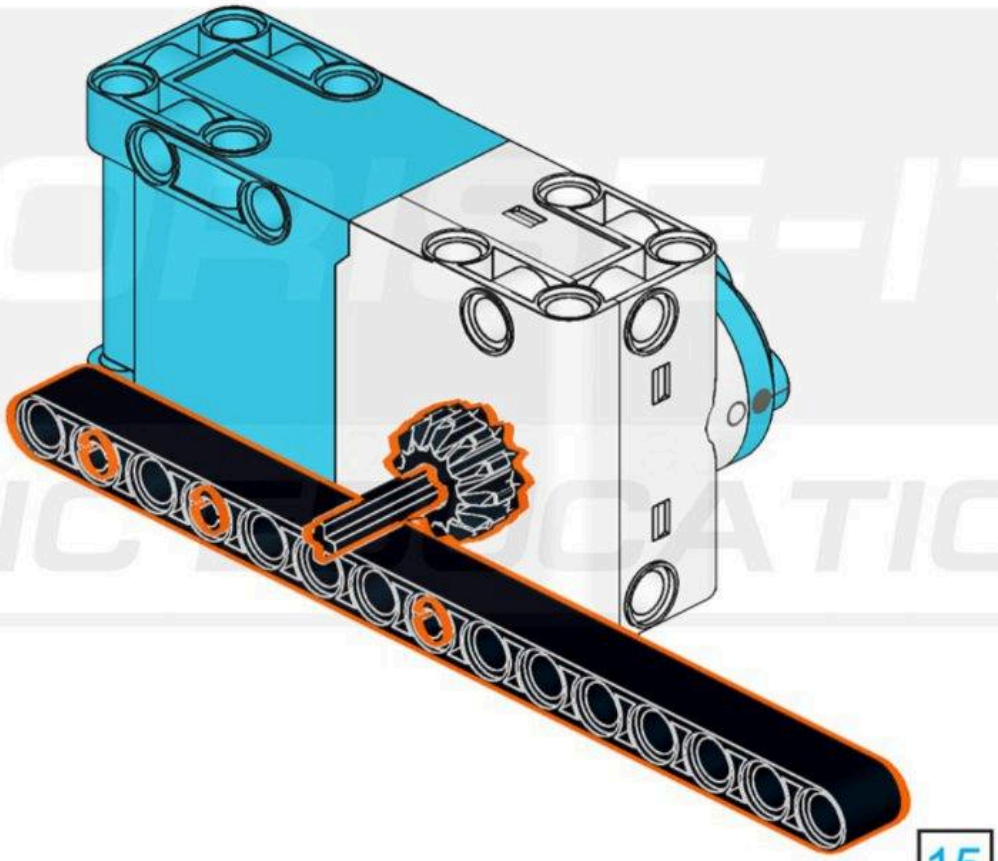


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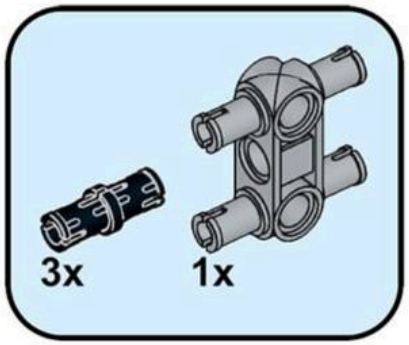




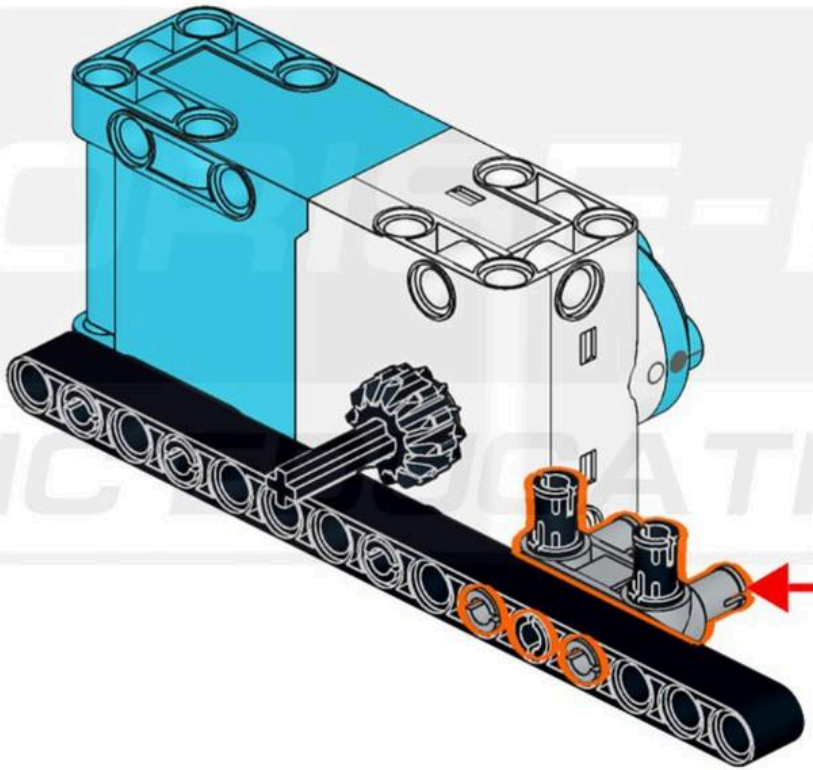
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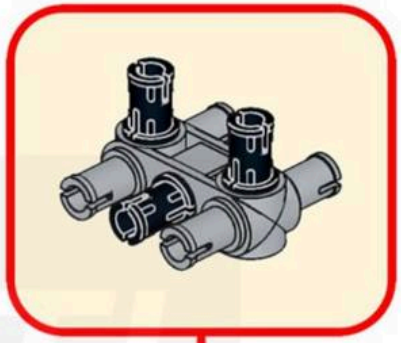


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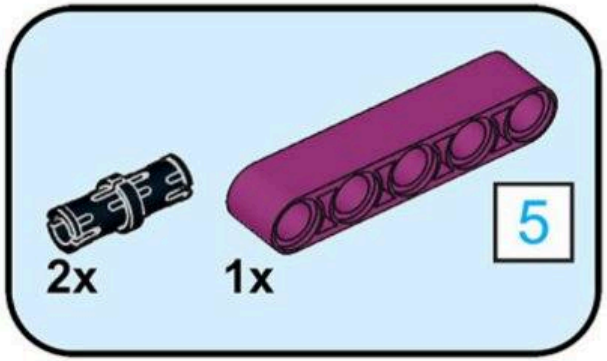


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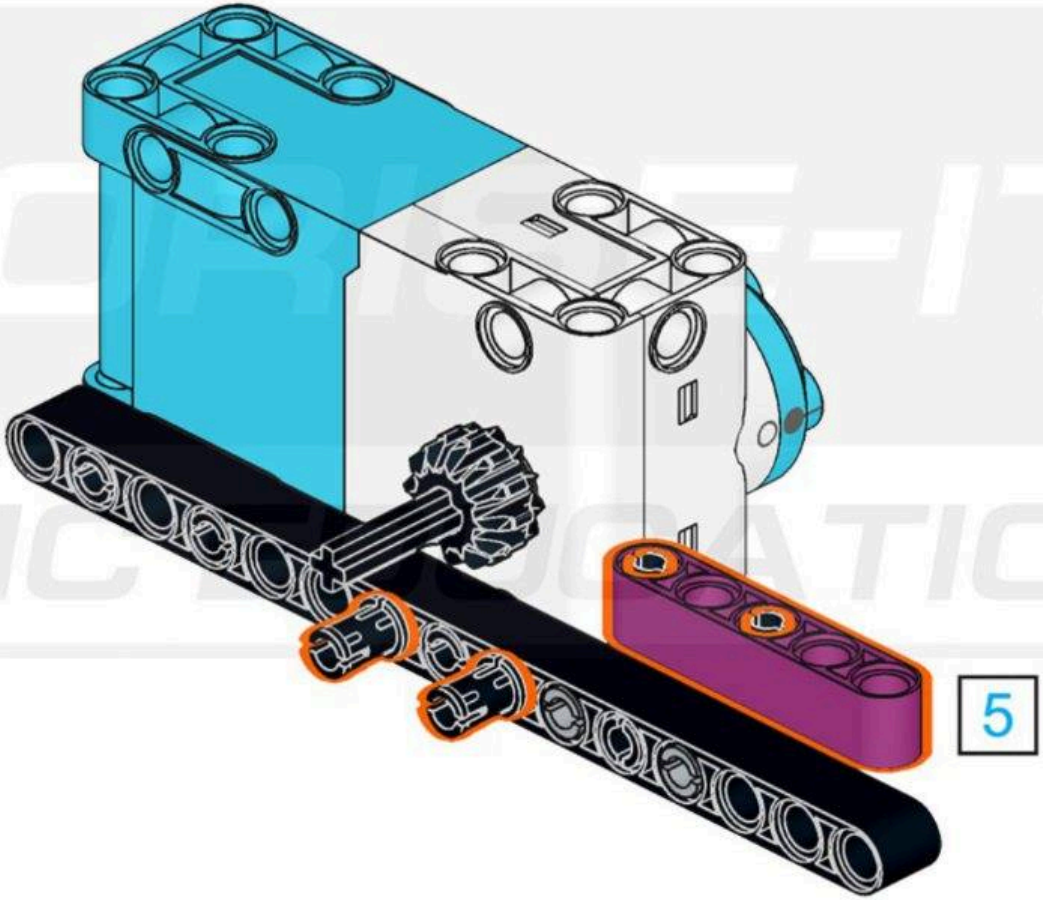




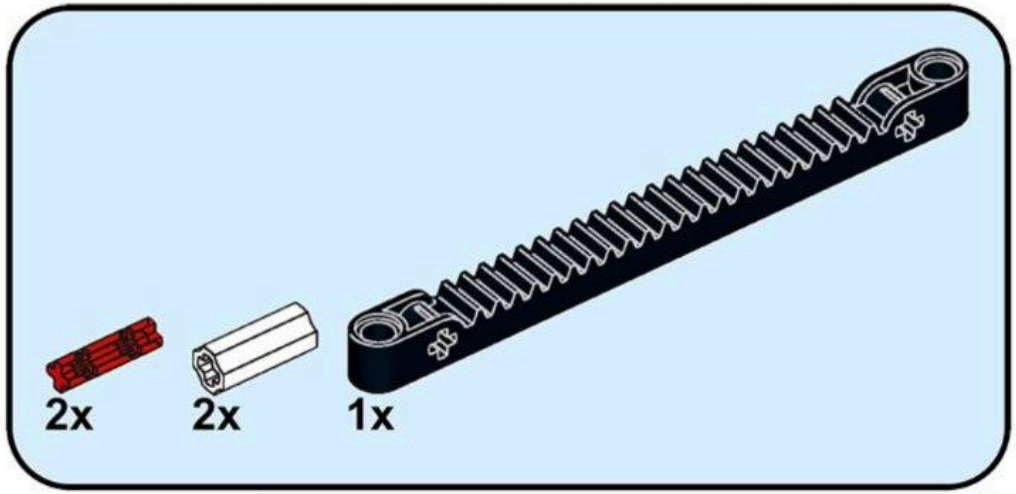
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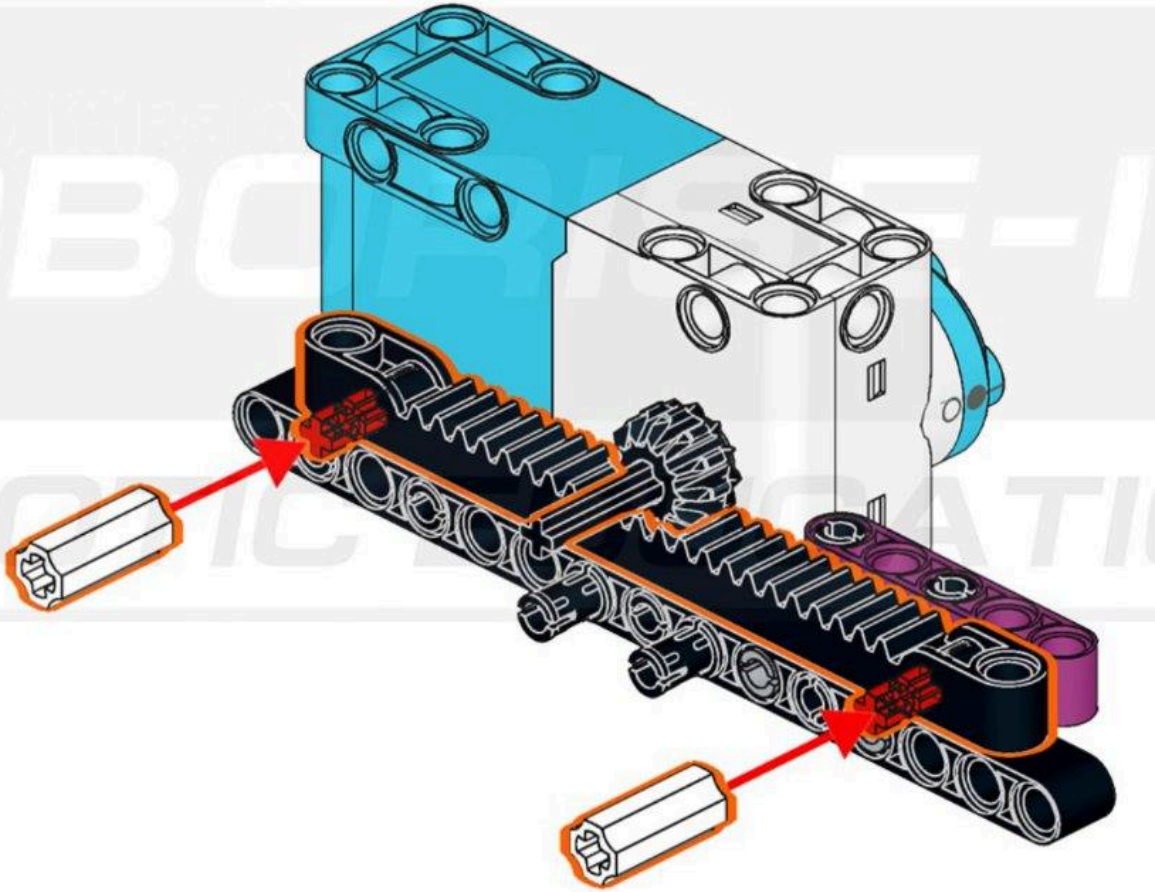
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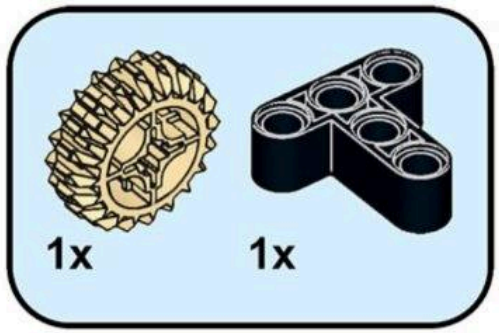


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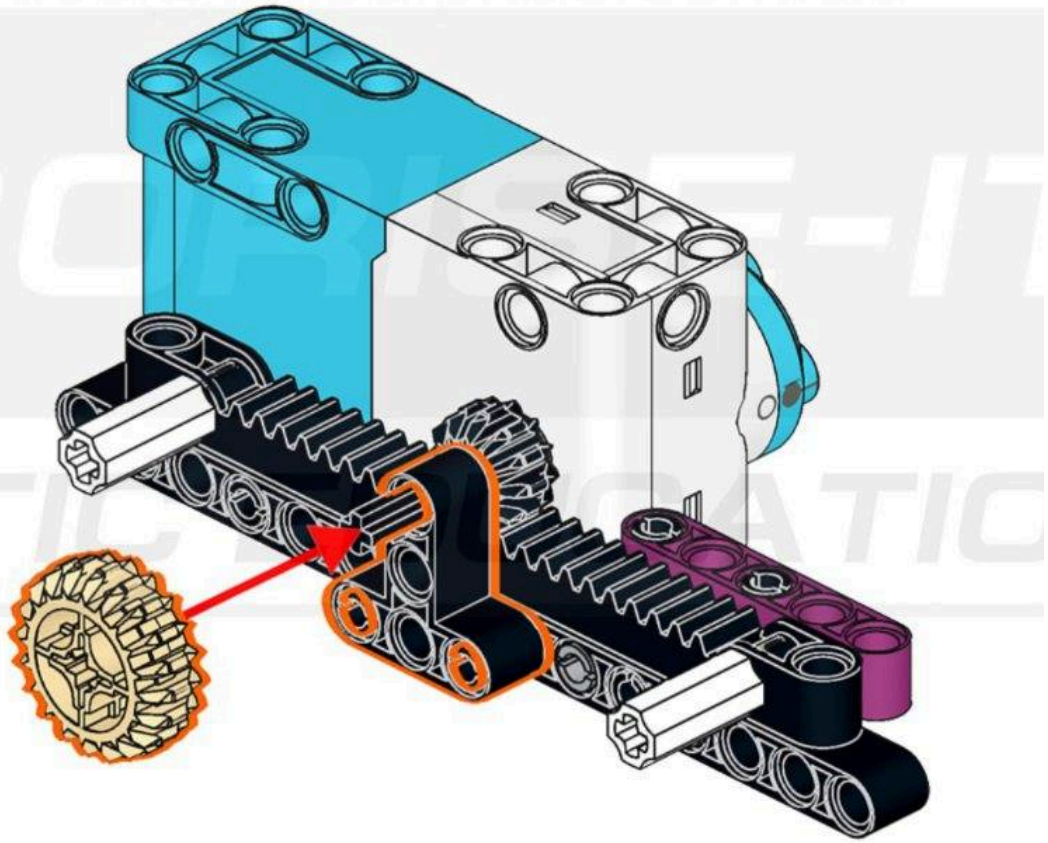


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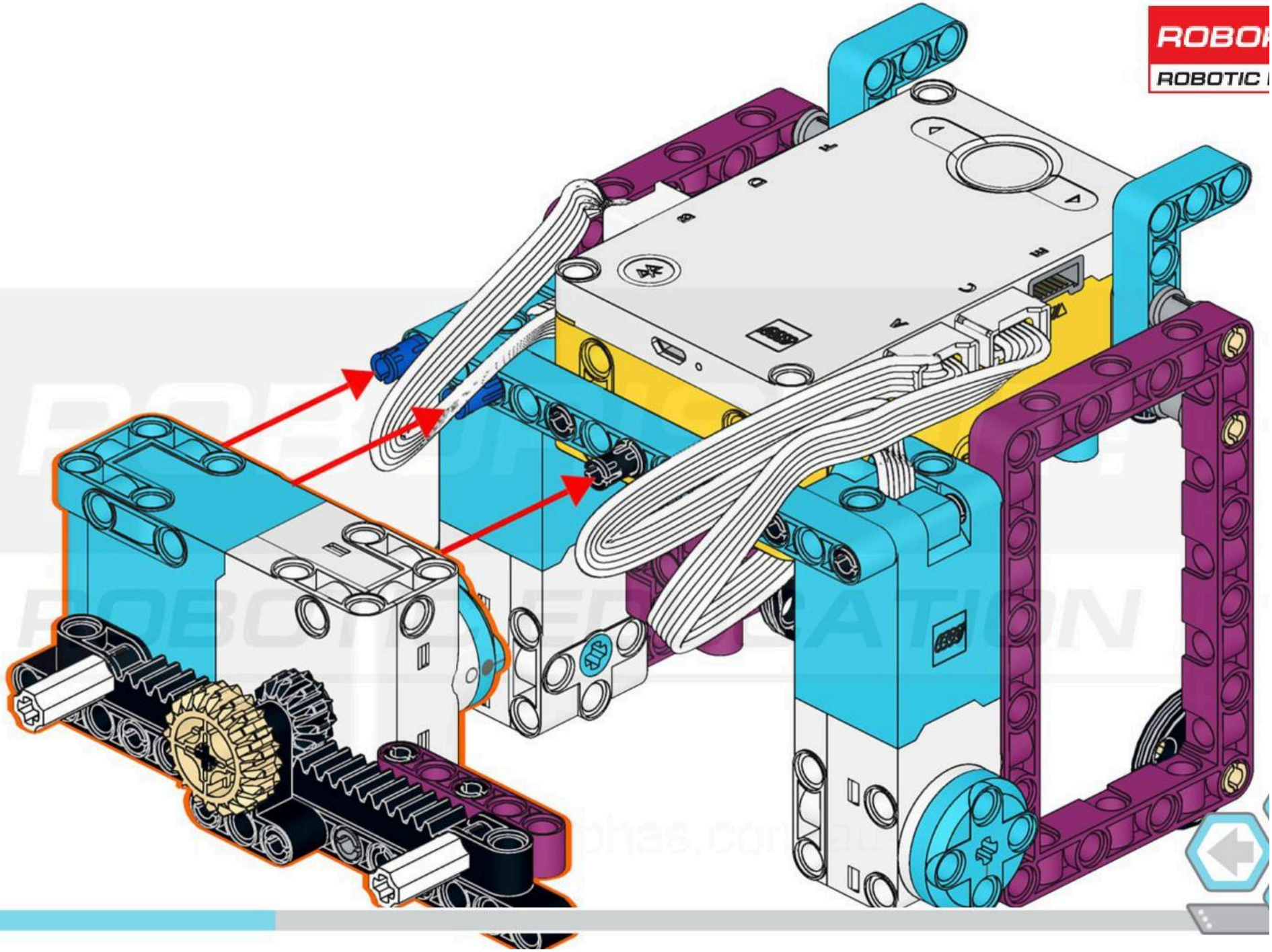


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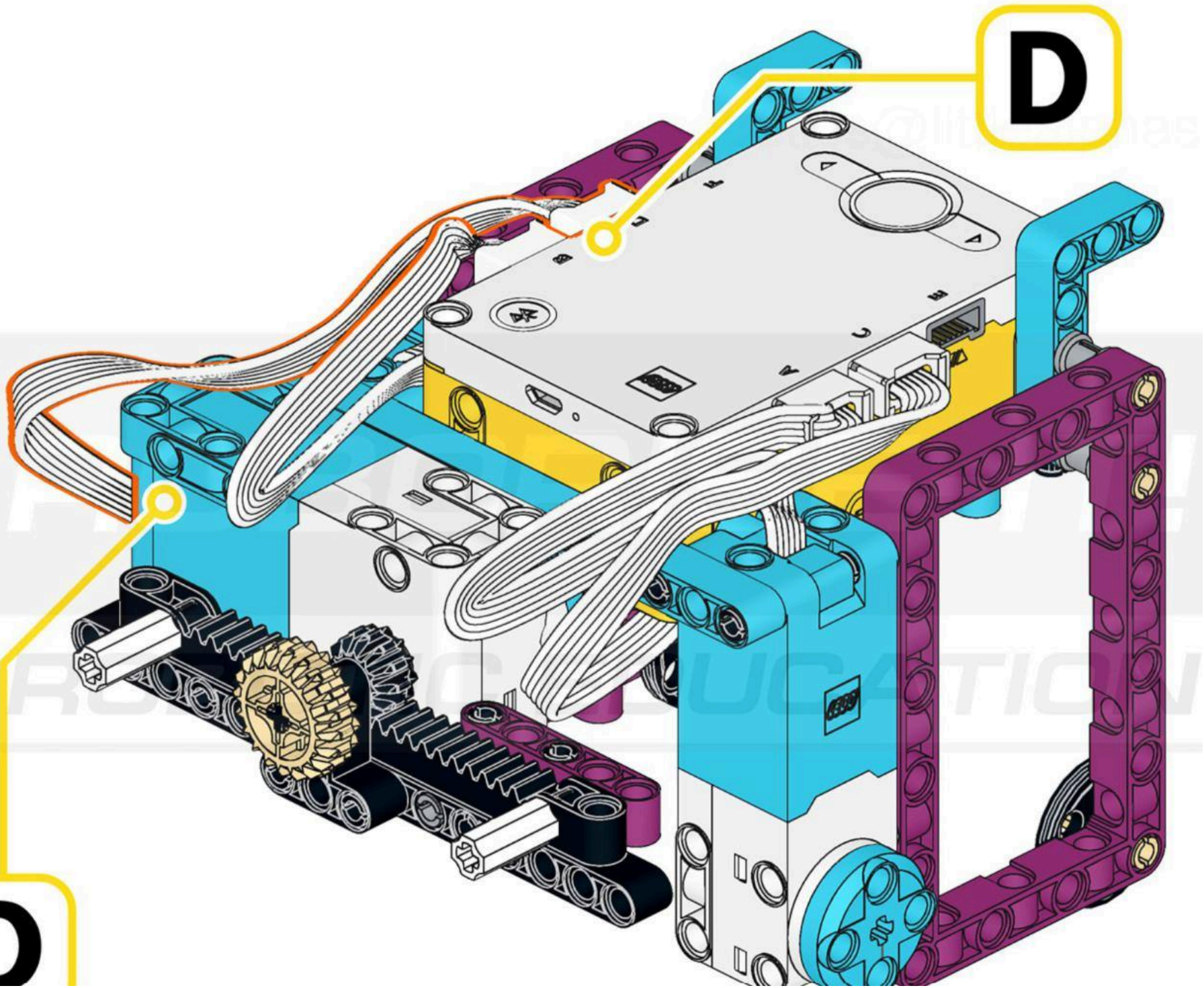


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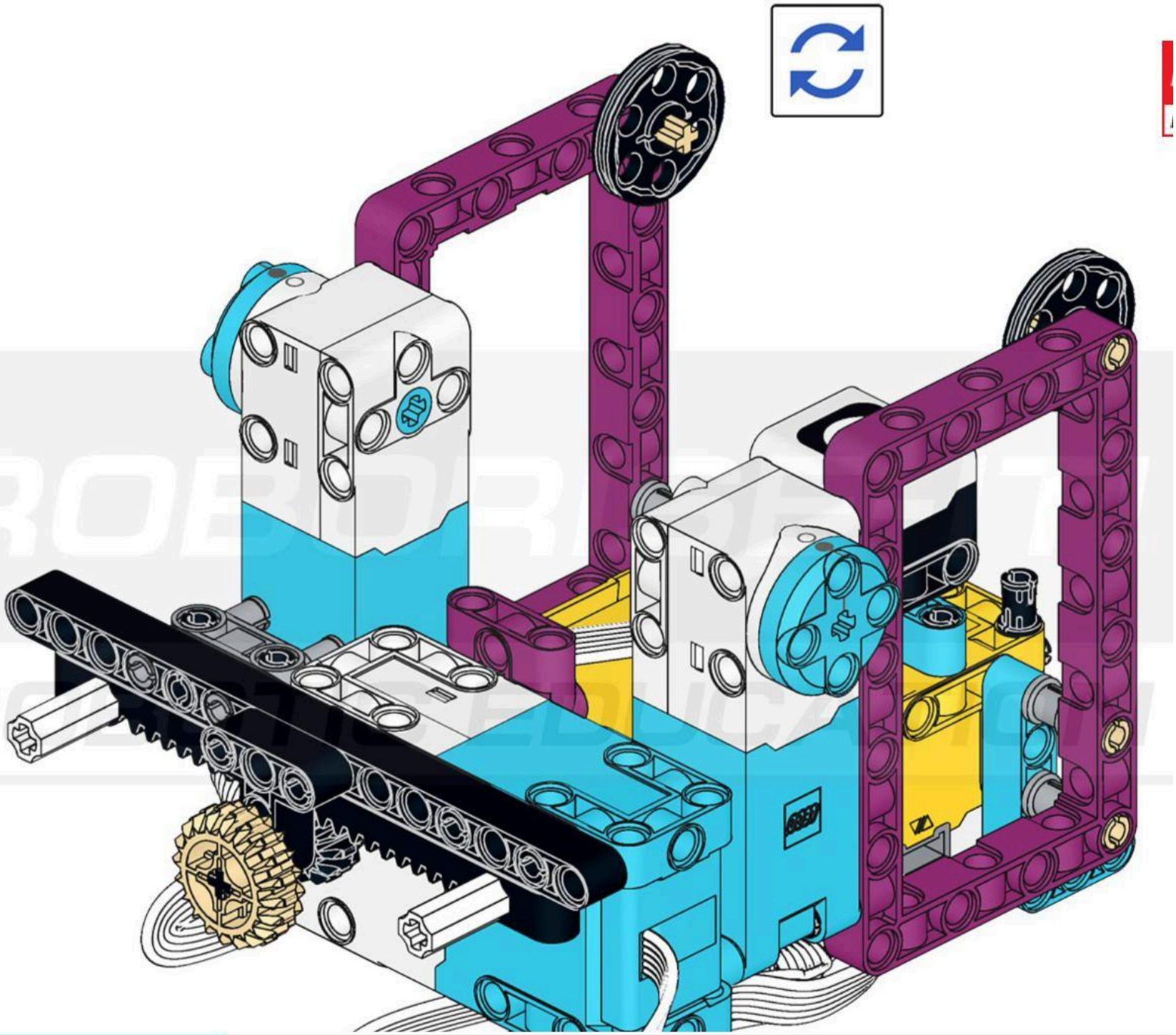
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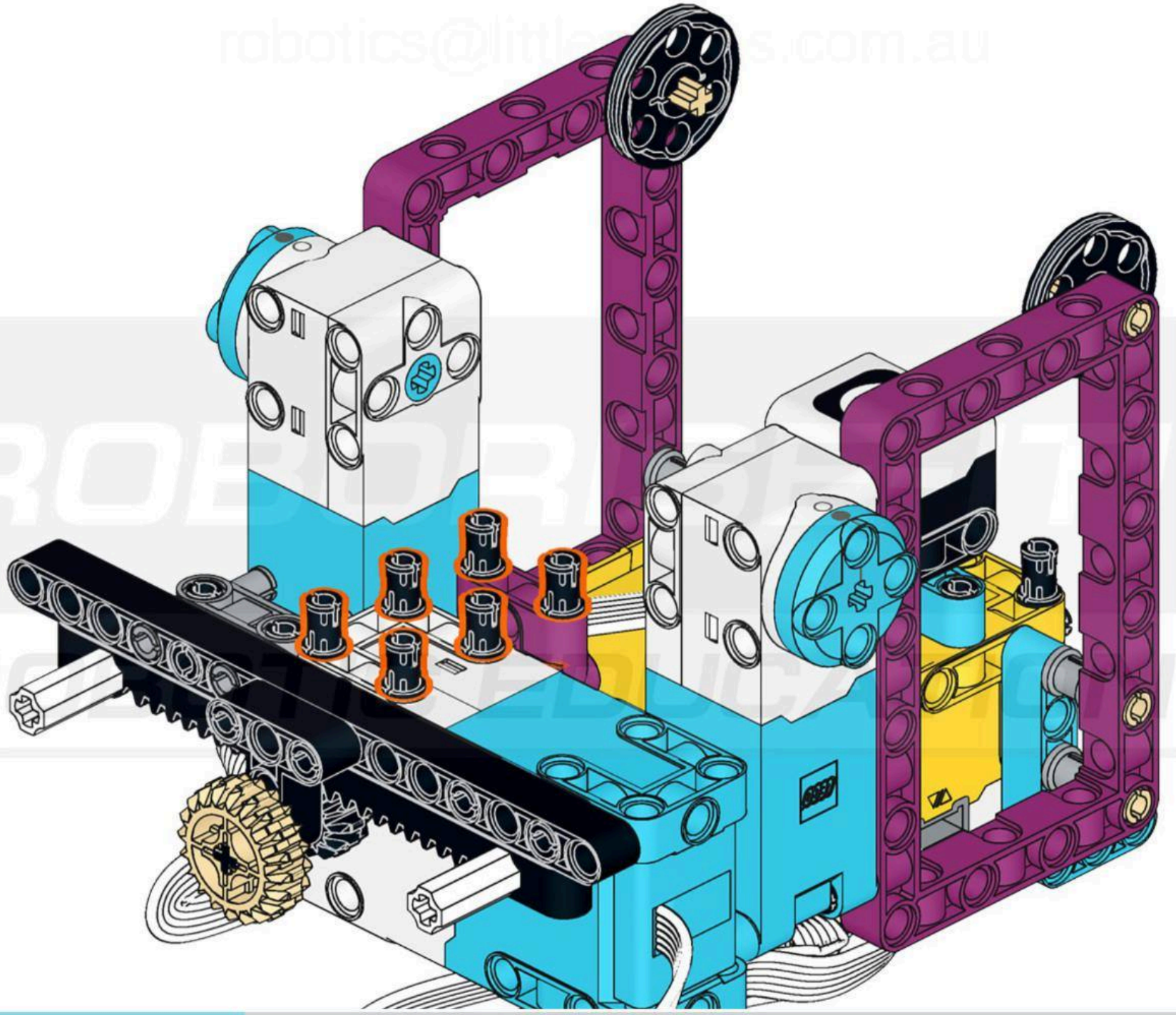


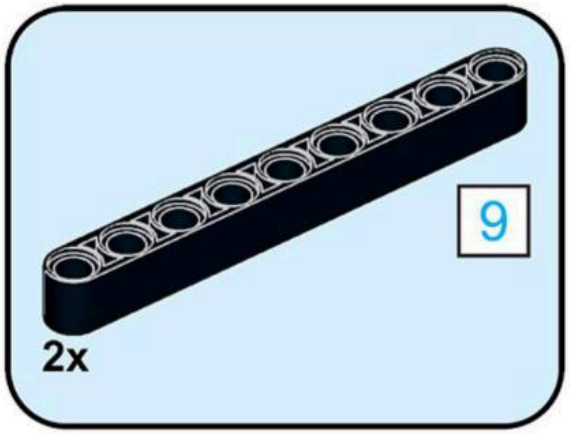
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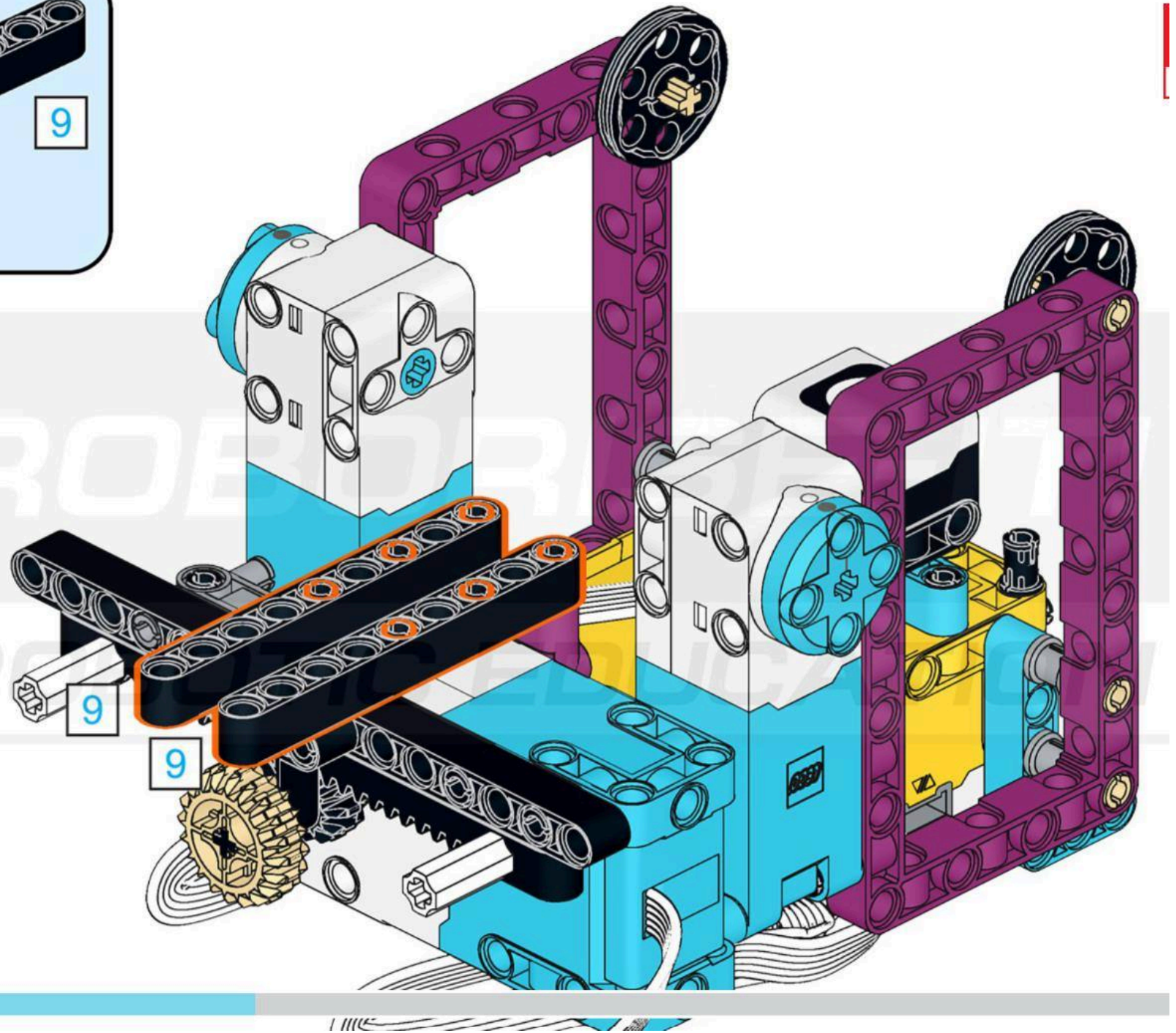


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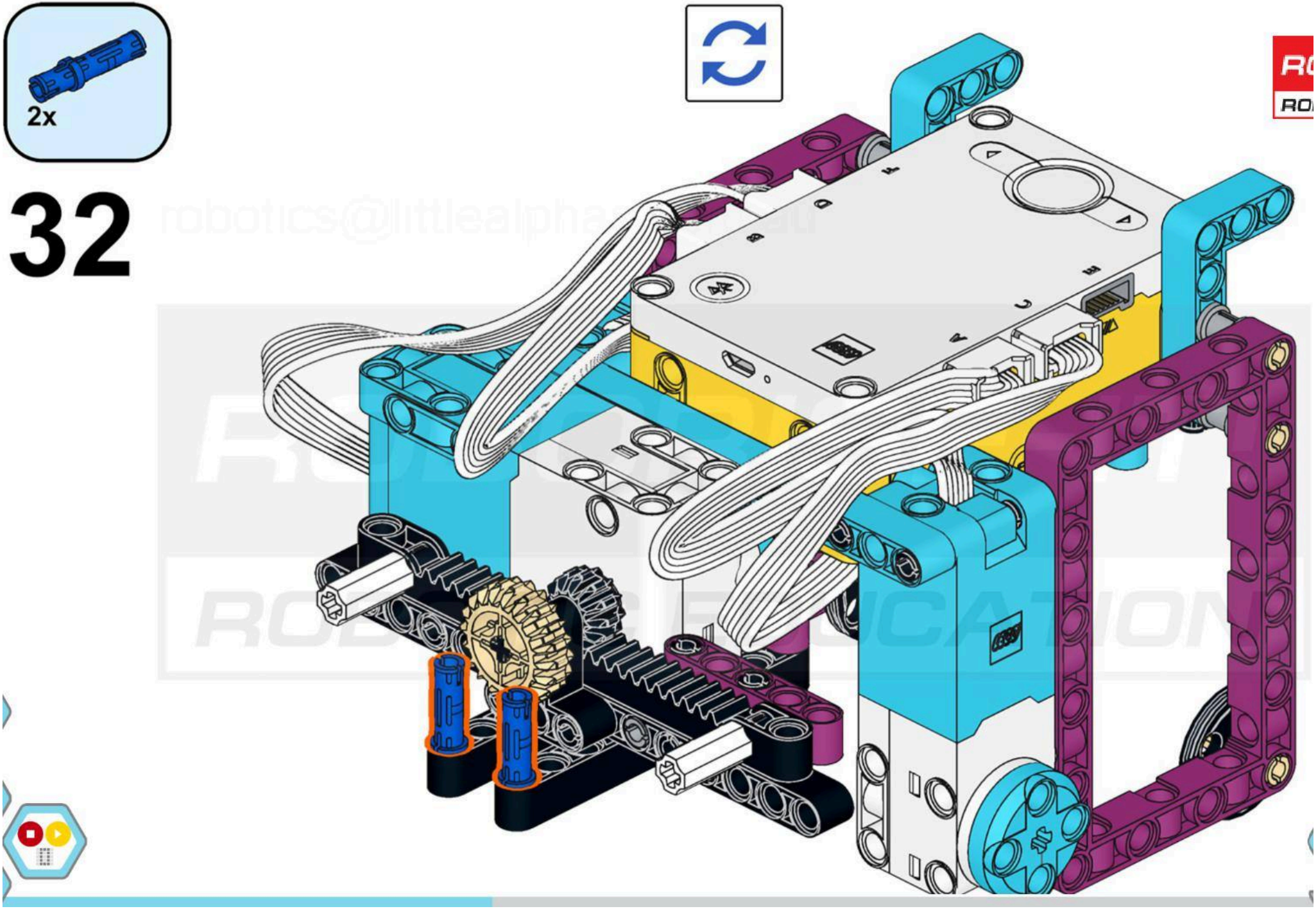


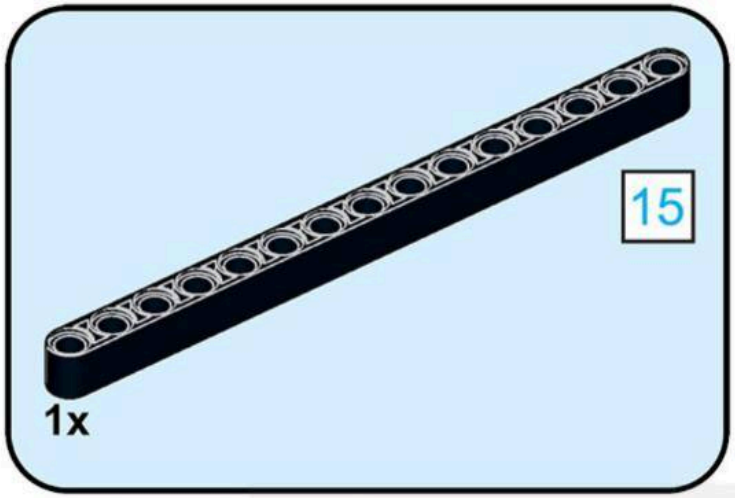


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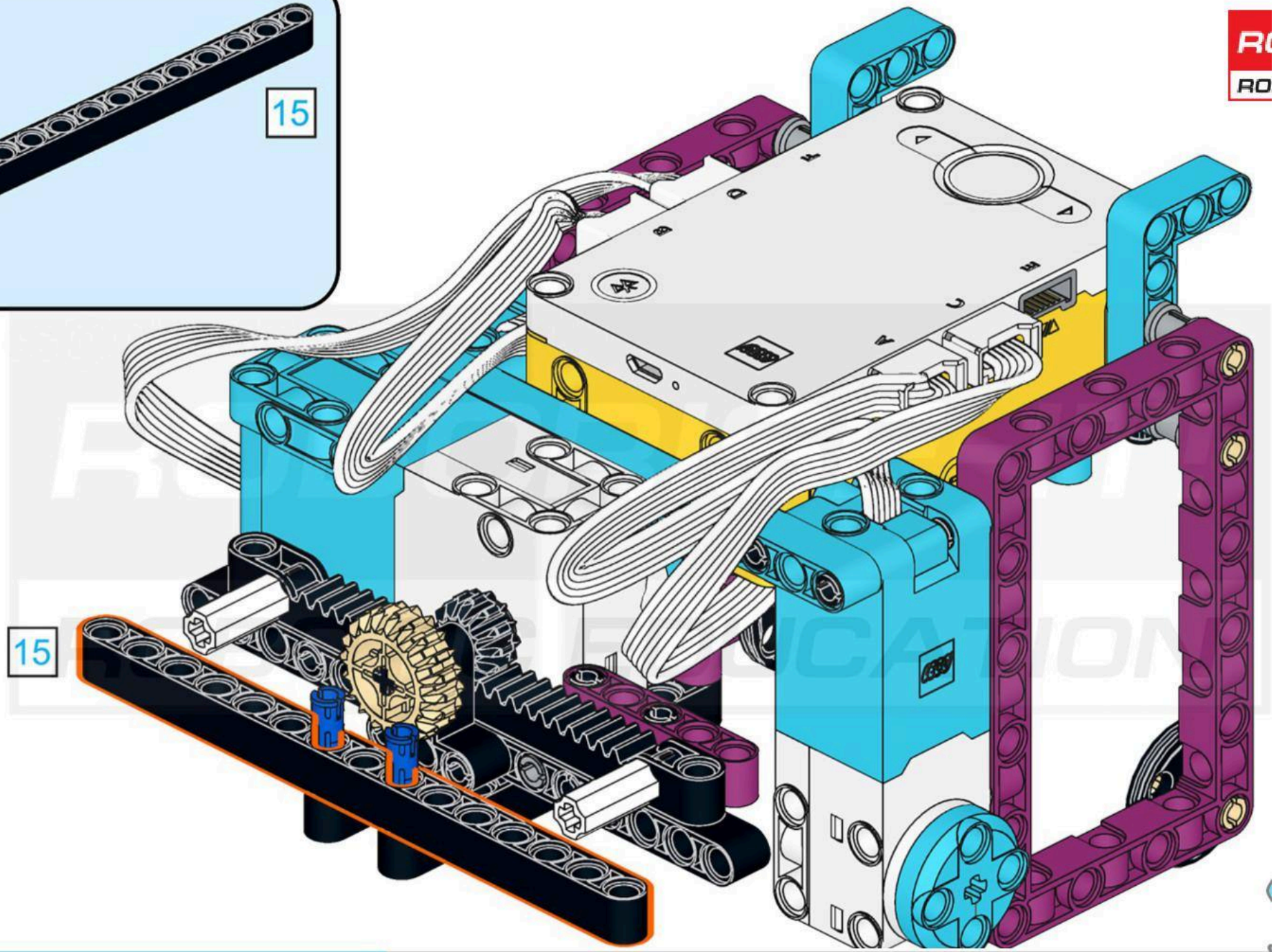


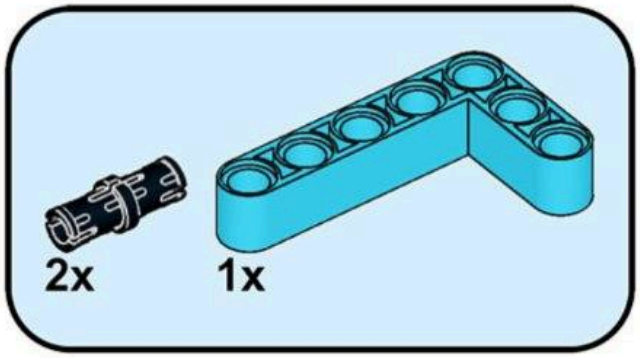
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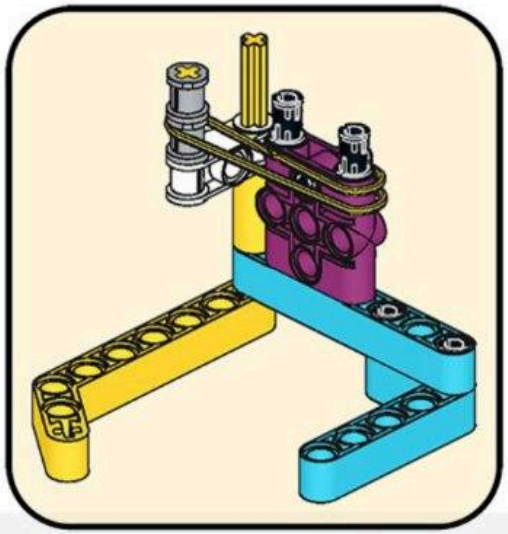


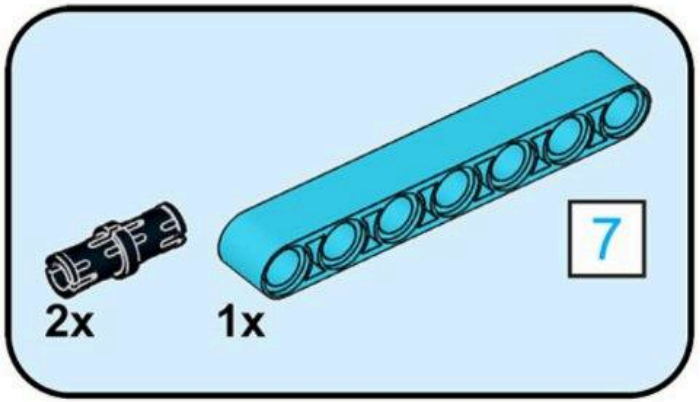
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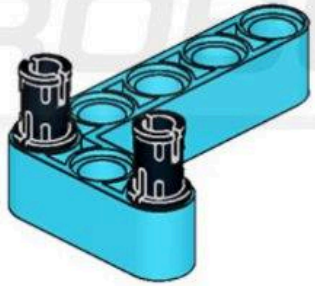


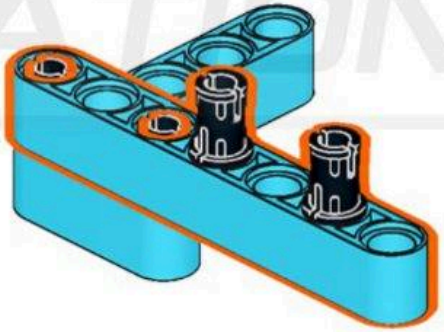
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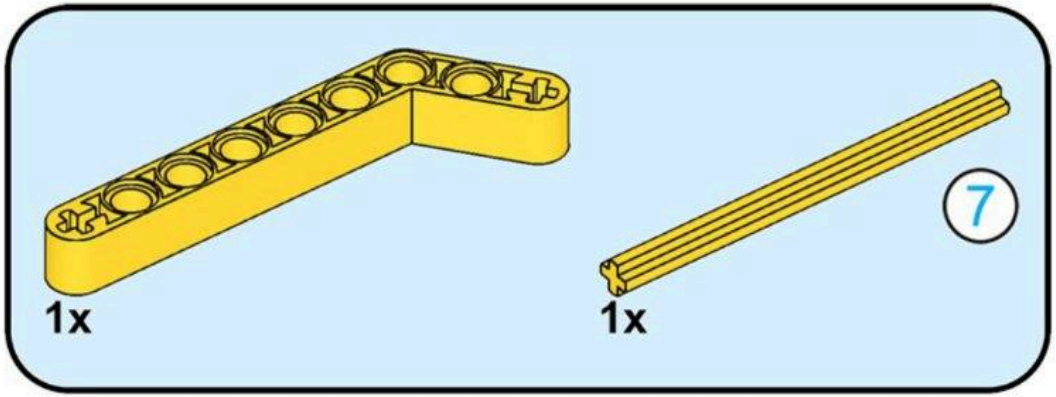


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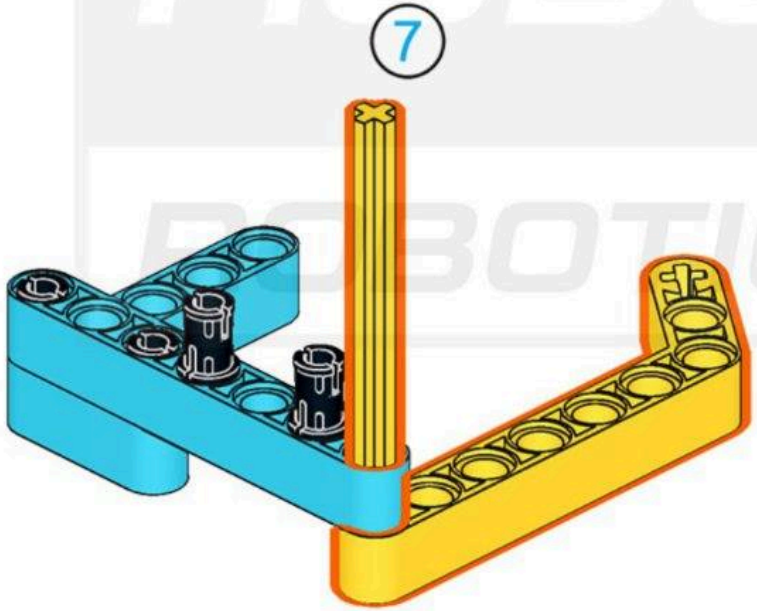


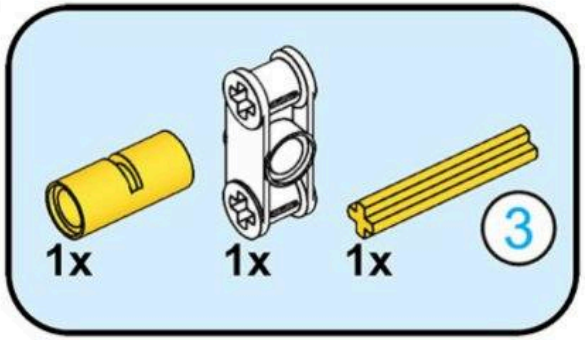


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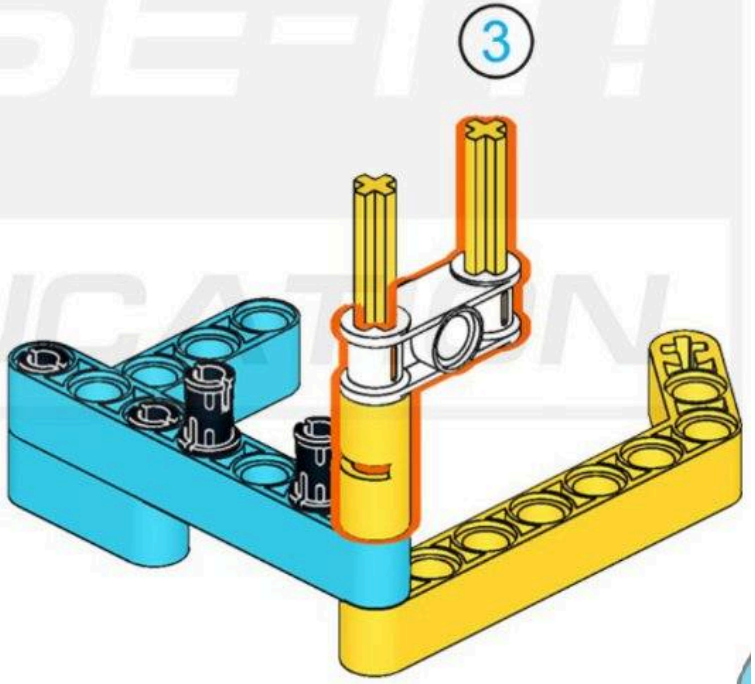


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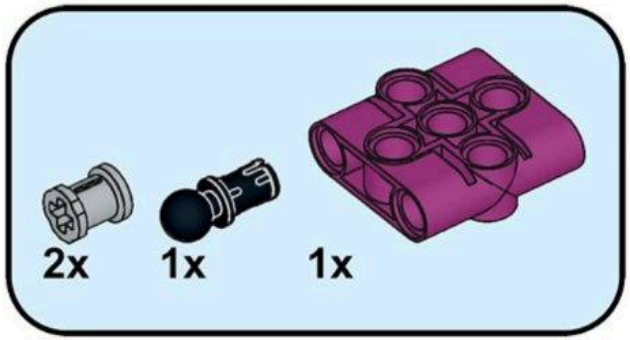




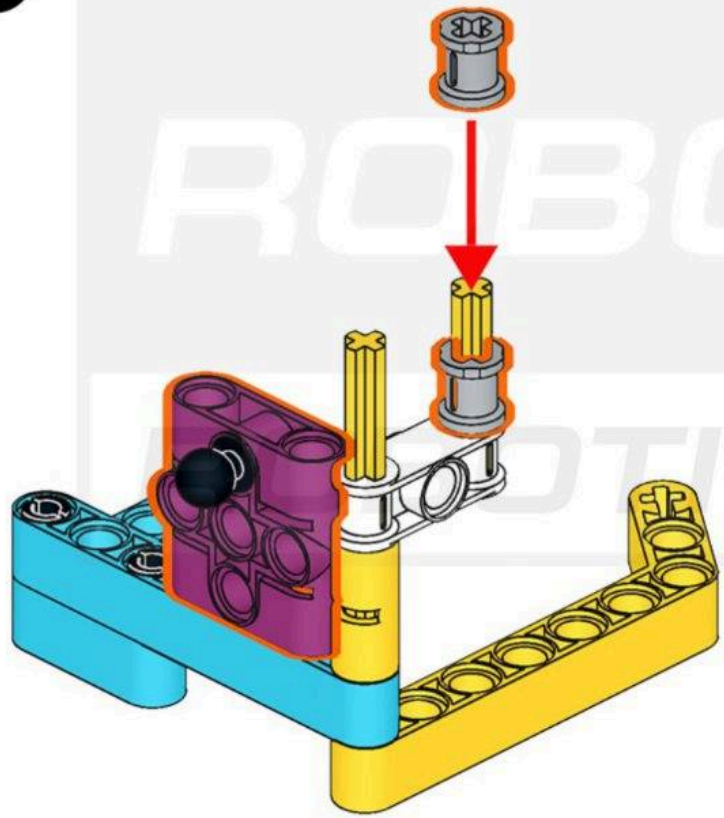
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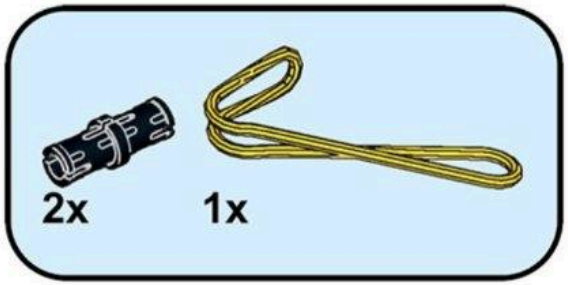


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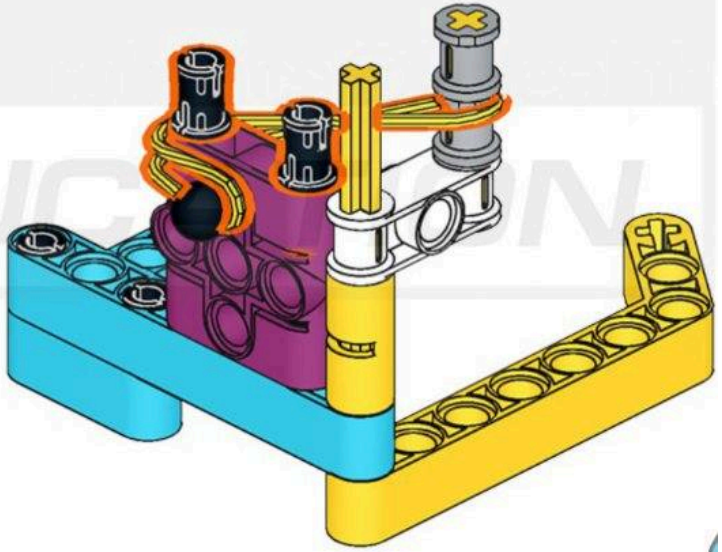


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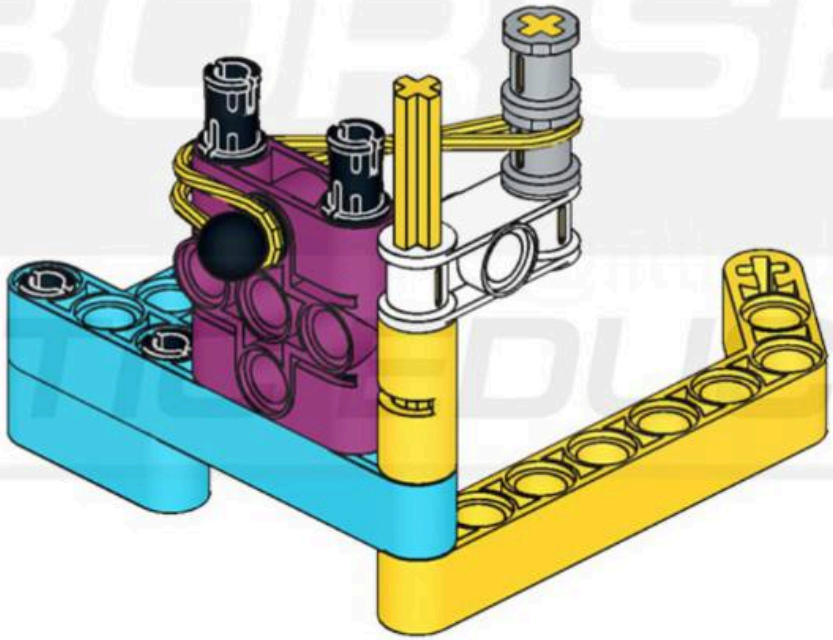


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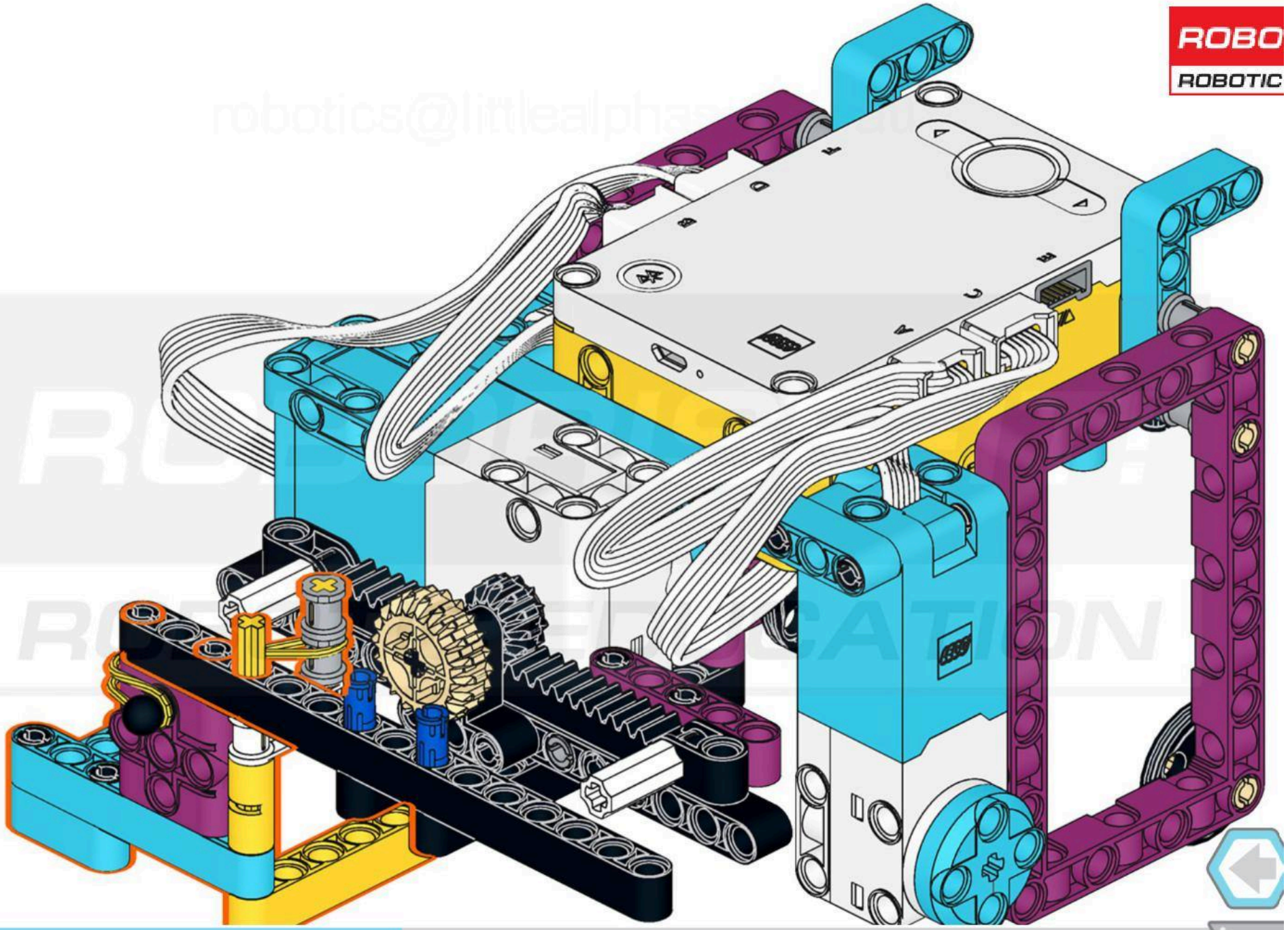


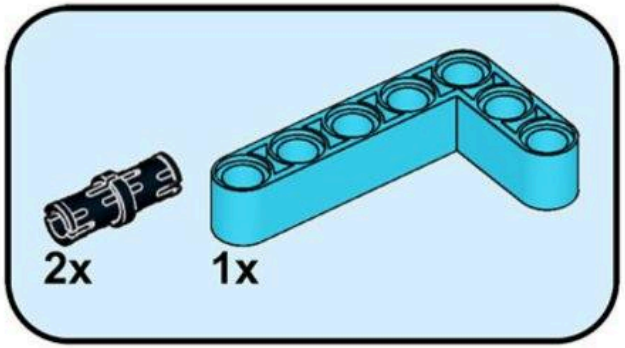
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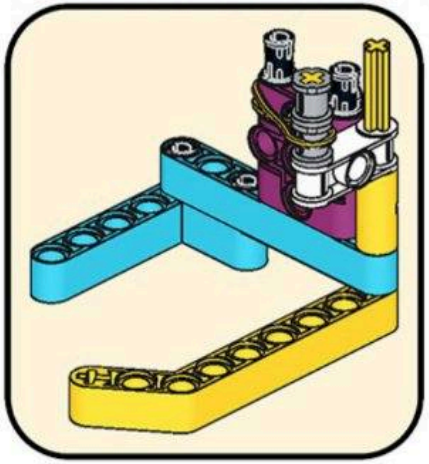


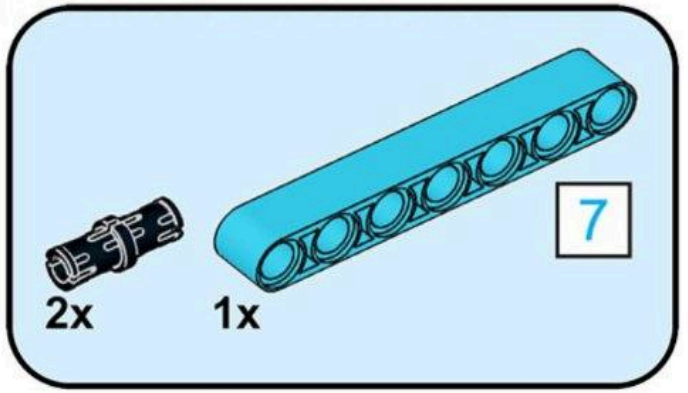
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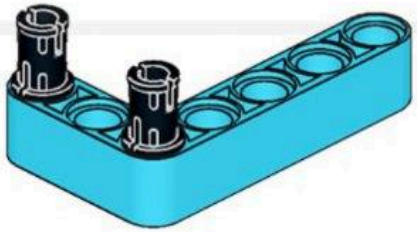


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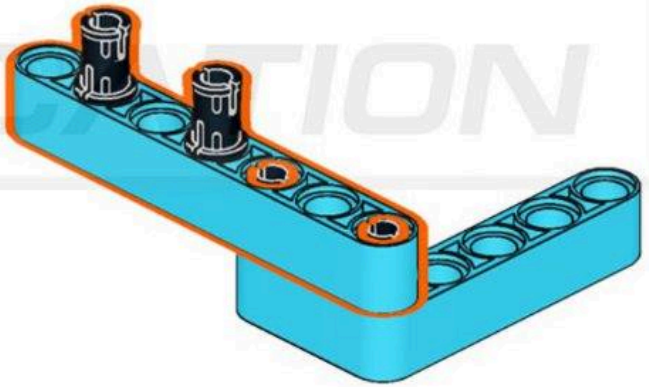




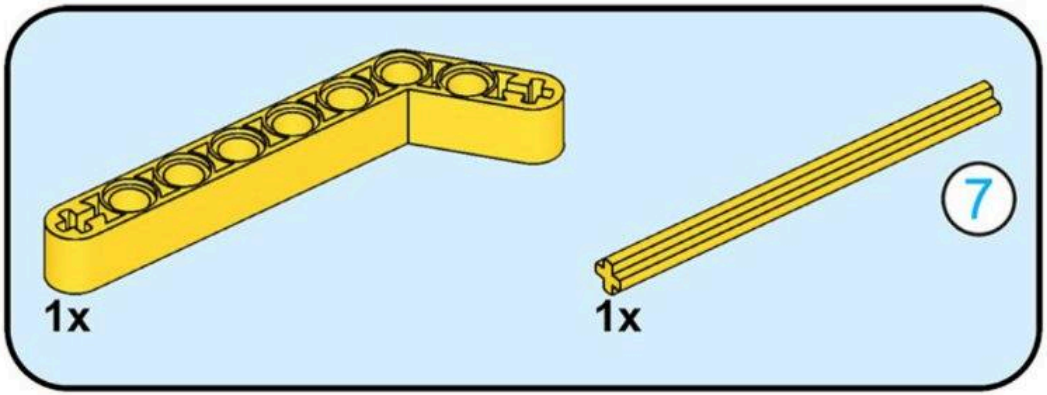
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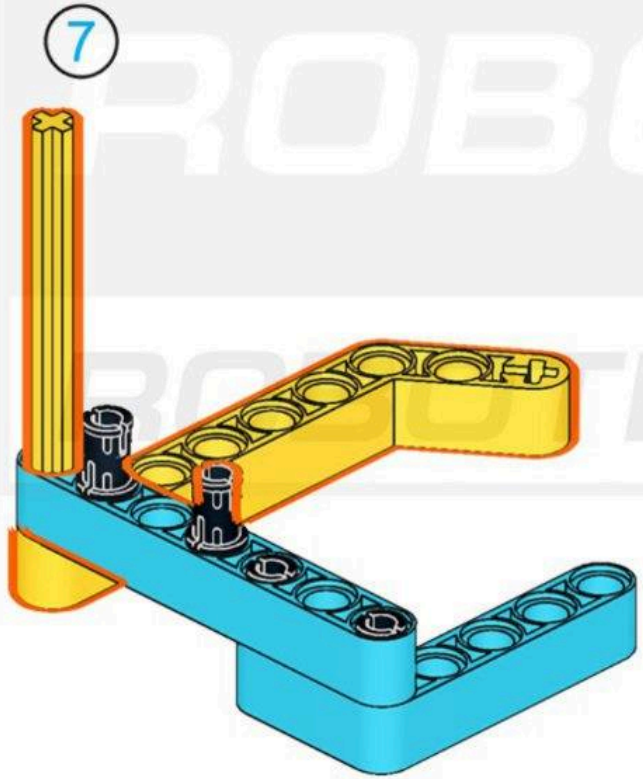
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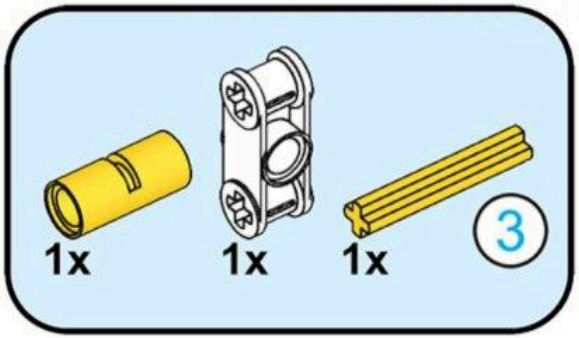


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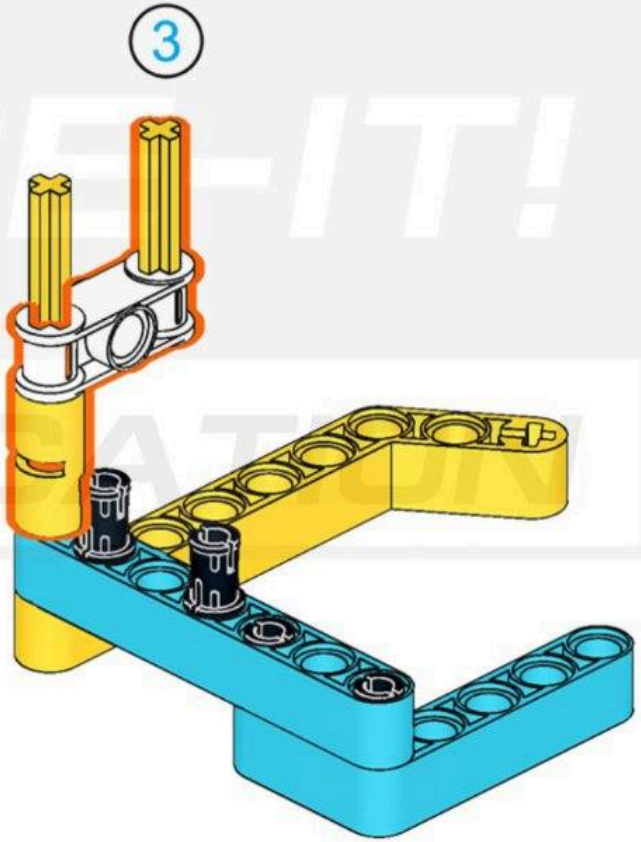


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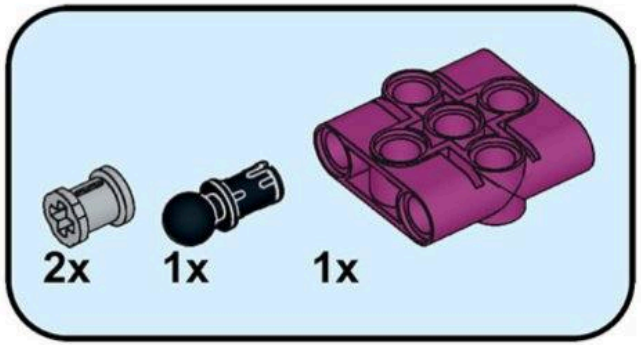




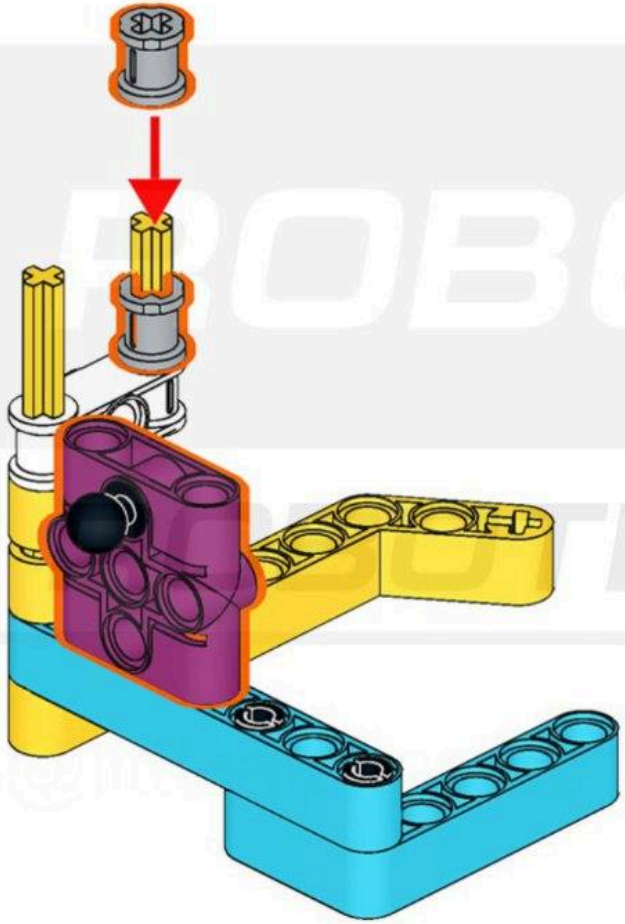
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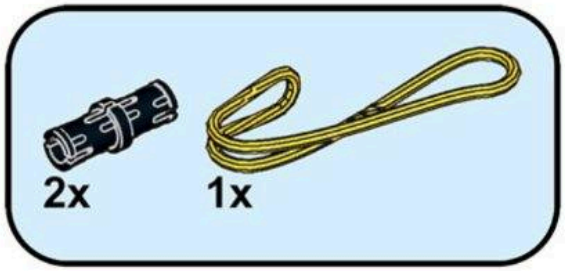


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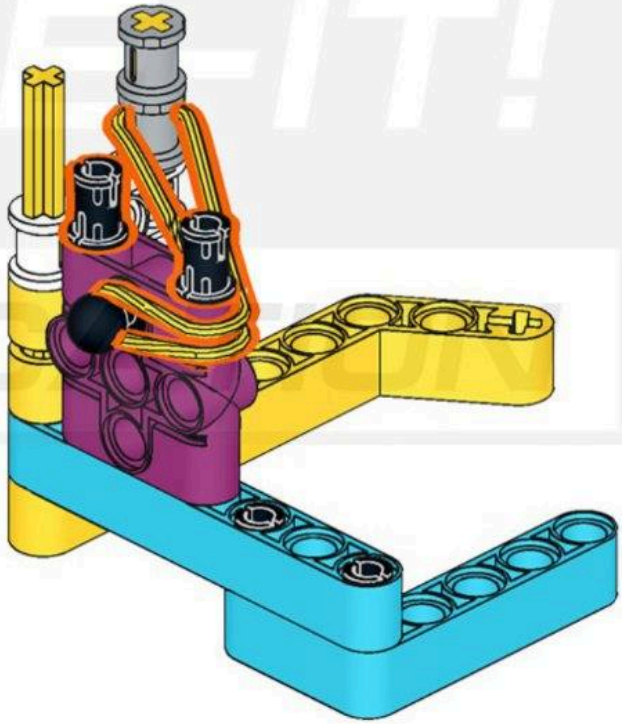


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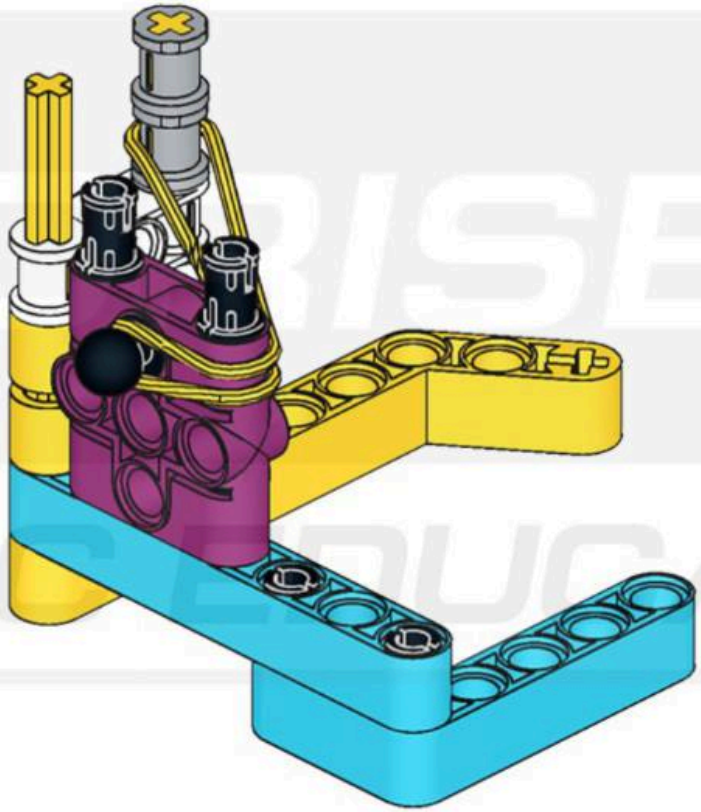


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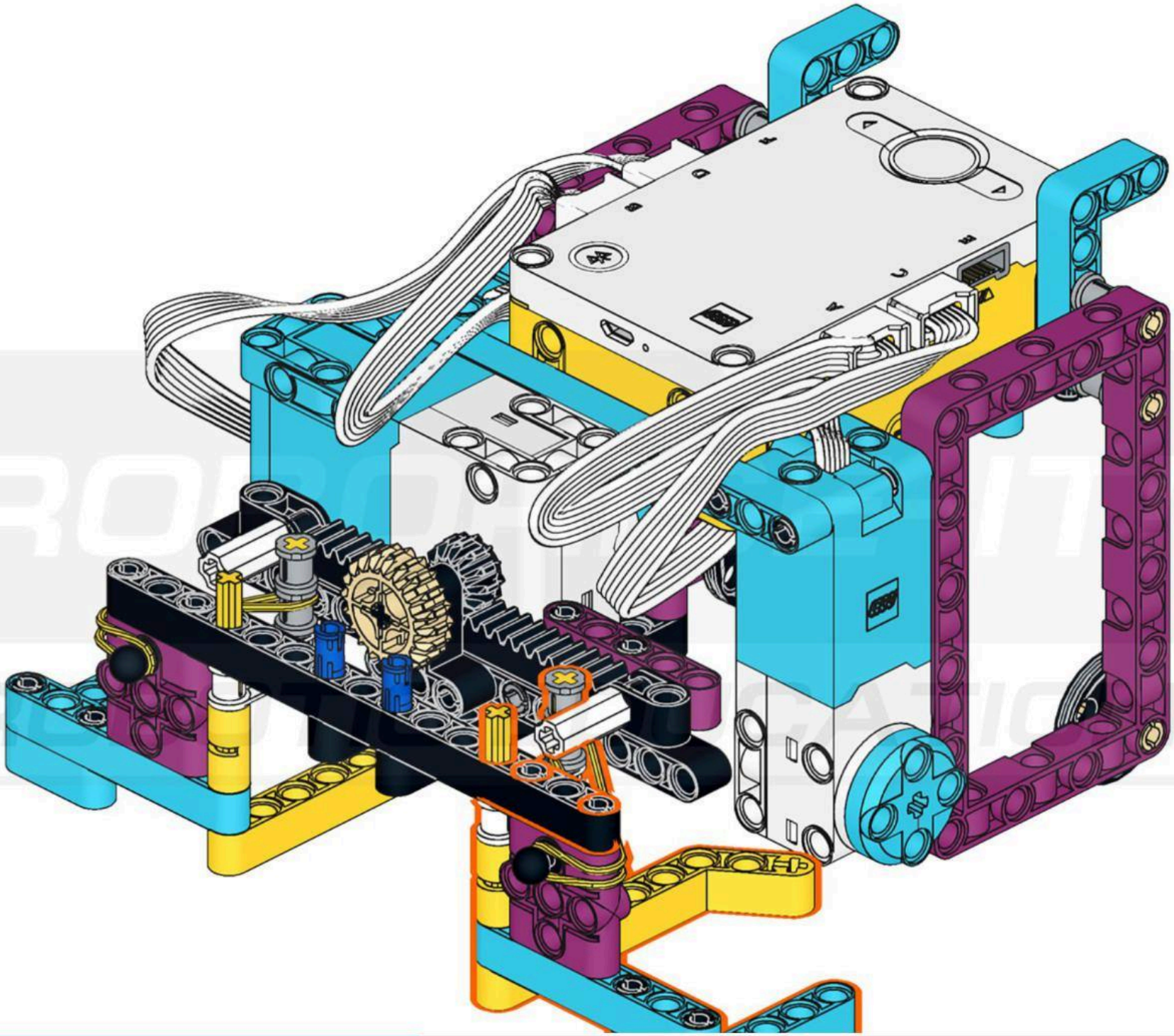


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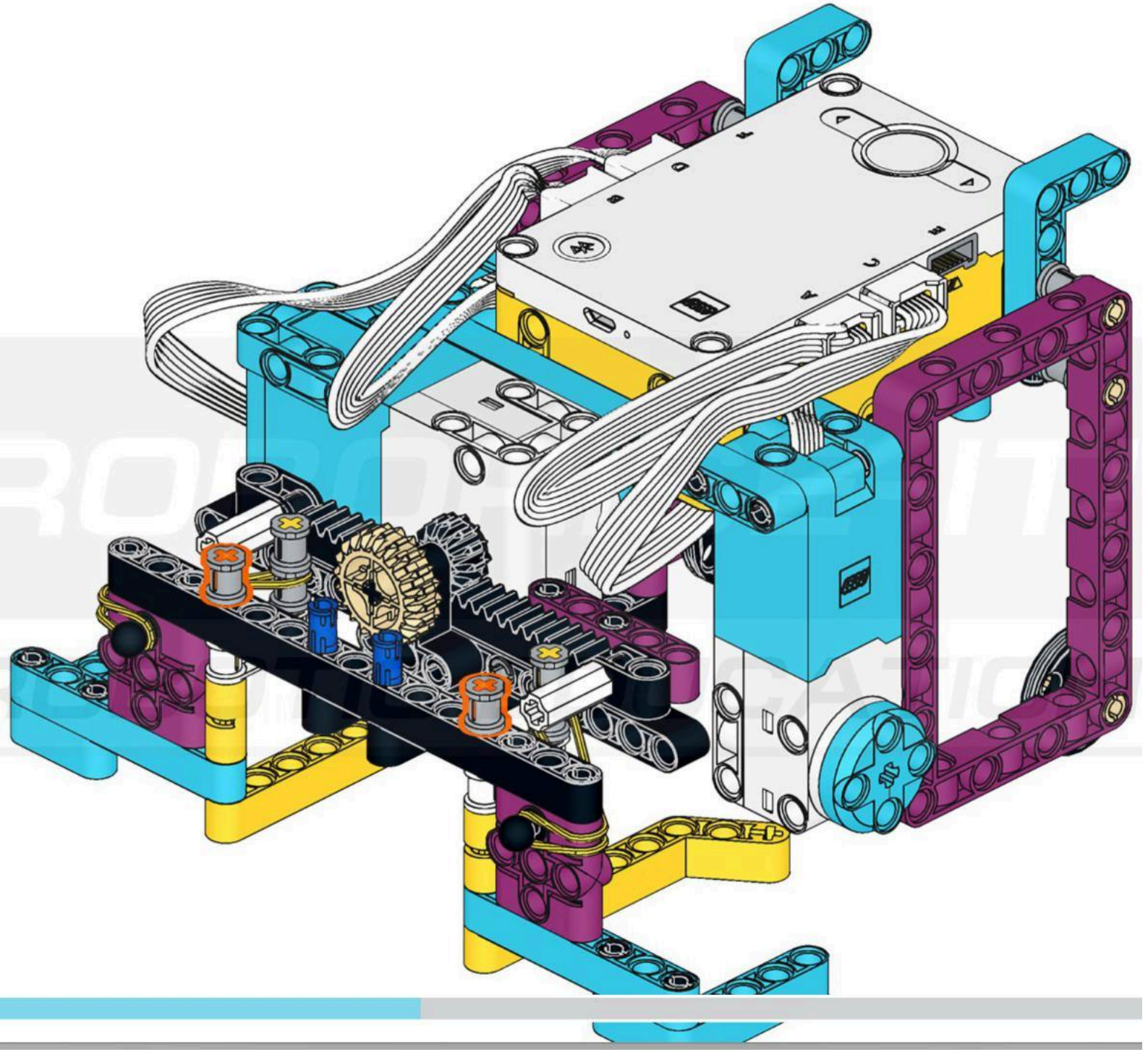


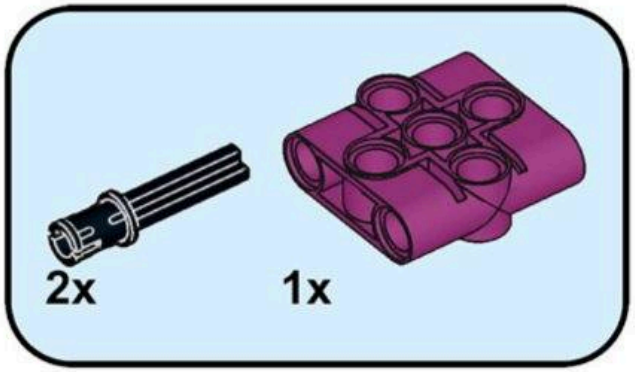
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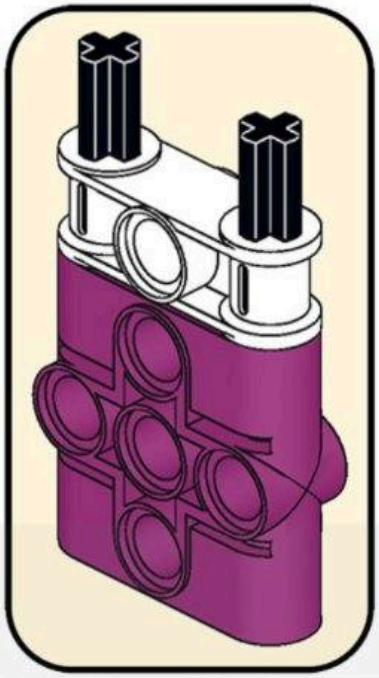


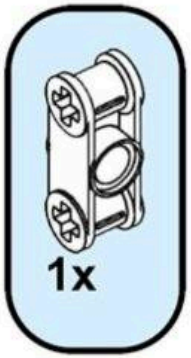
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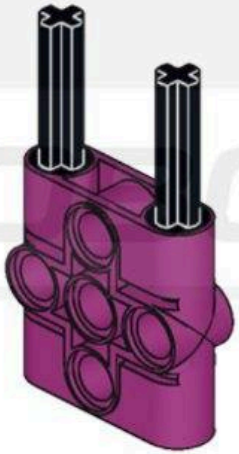


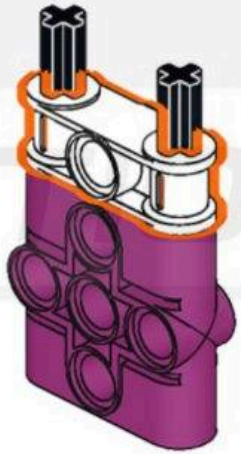
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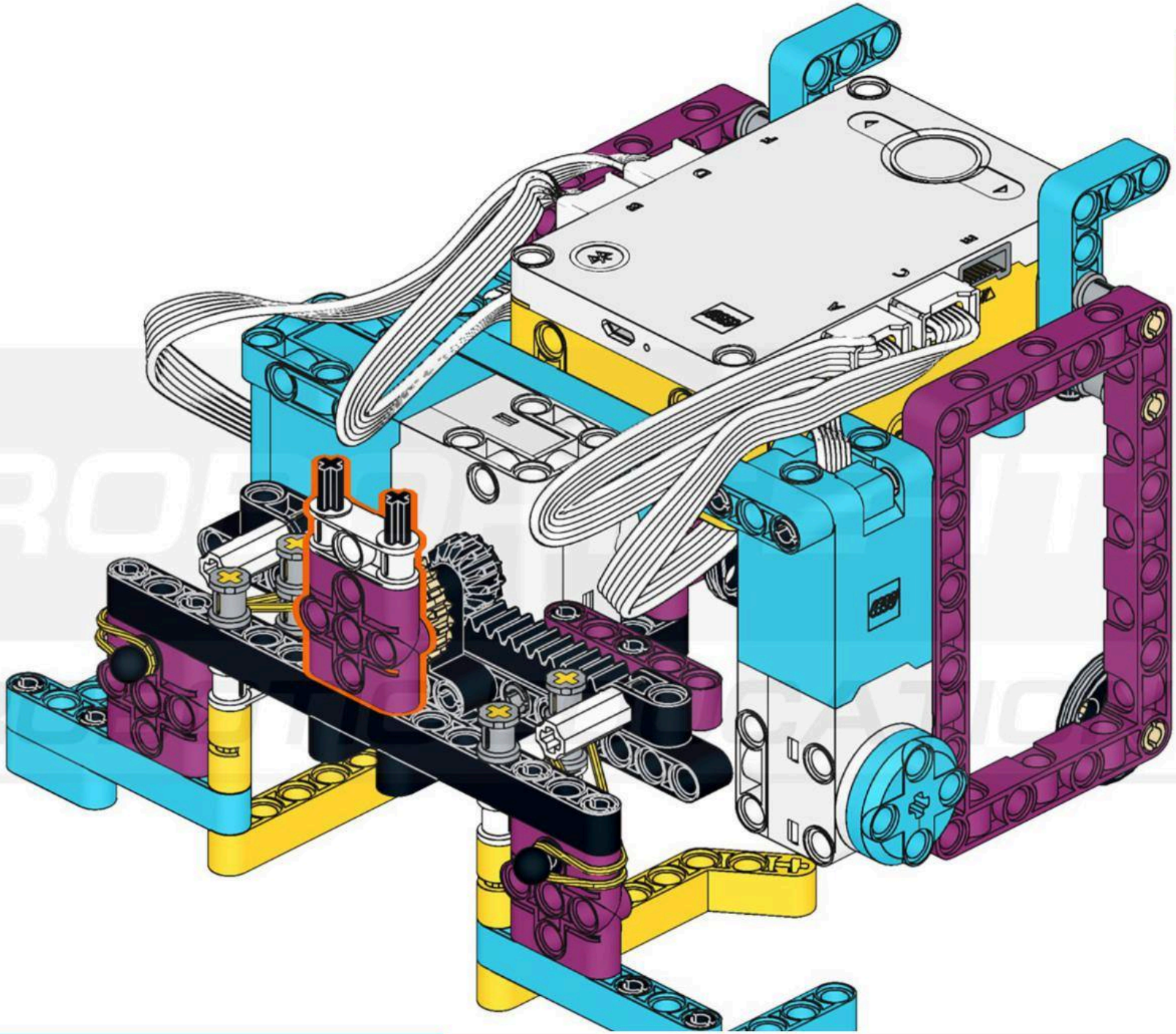


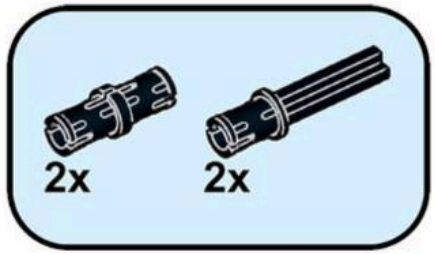
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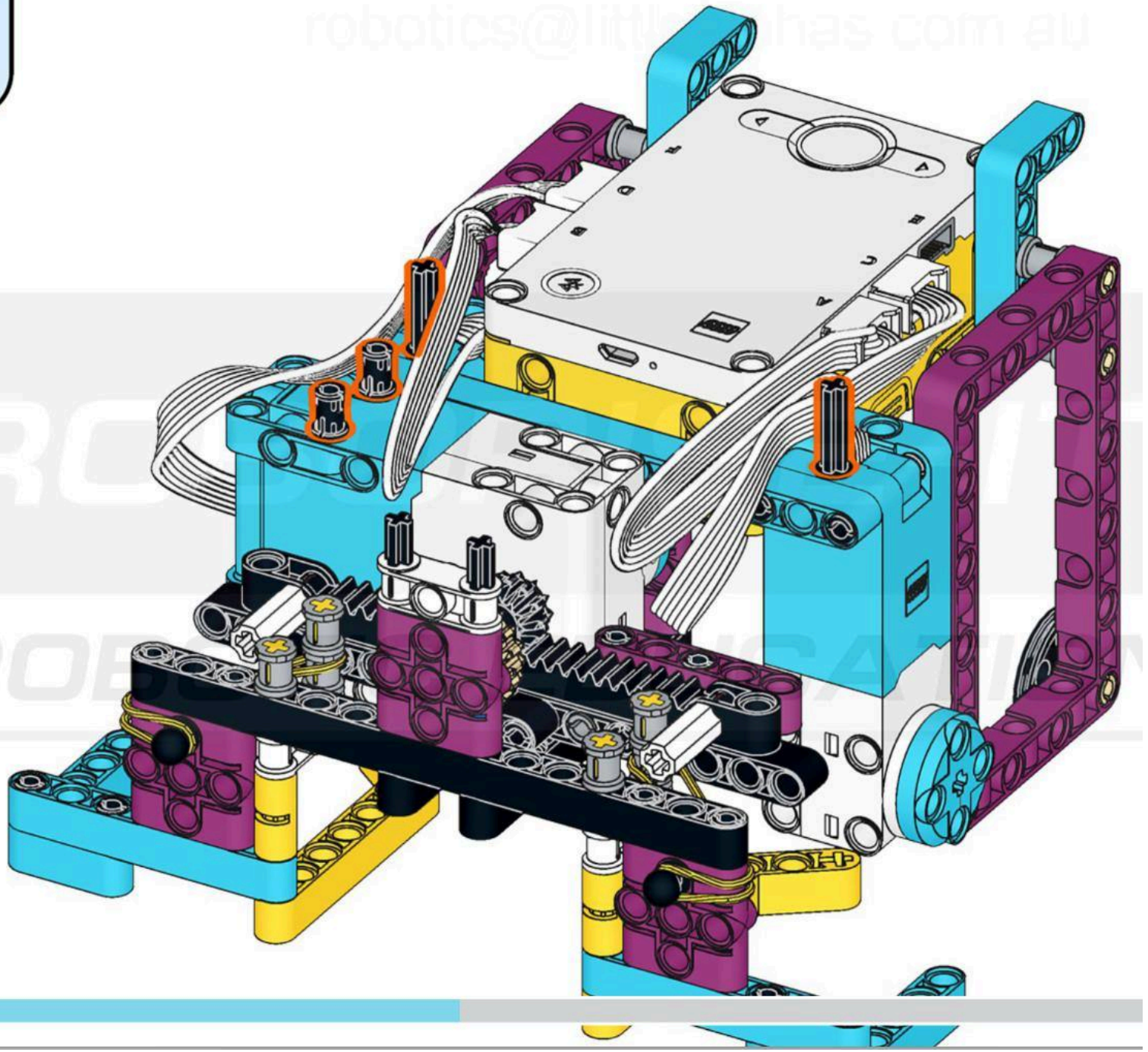


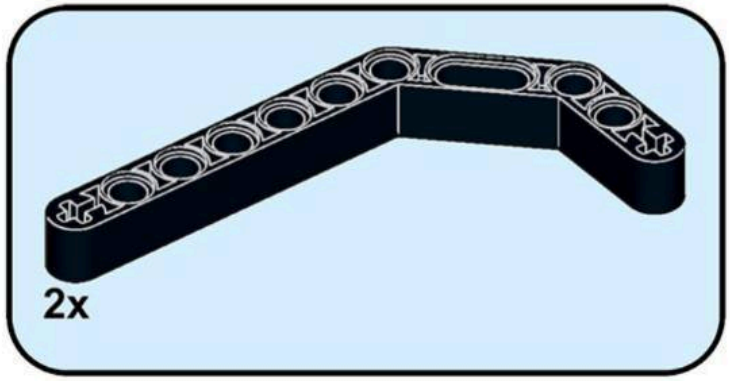
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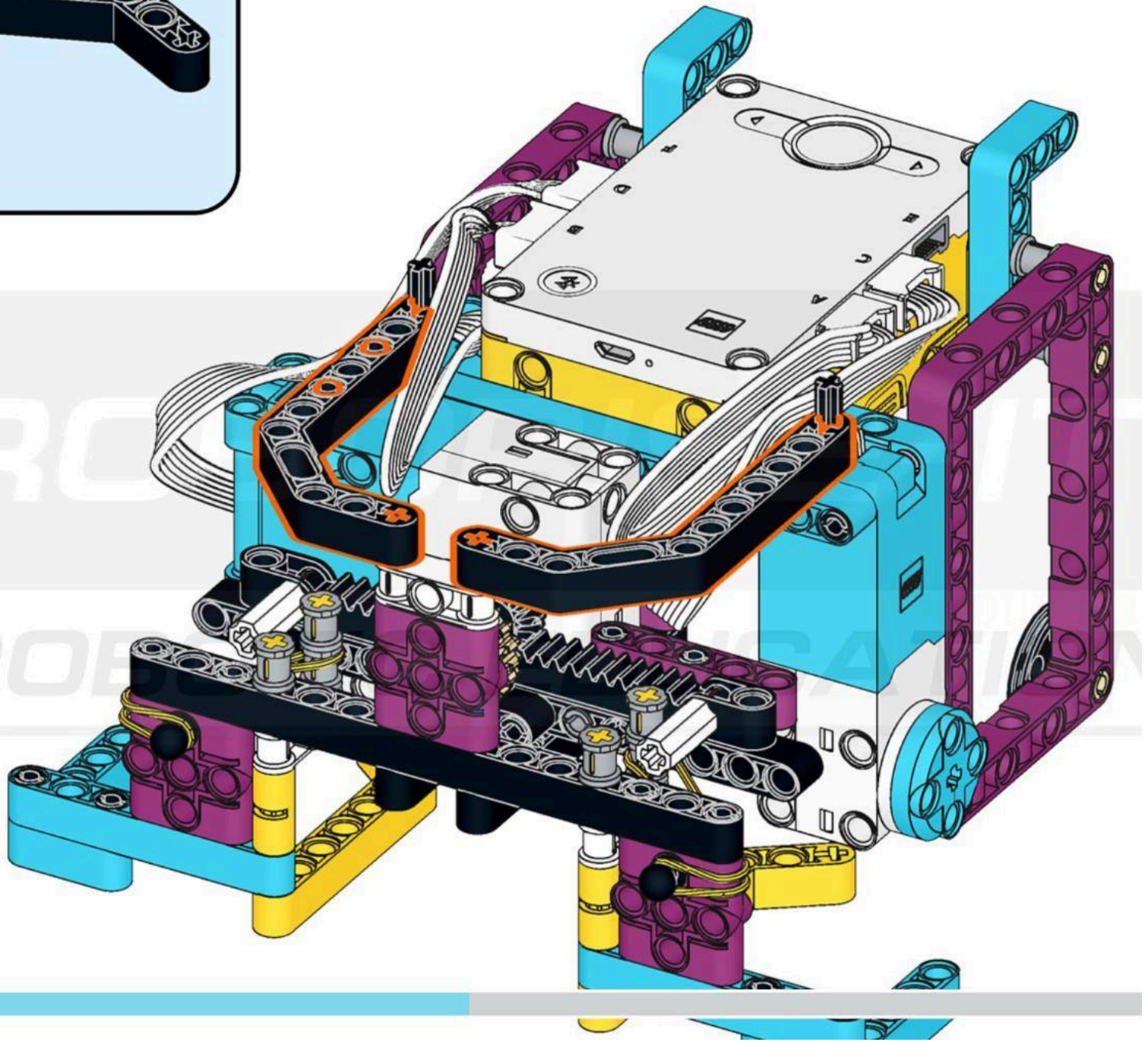


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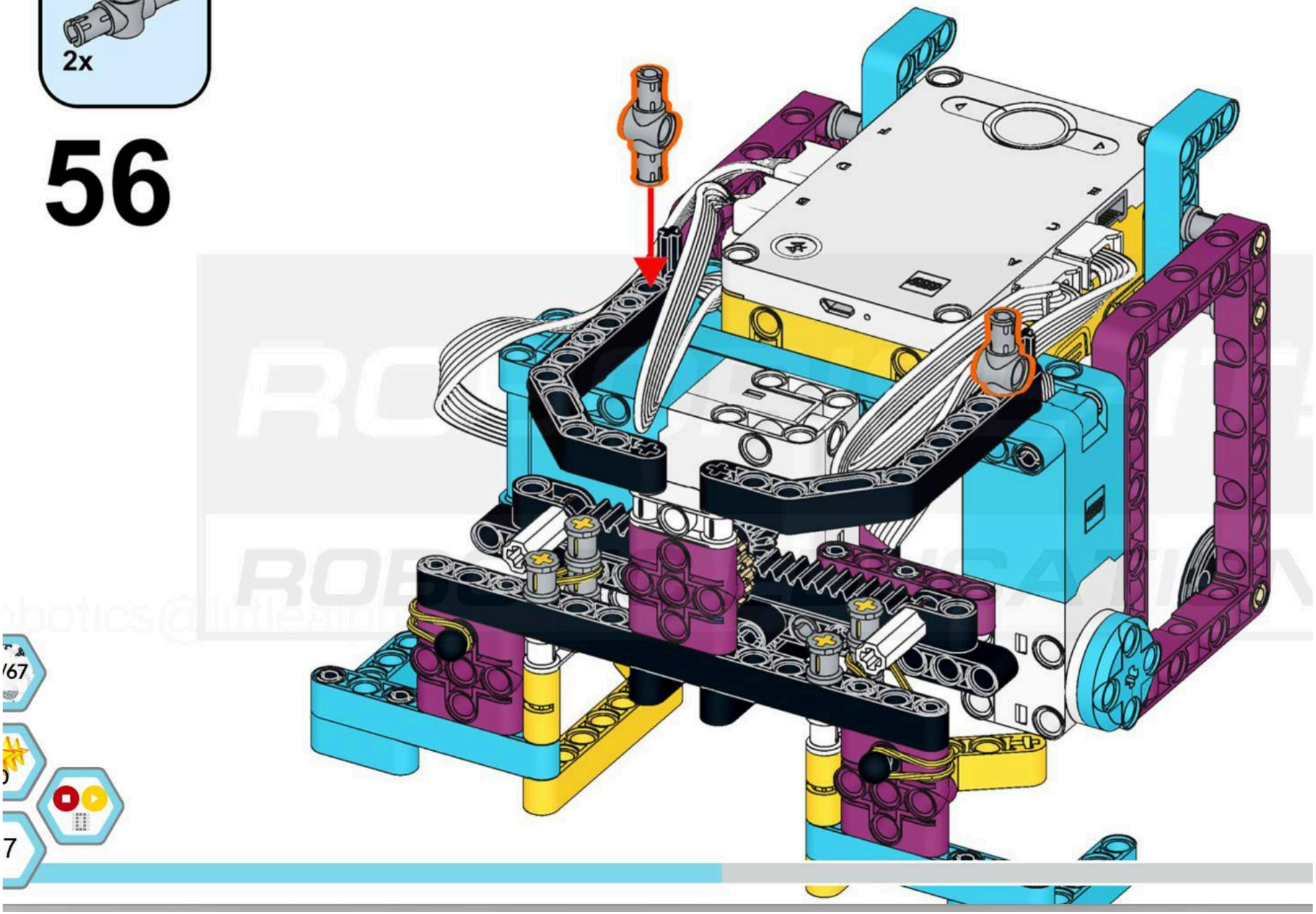


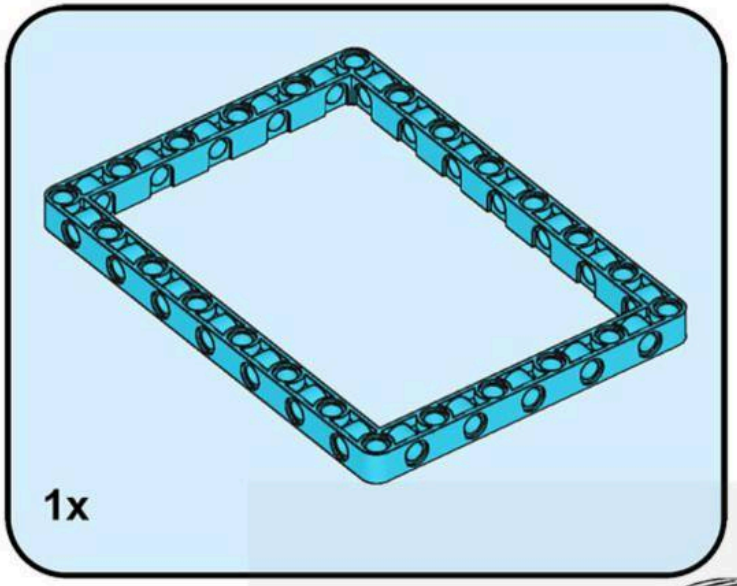
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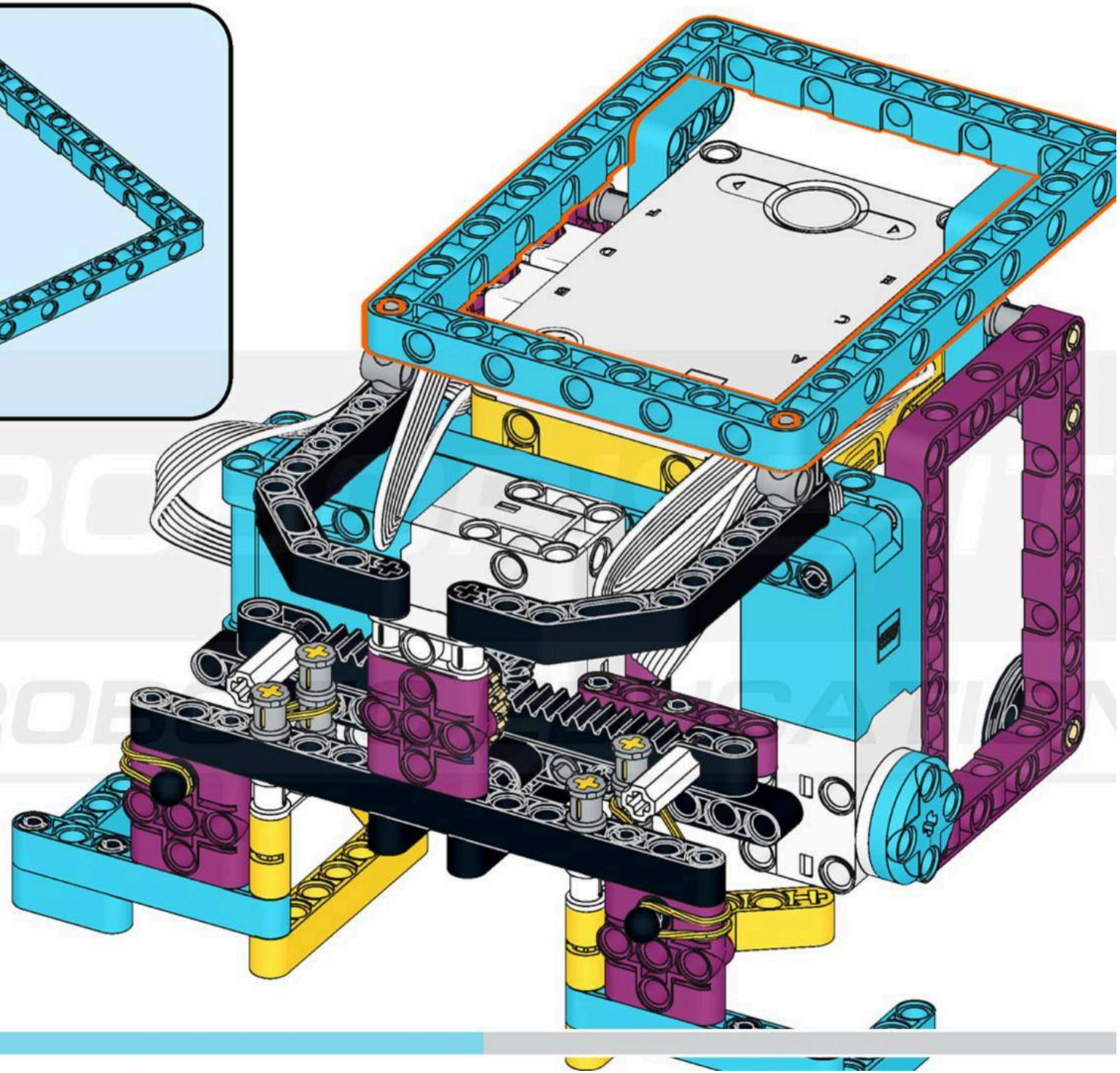


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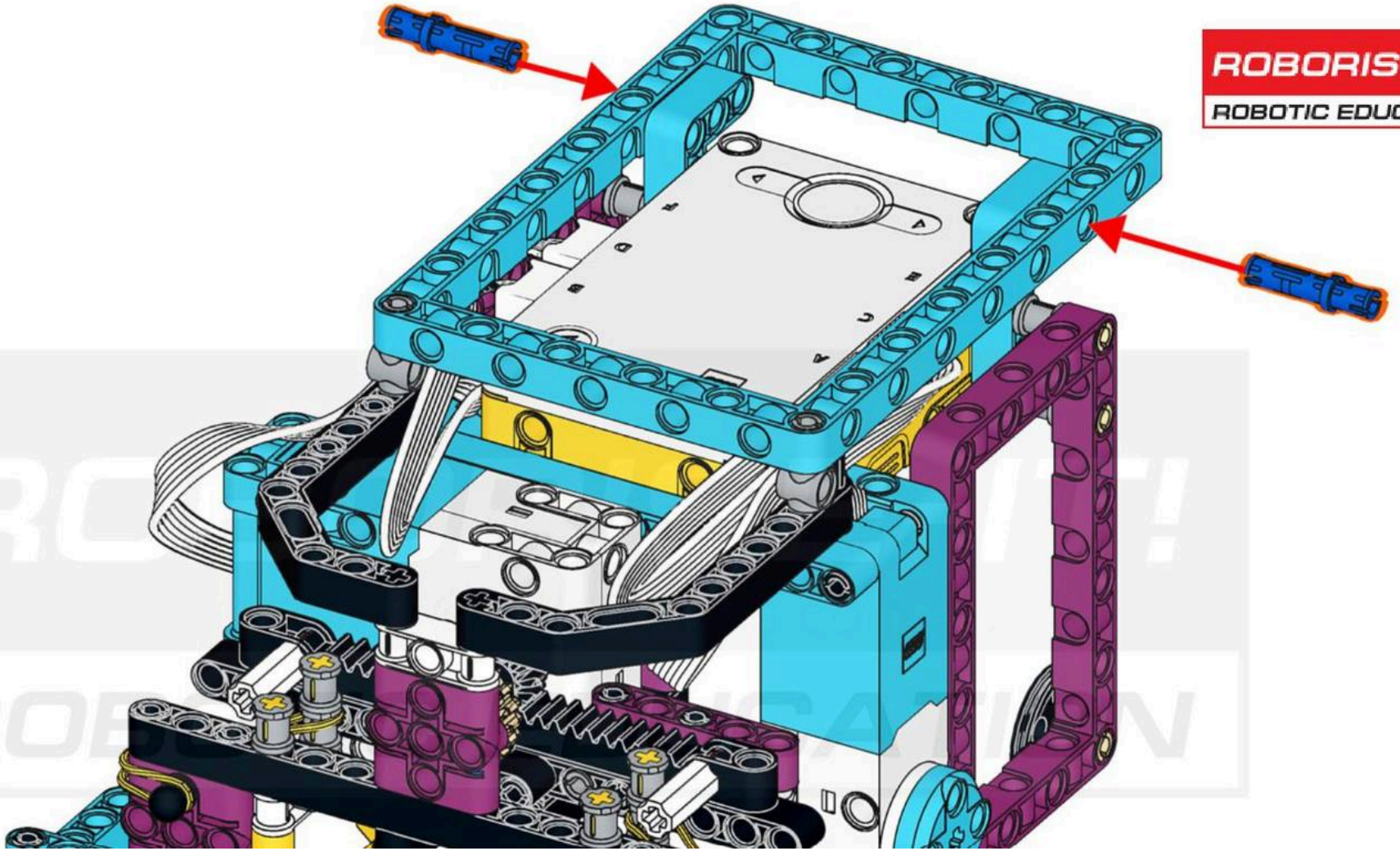


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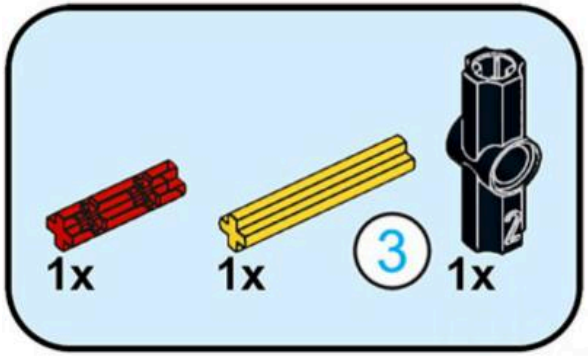




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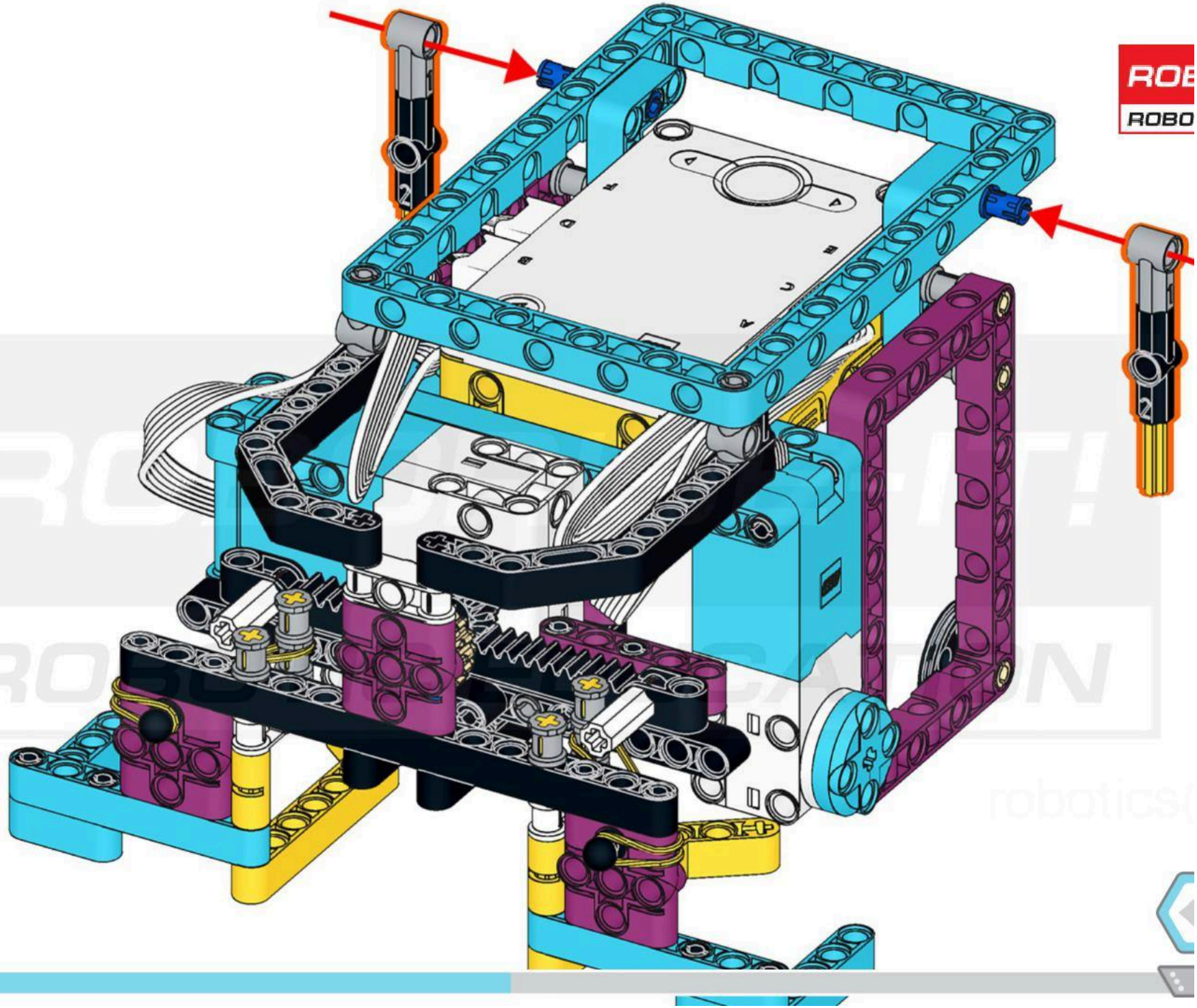
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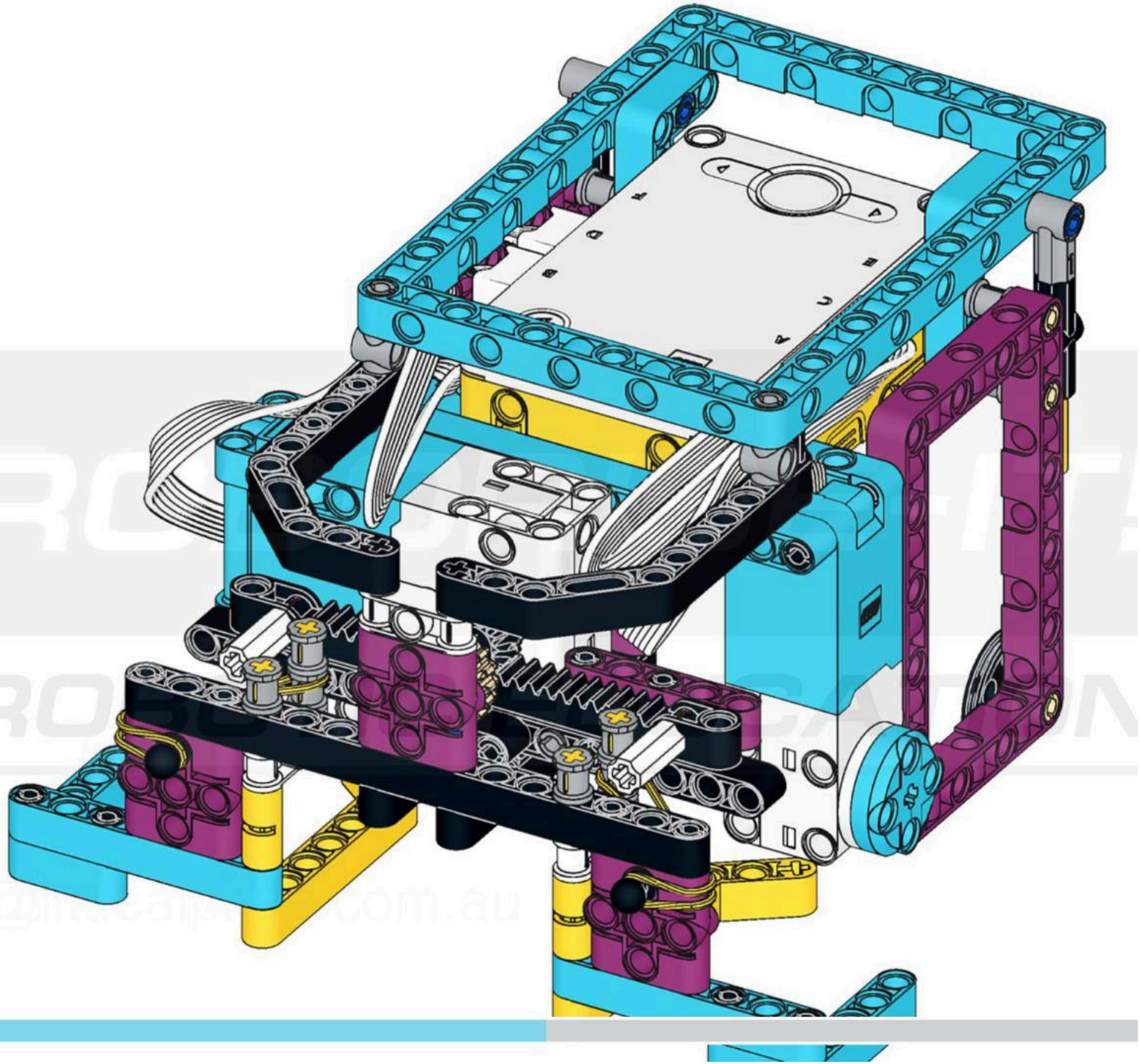


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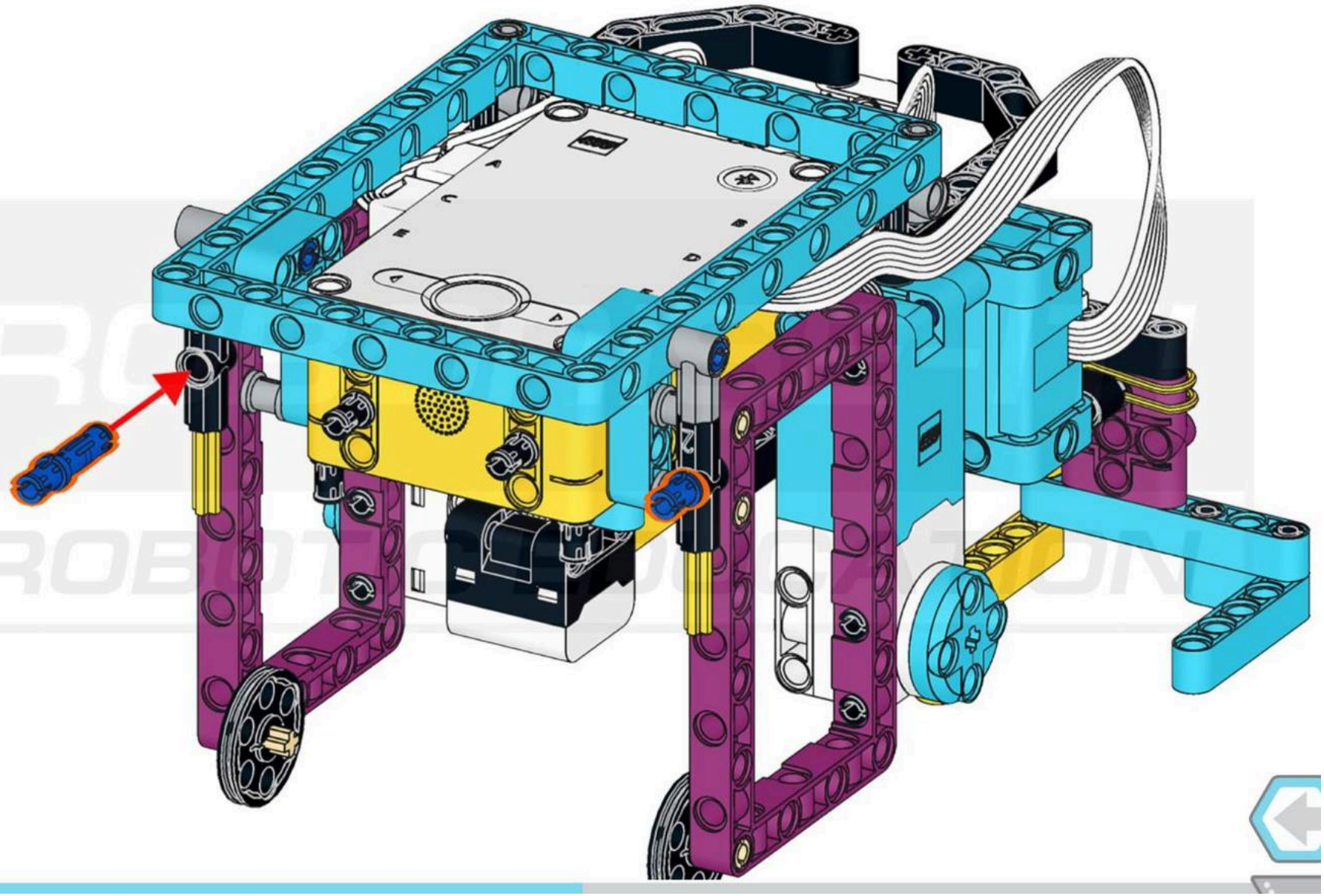


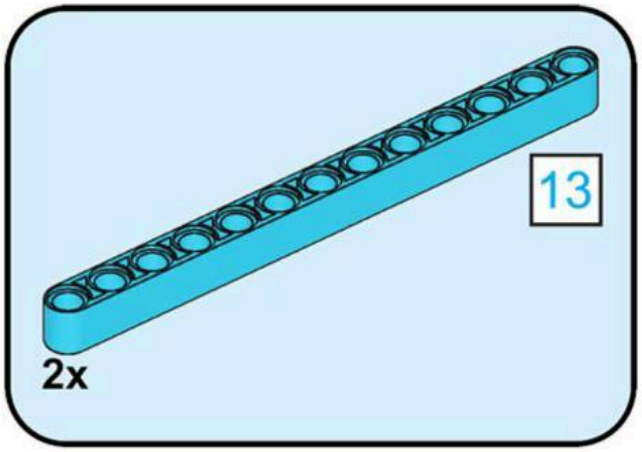


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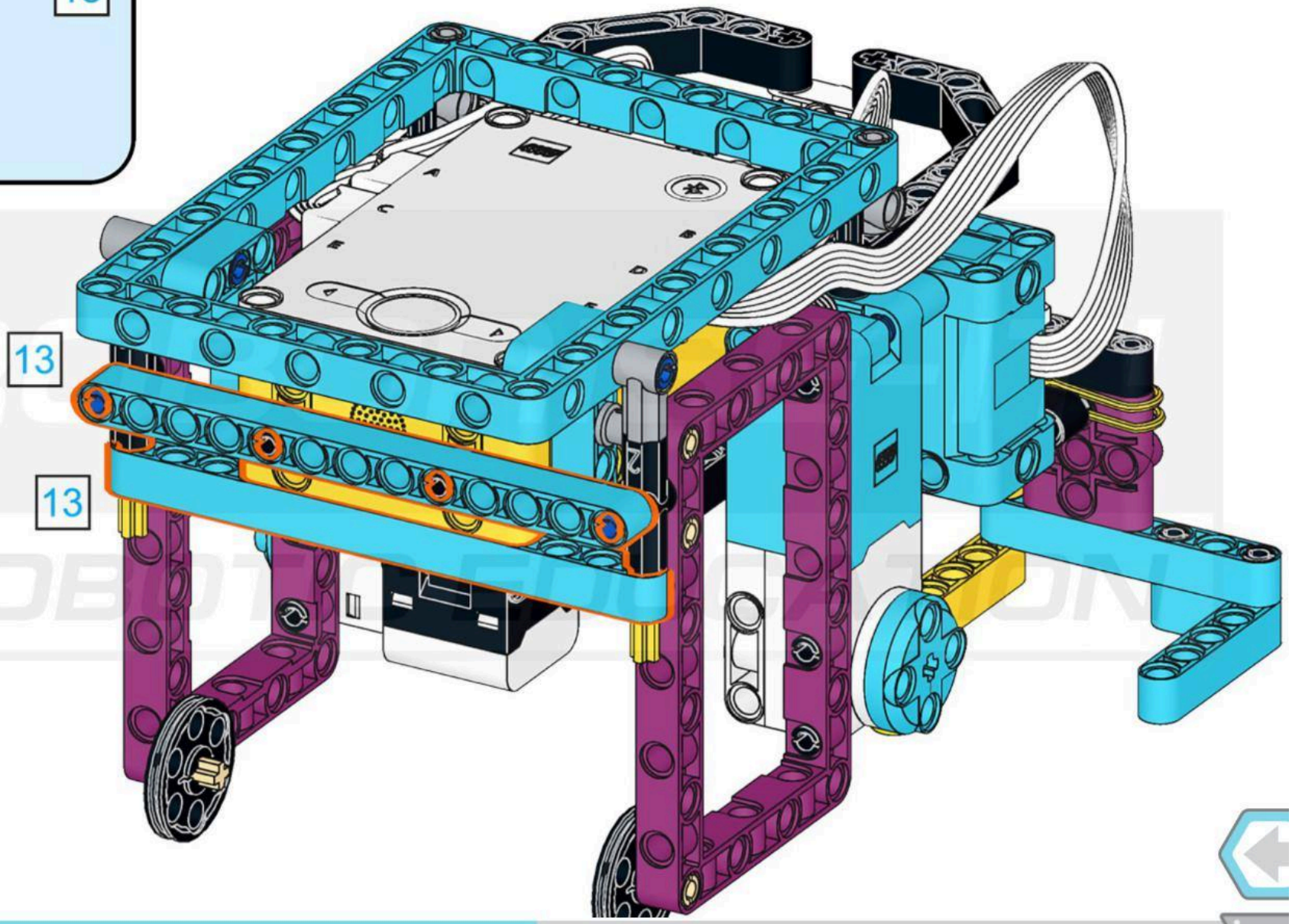


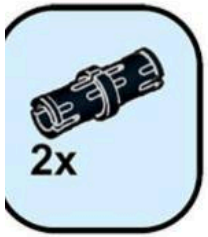
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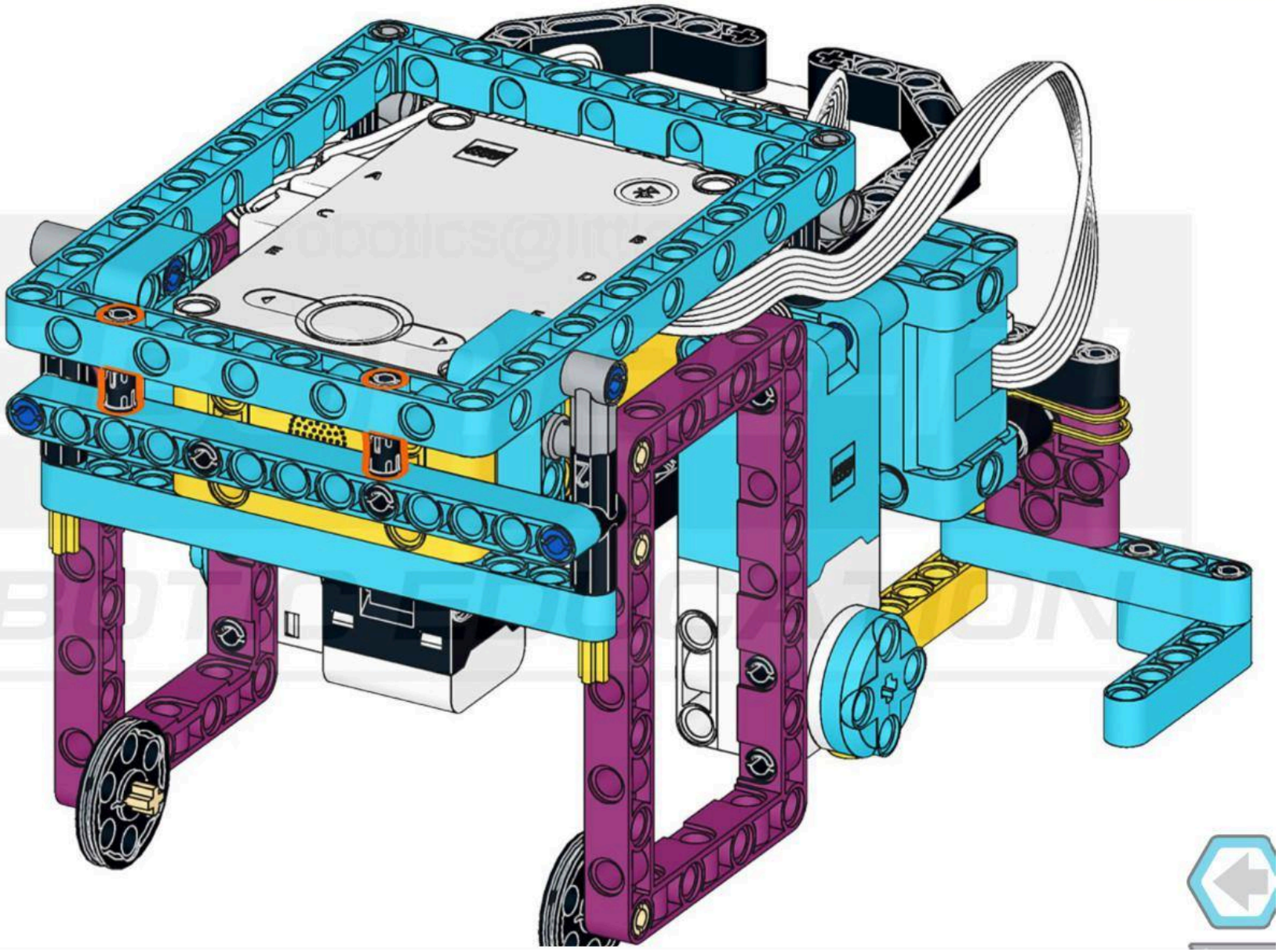


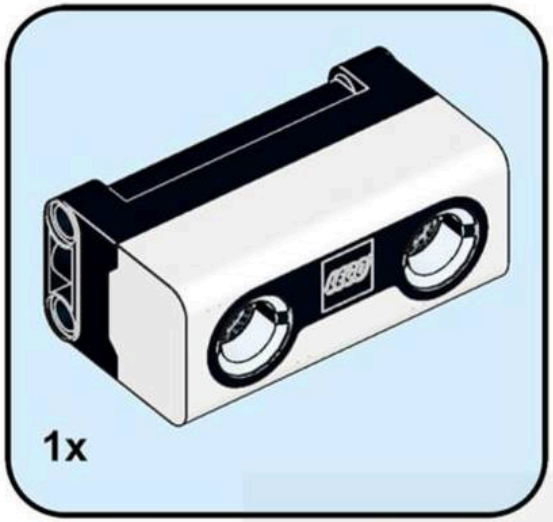
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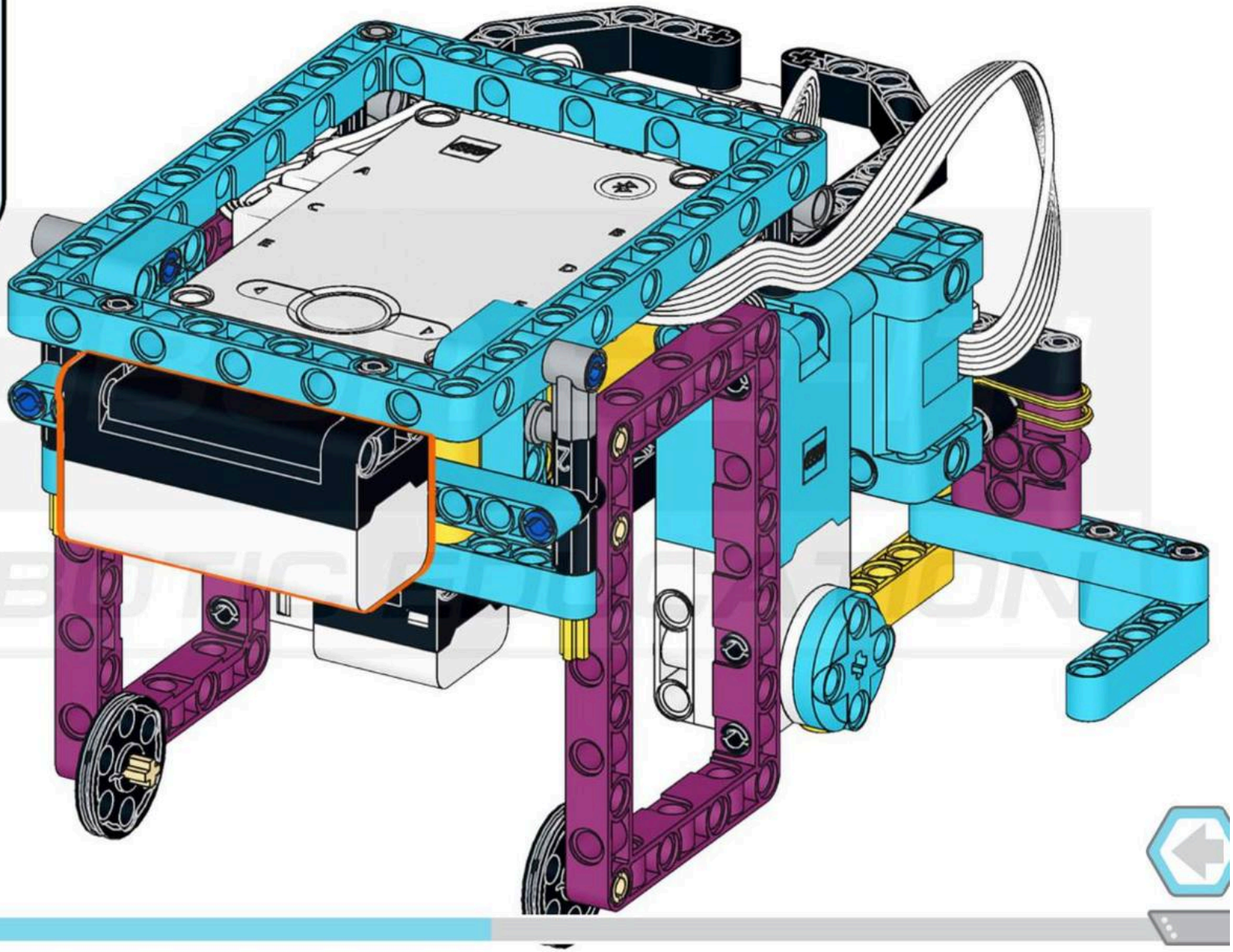


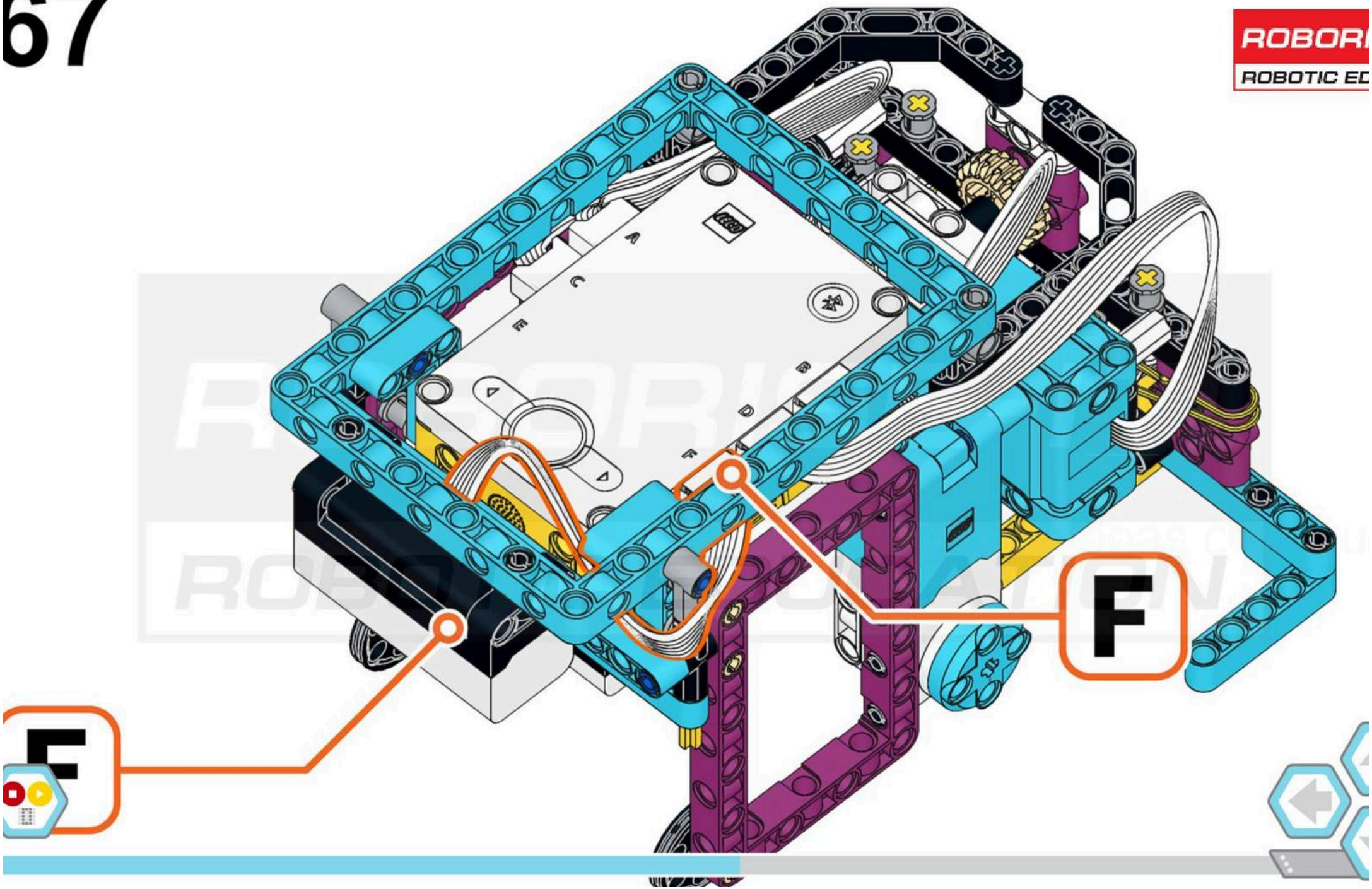
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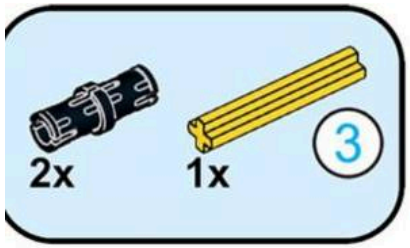




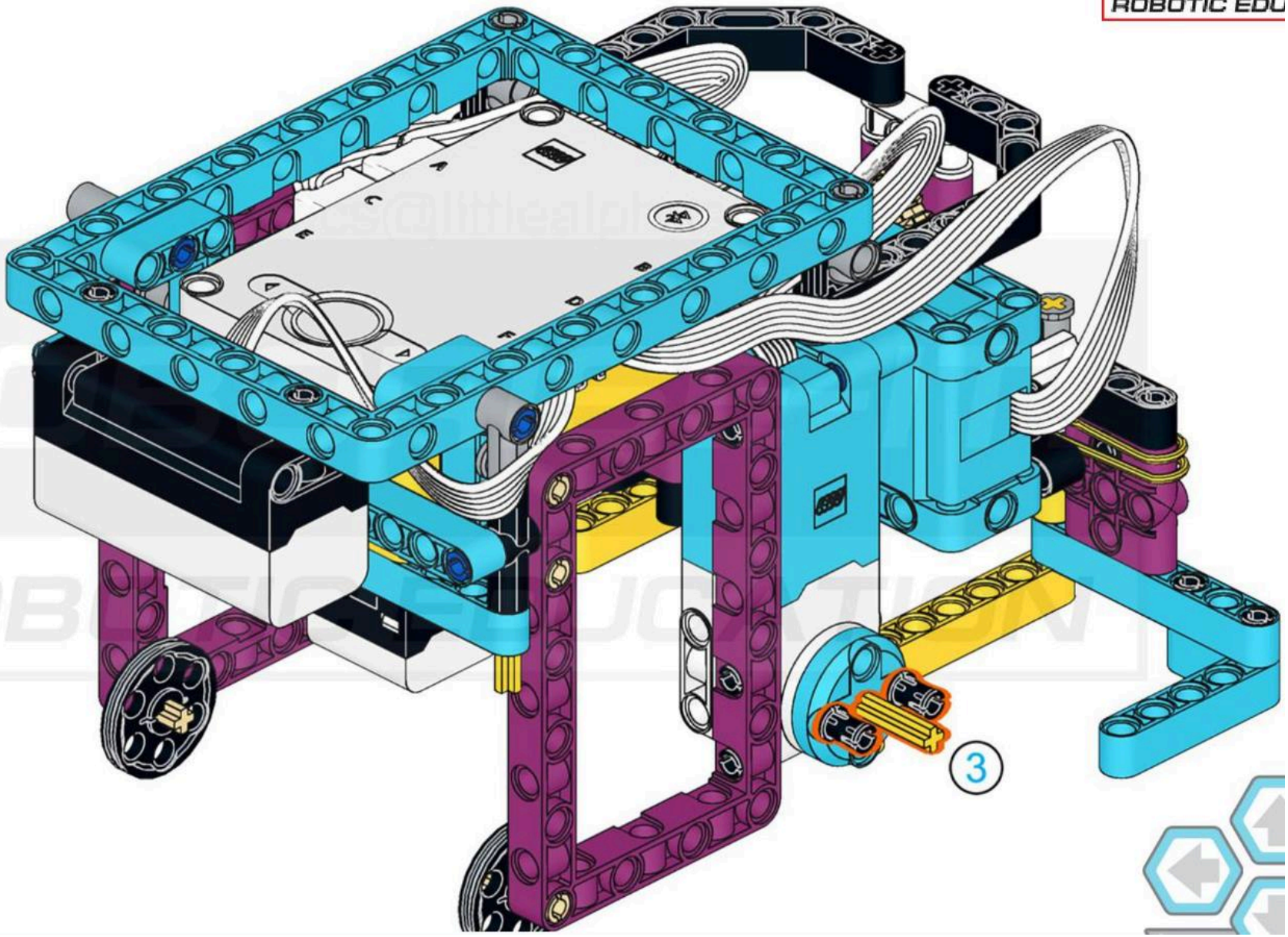
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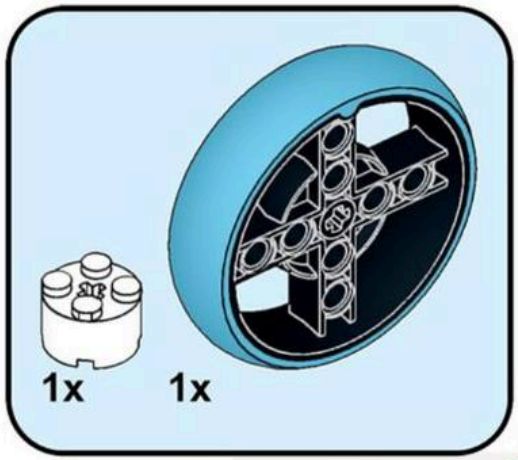



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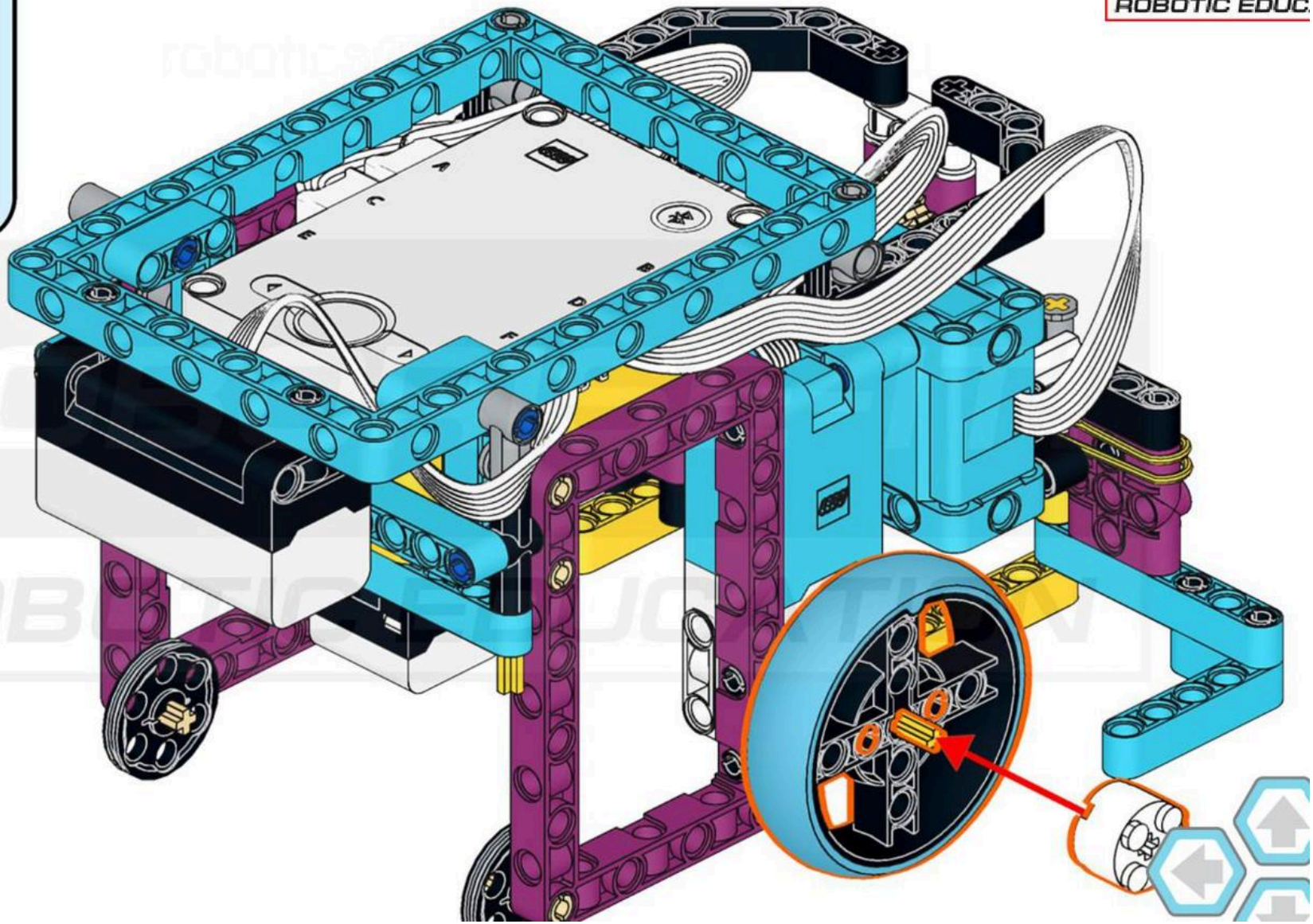


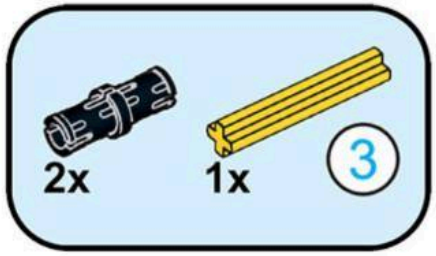
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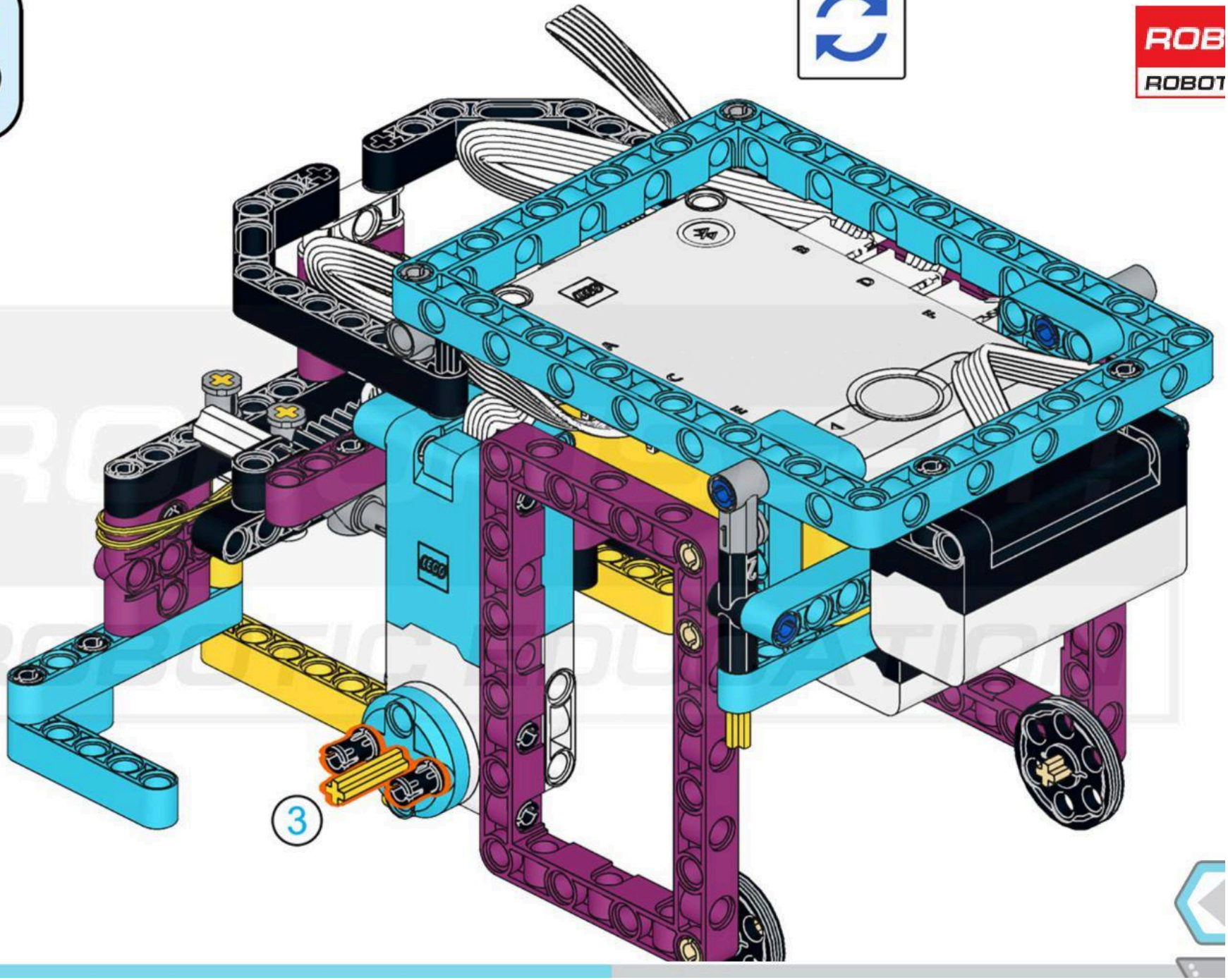


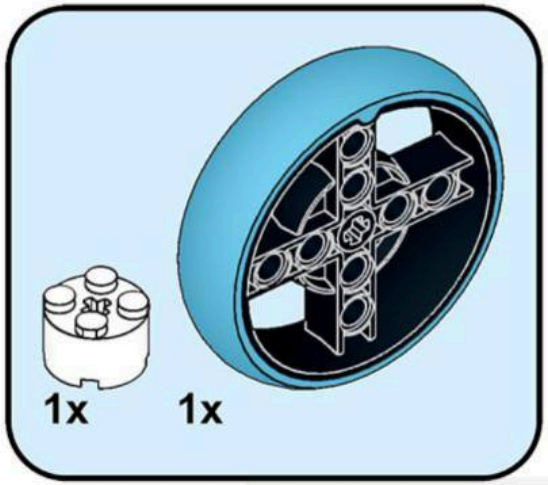
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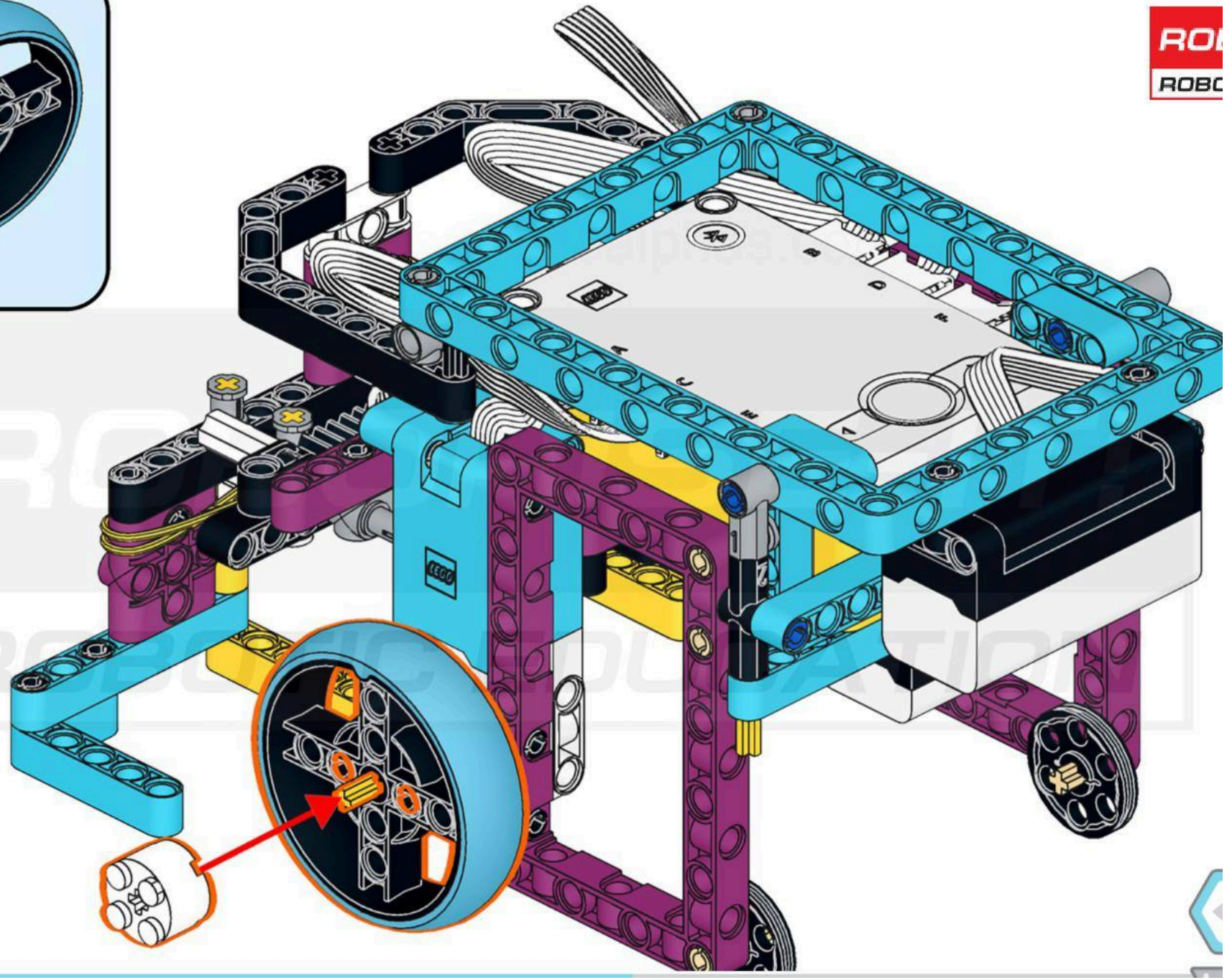


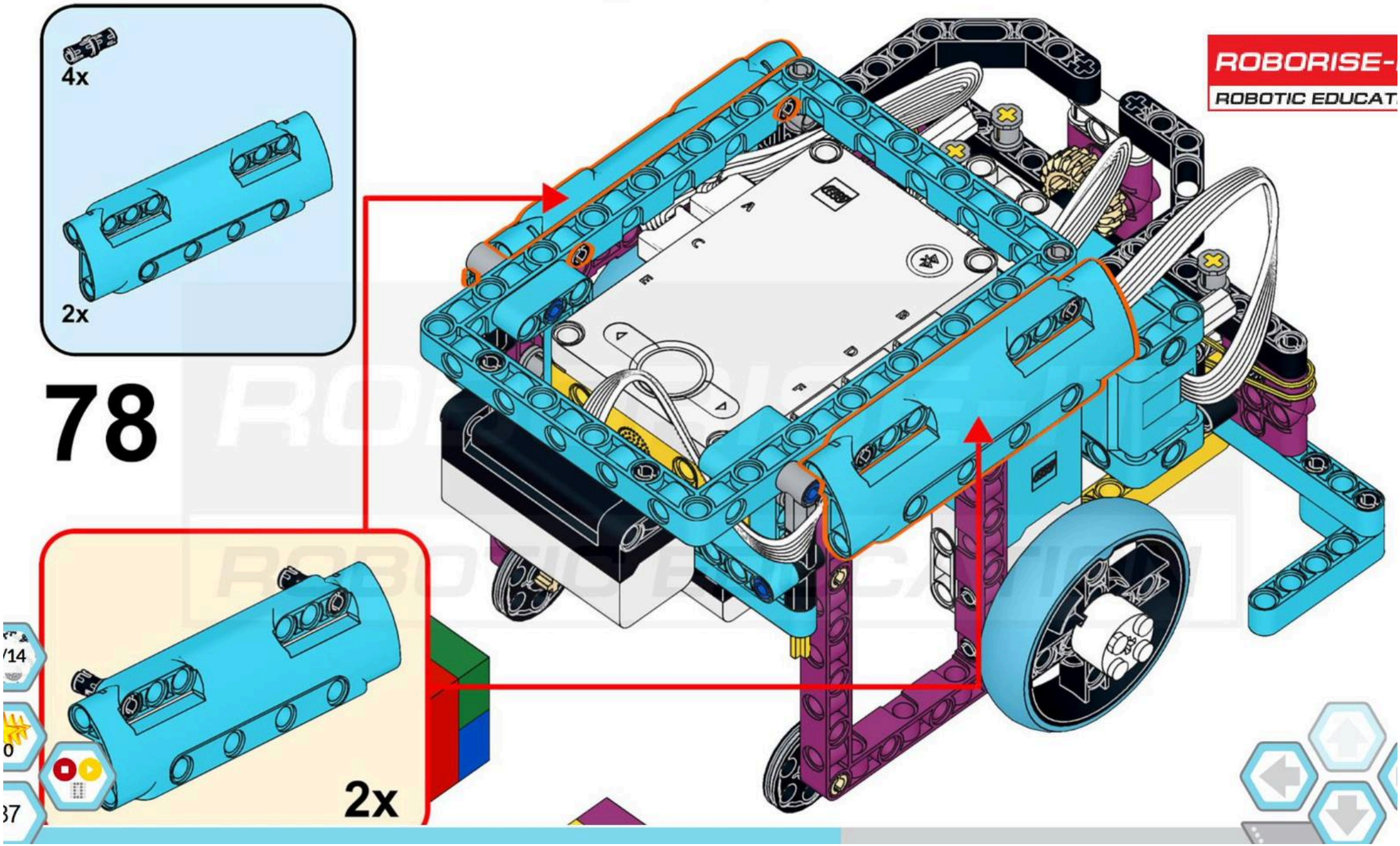
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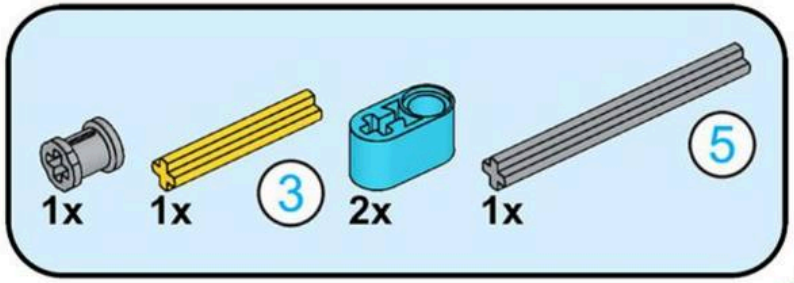




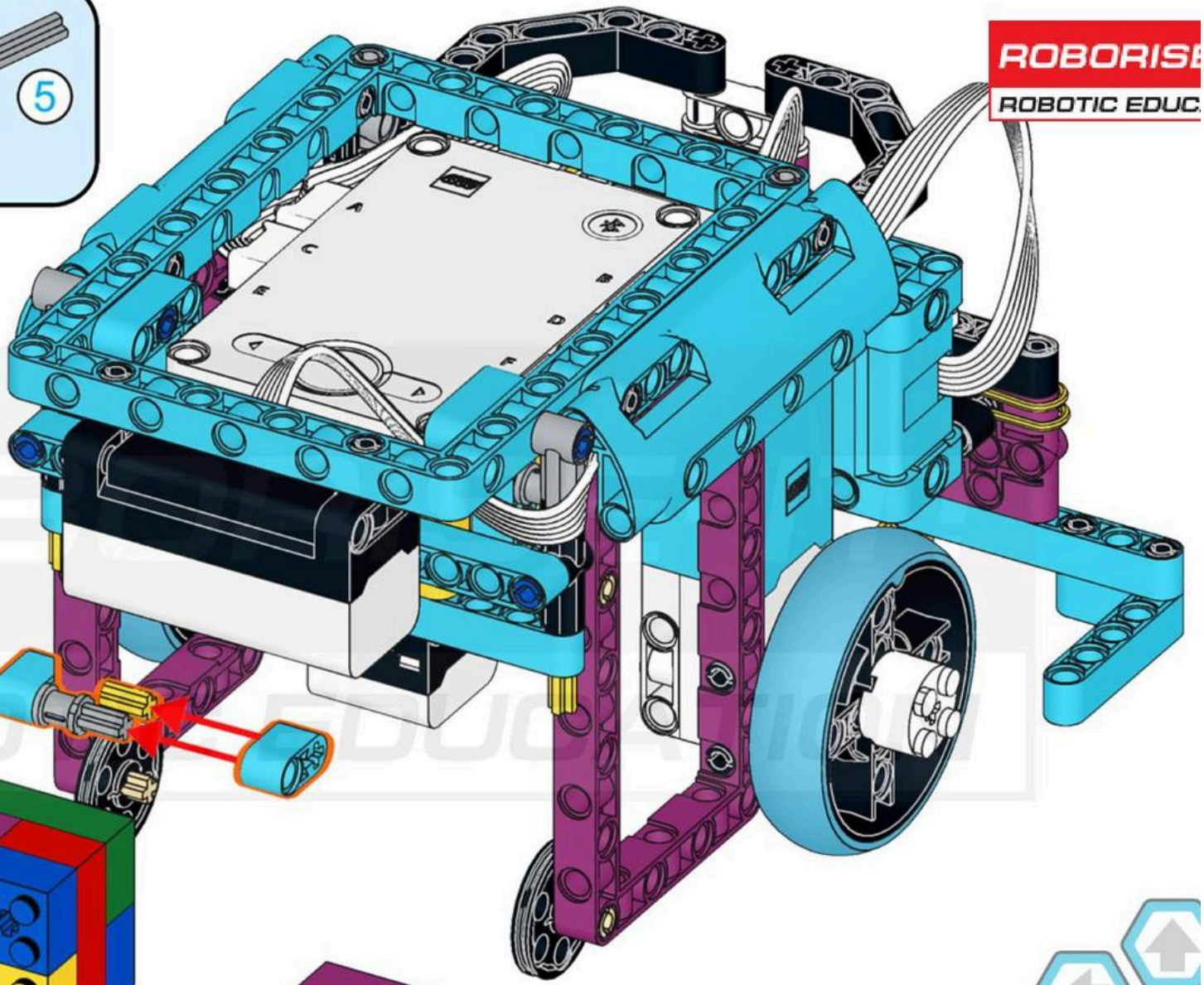
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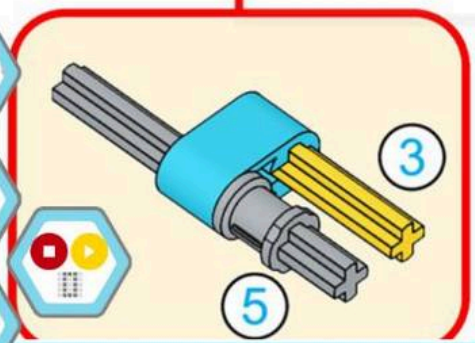


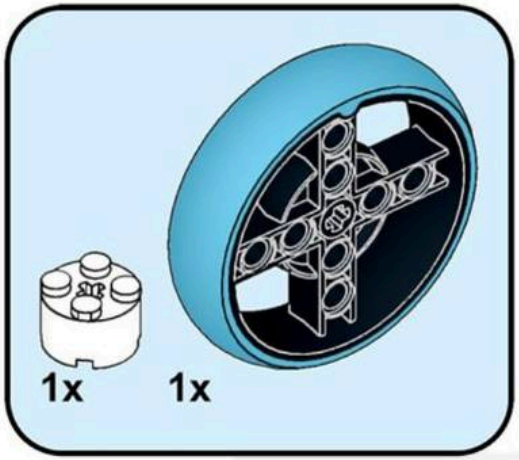

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2x
78
2x
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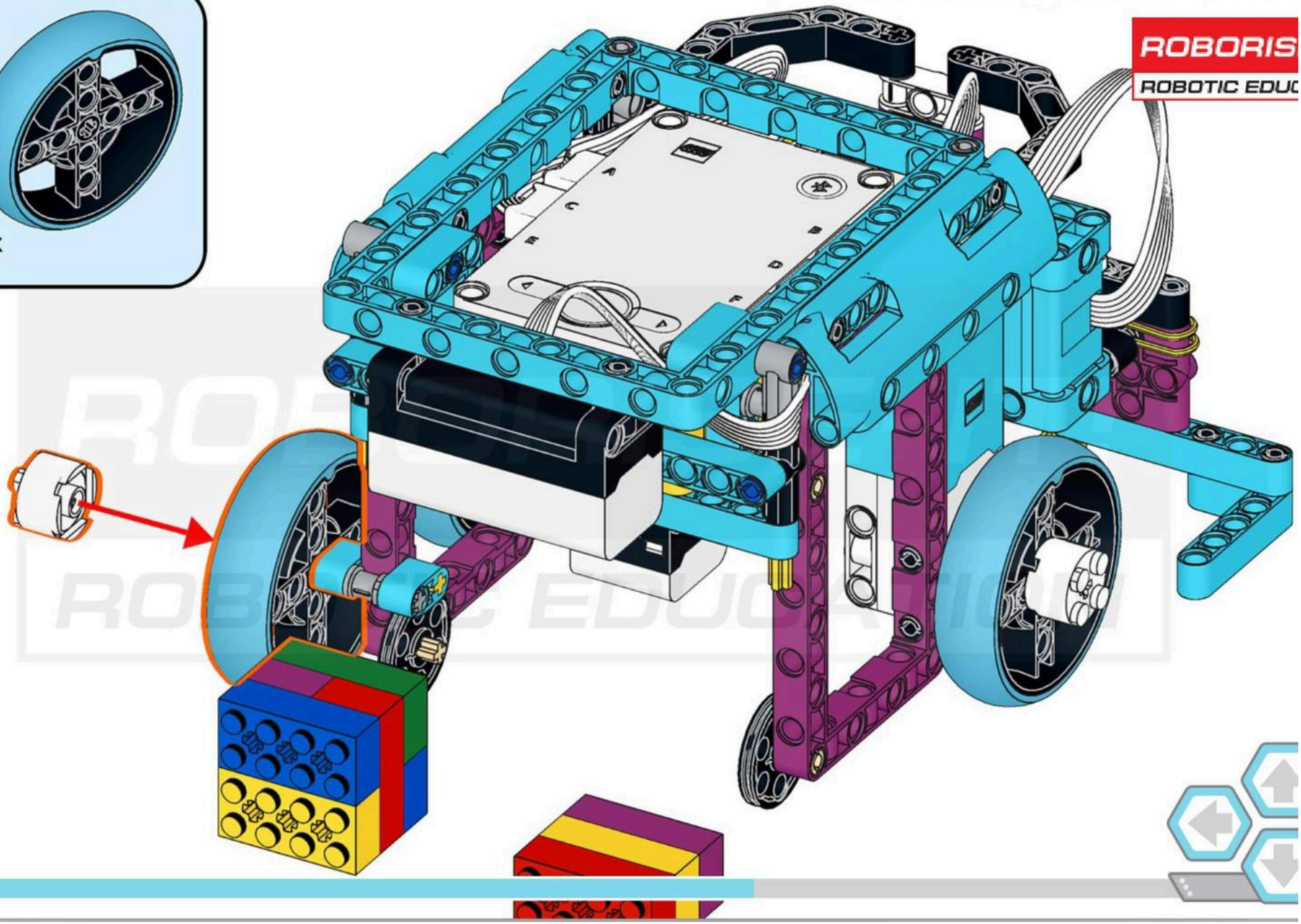
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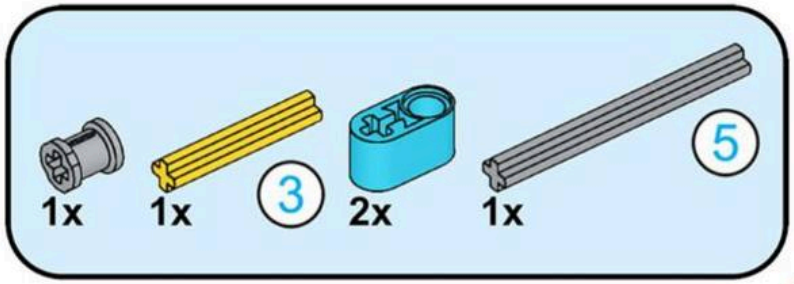






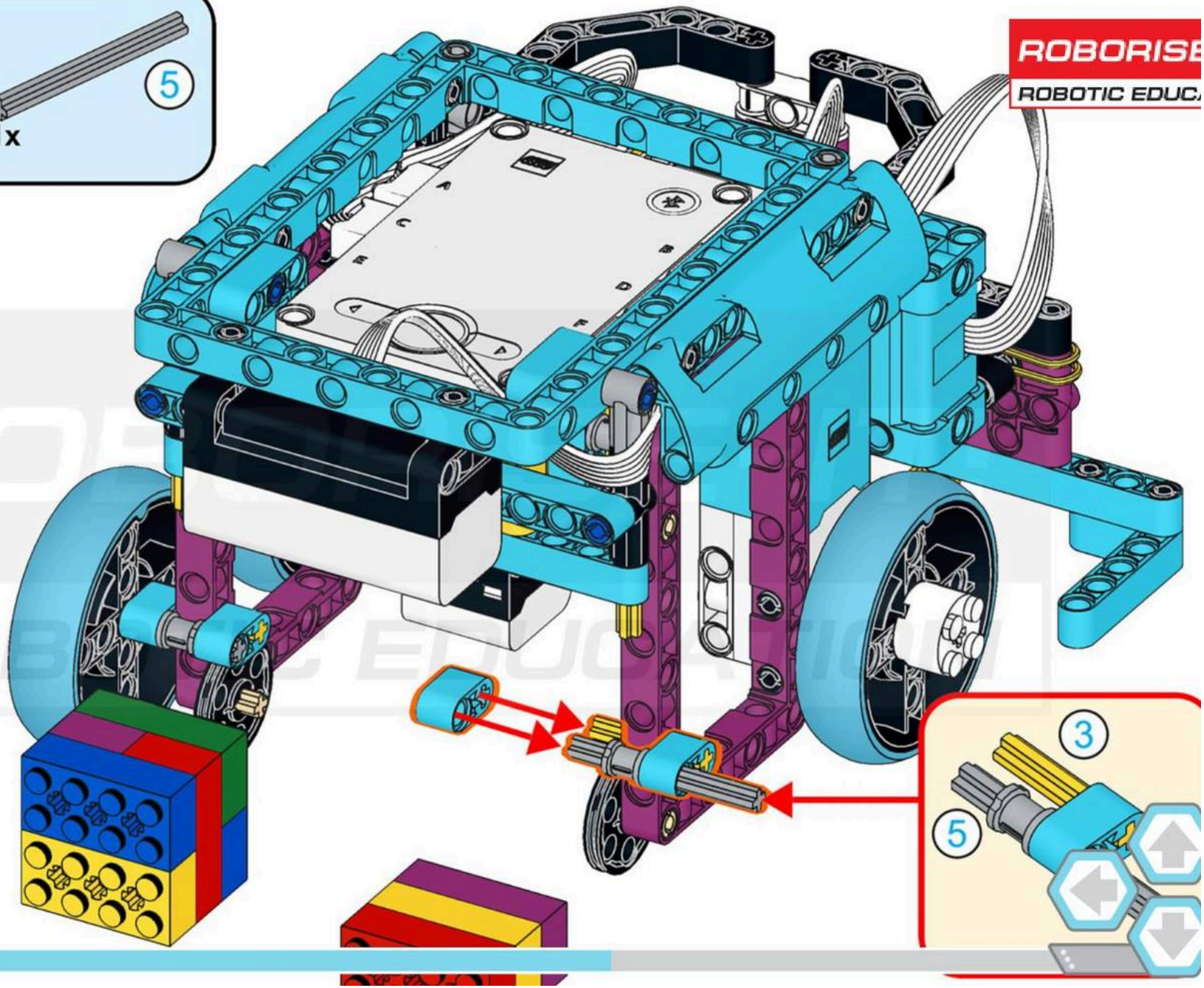
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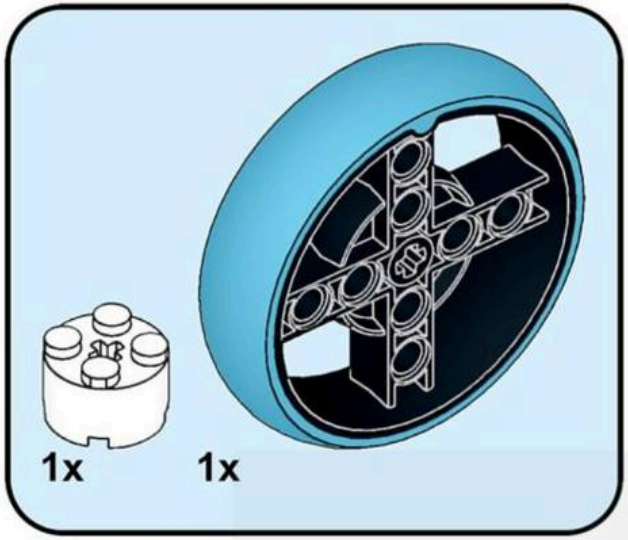




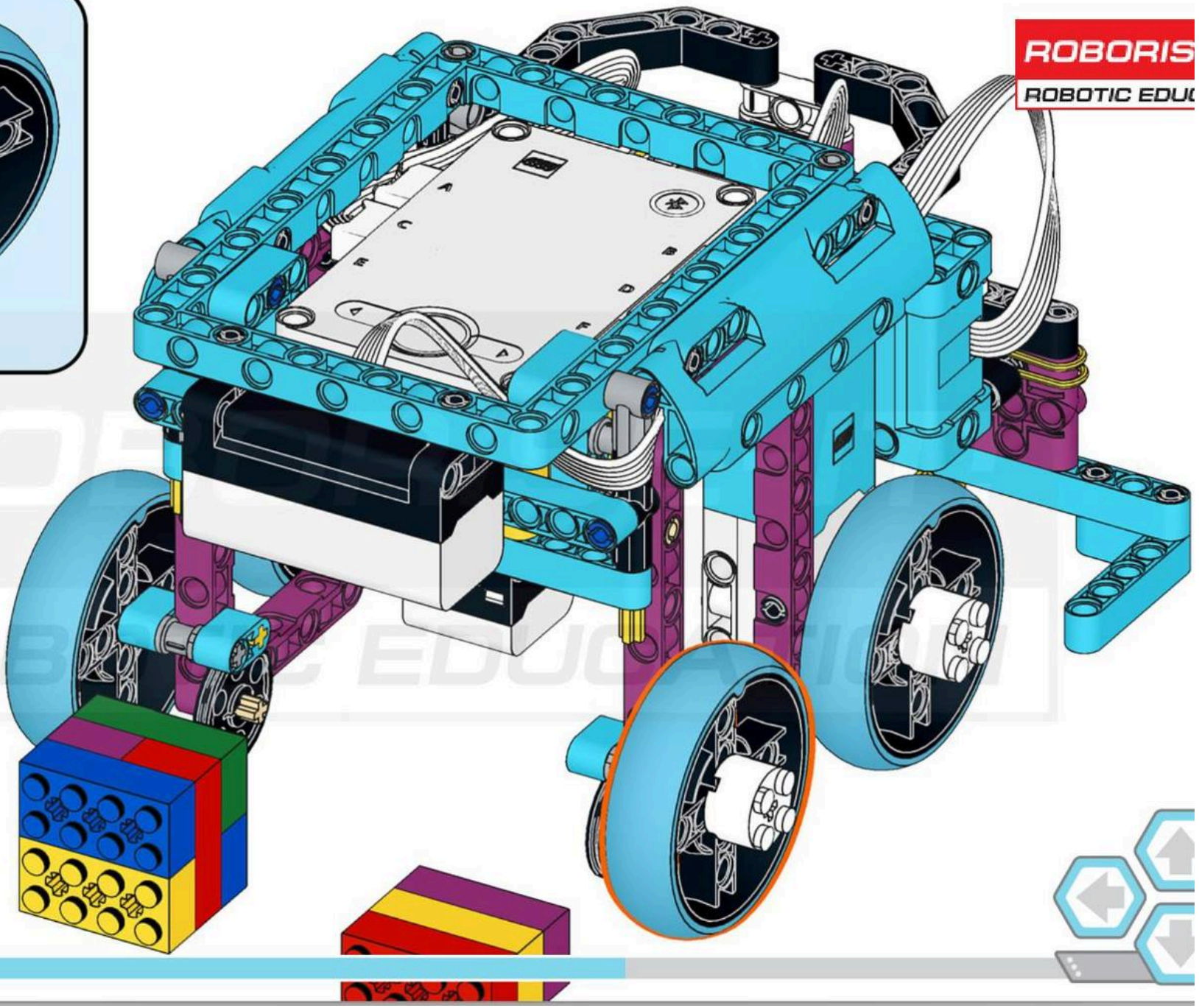
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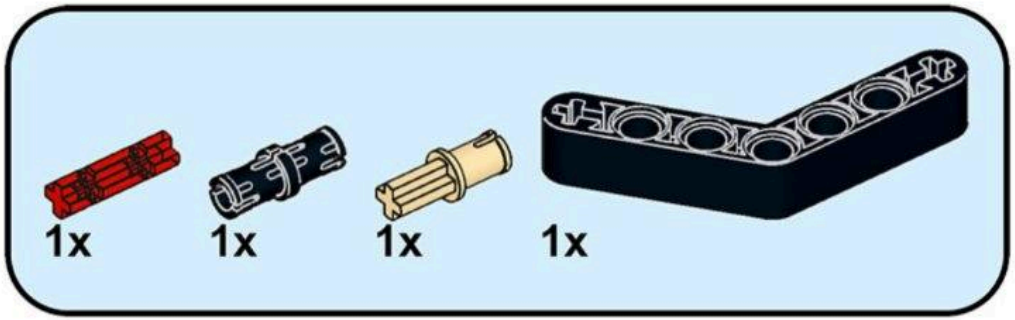
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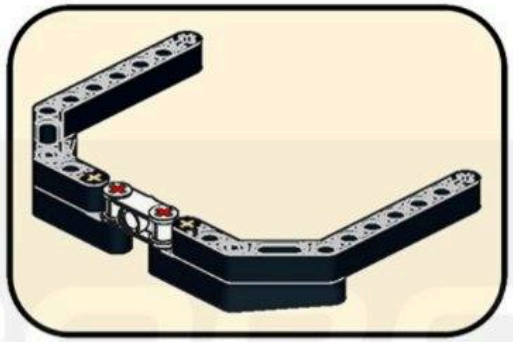


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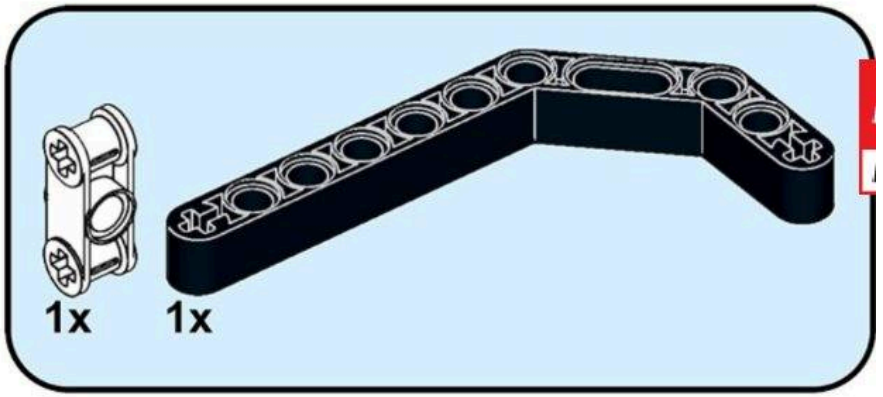




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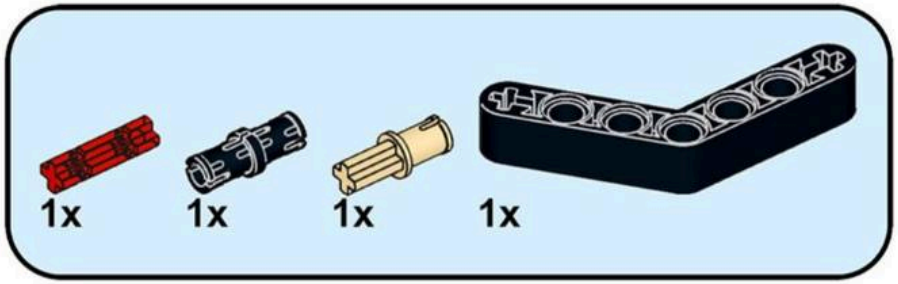




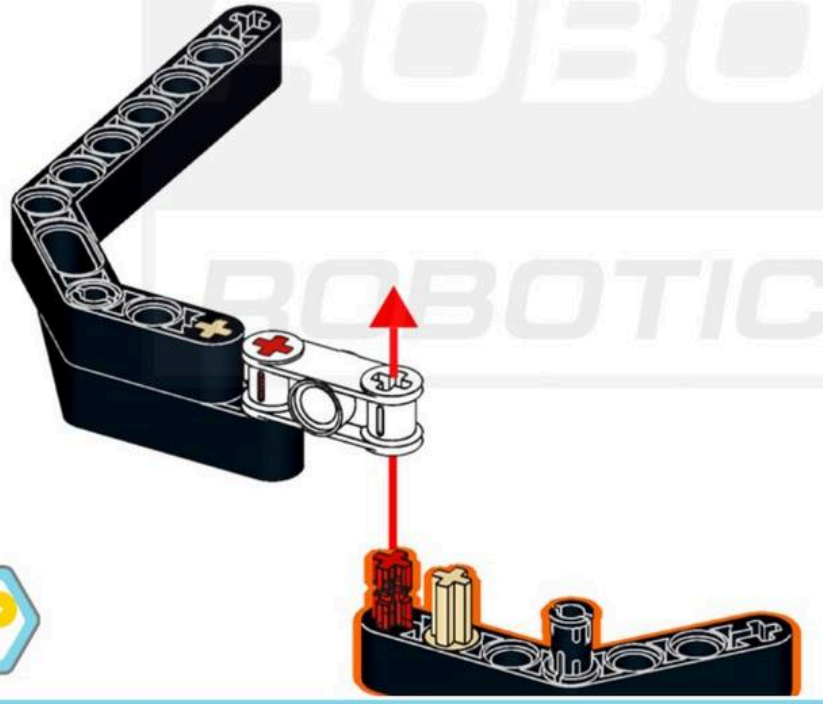
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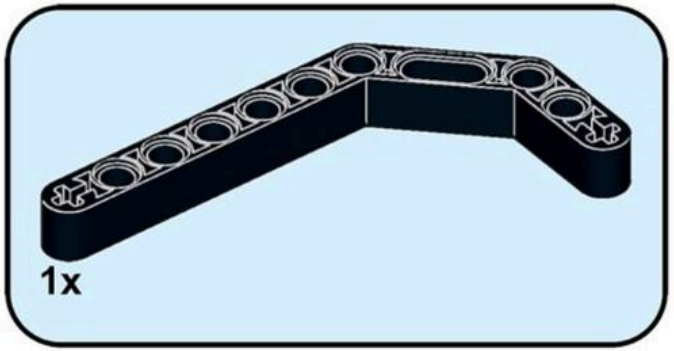


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85

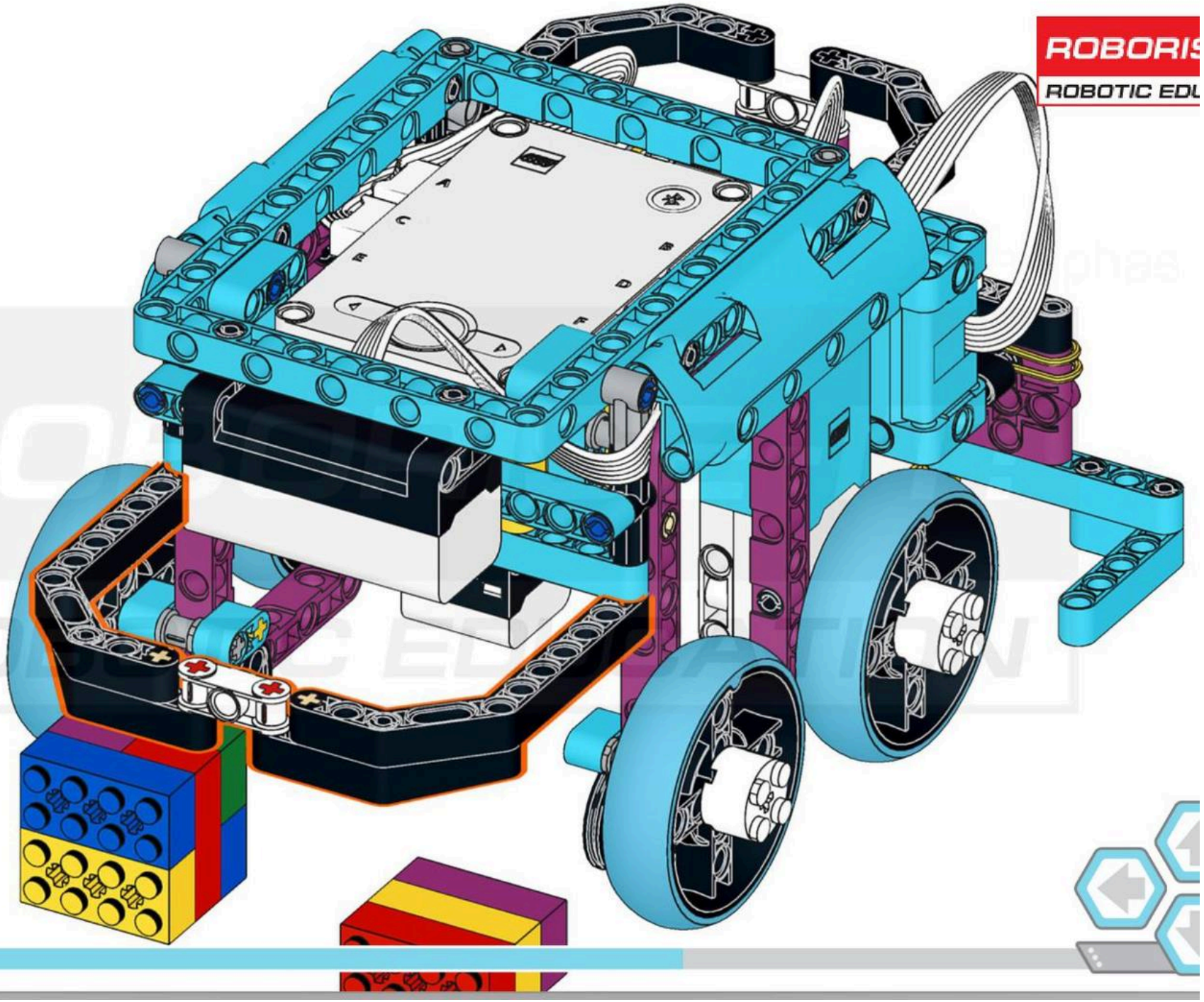




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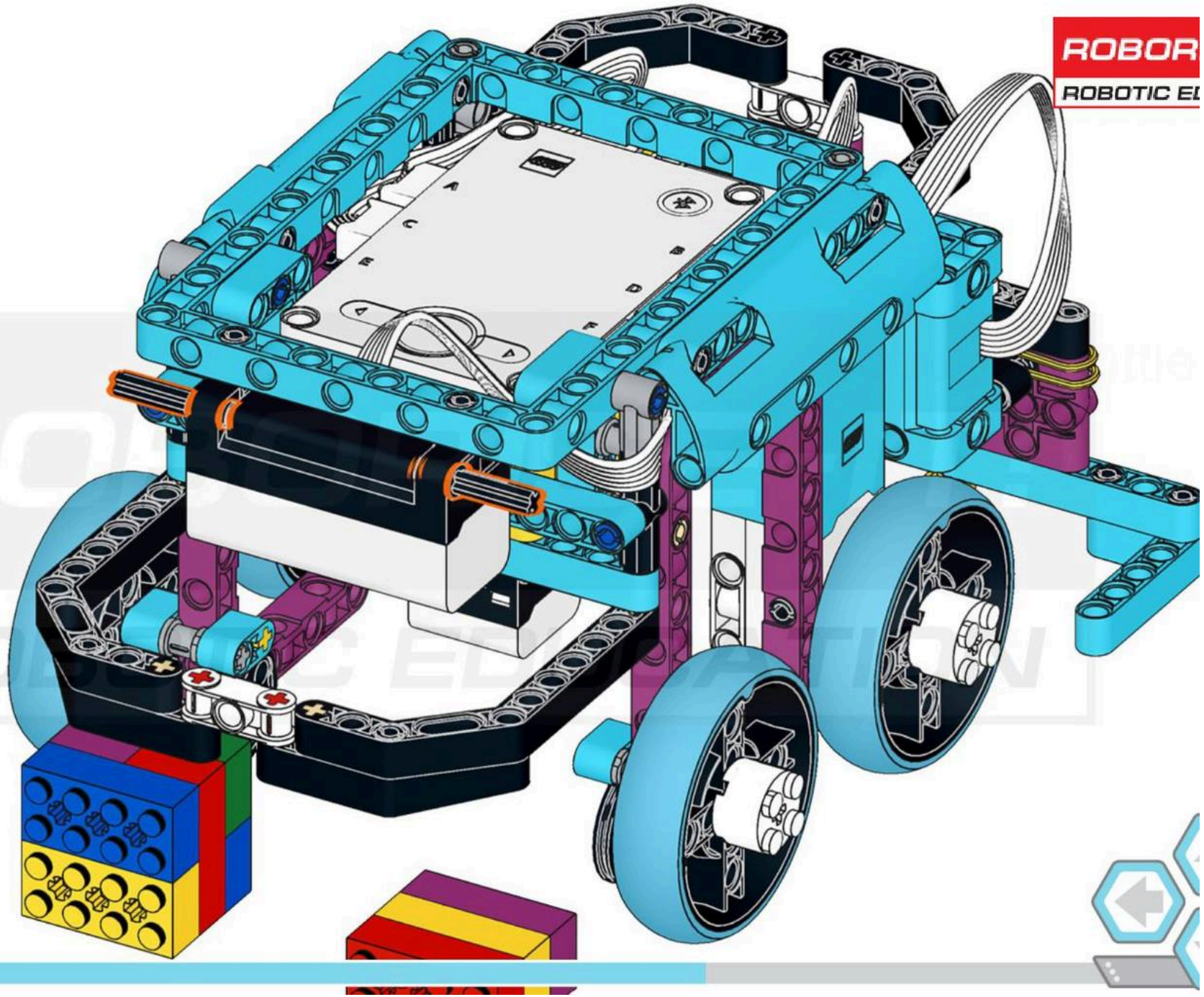


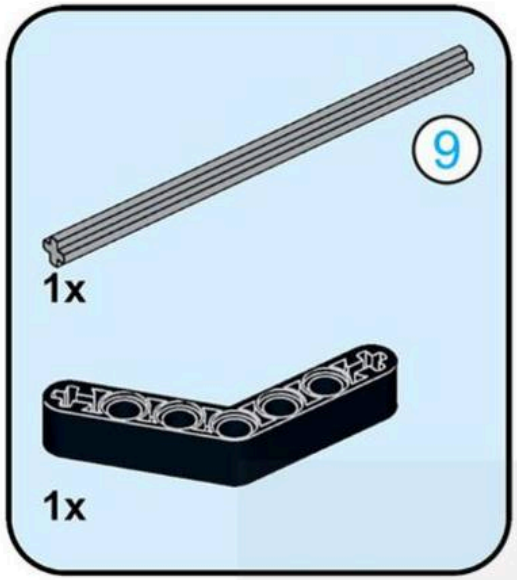
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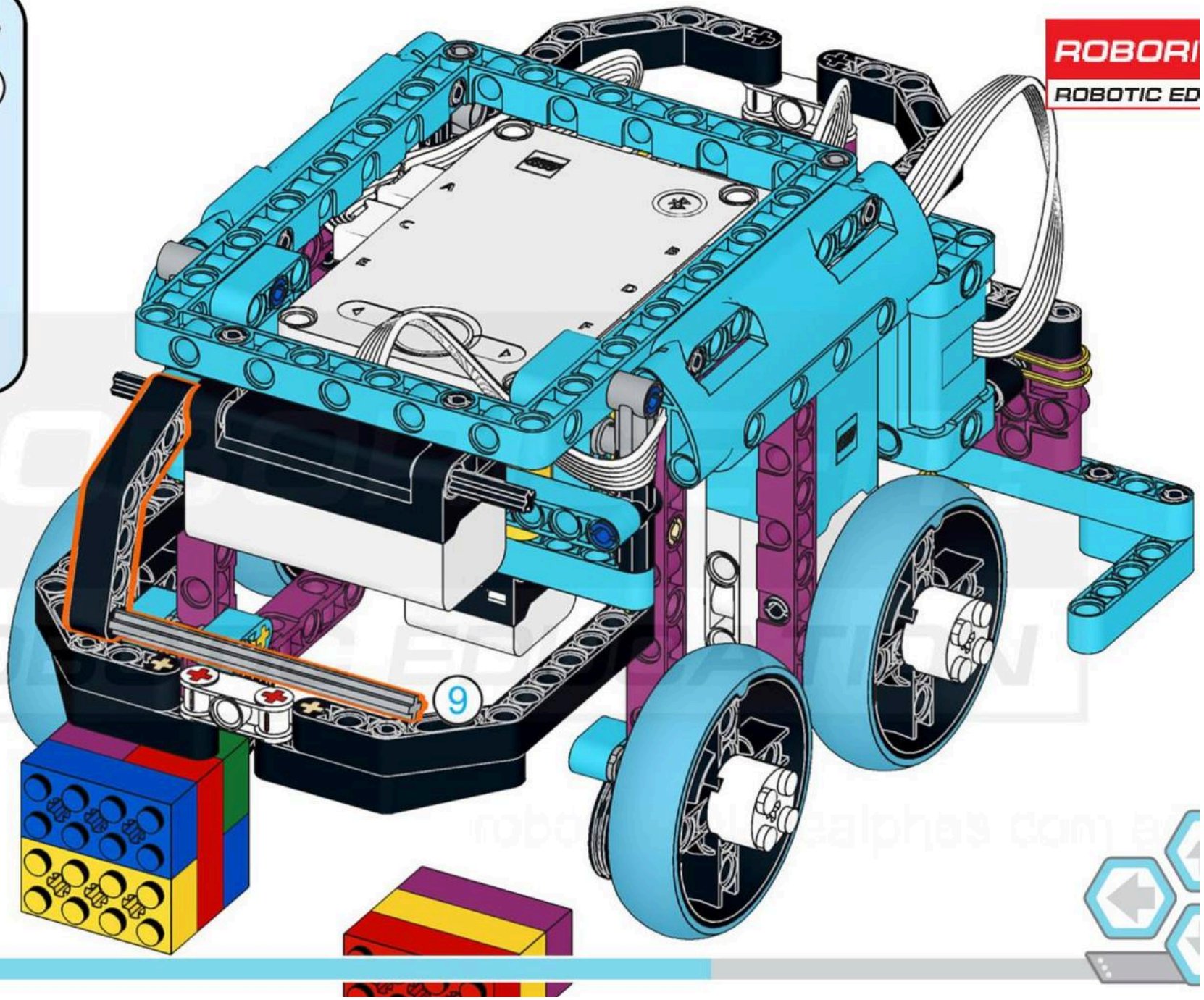


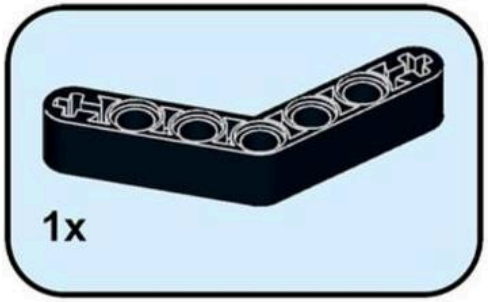
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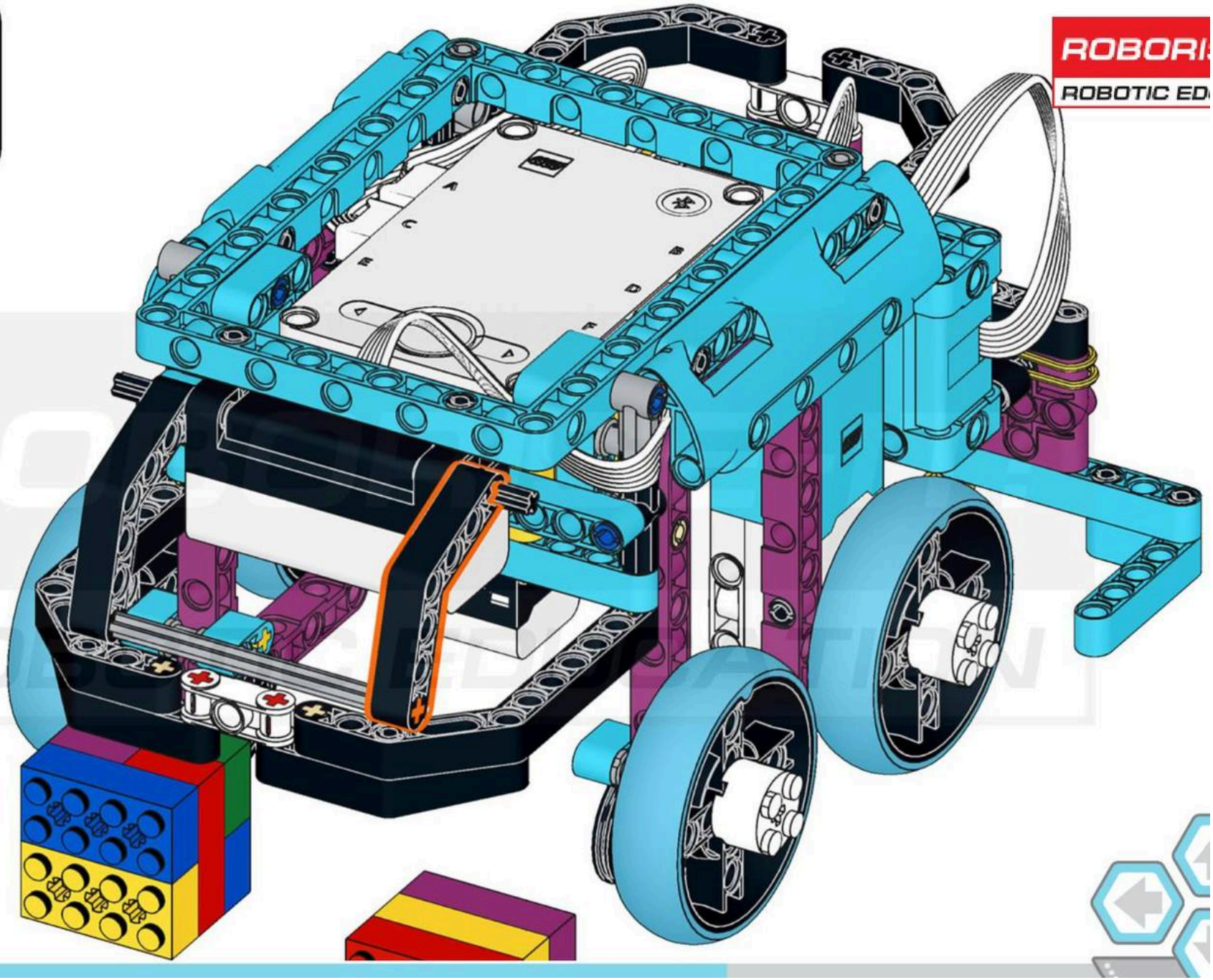


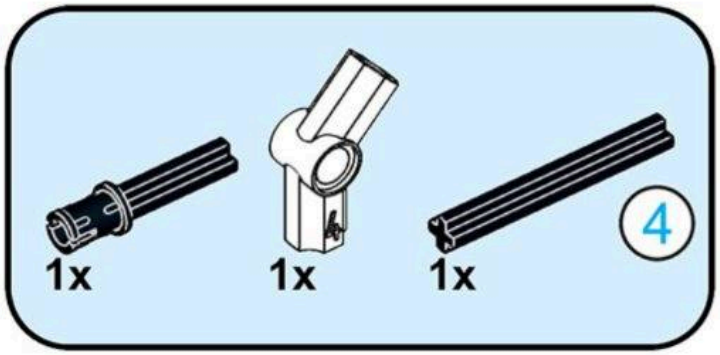
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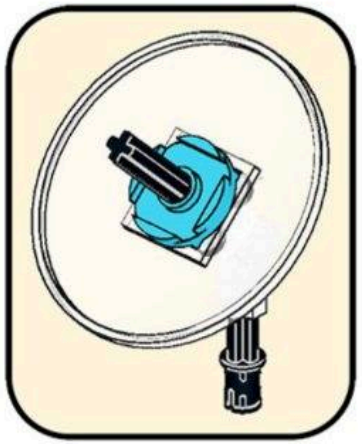


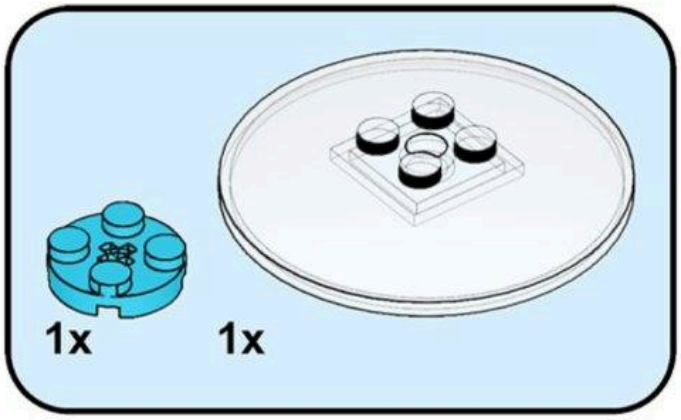


90





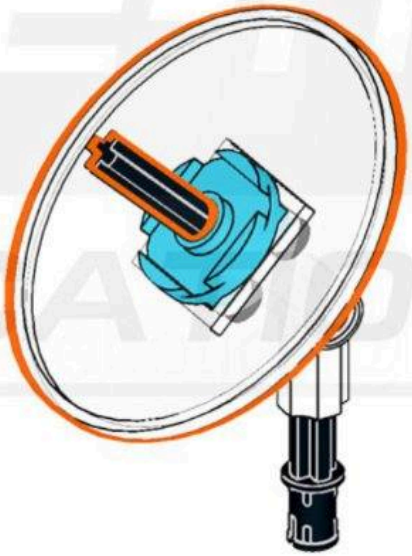




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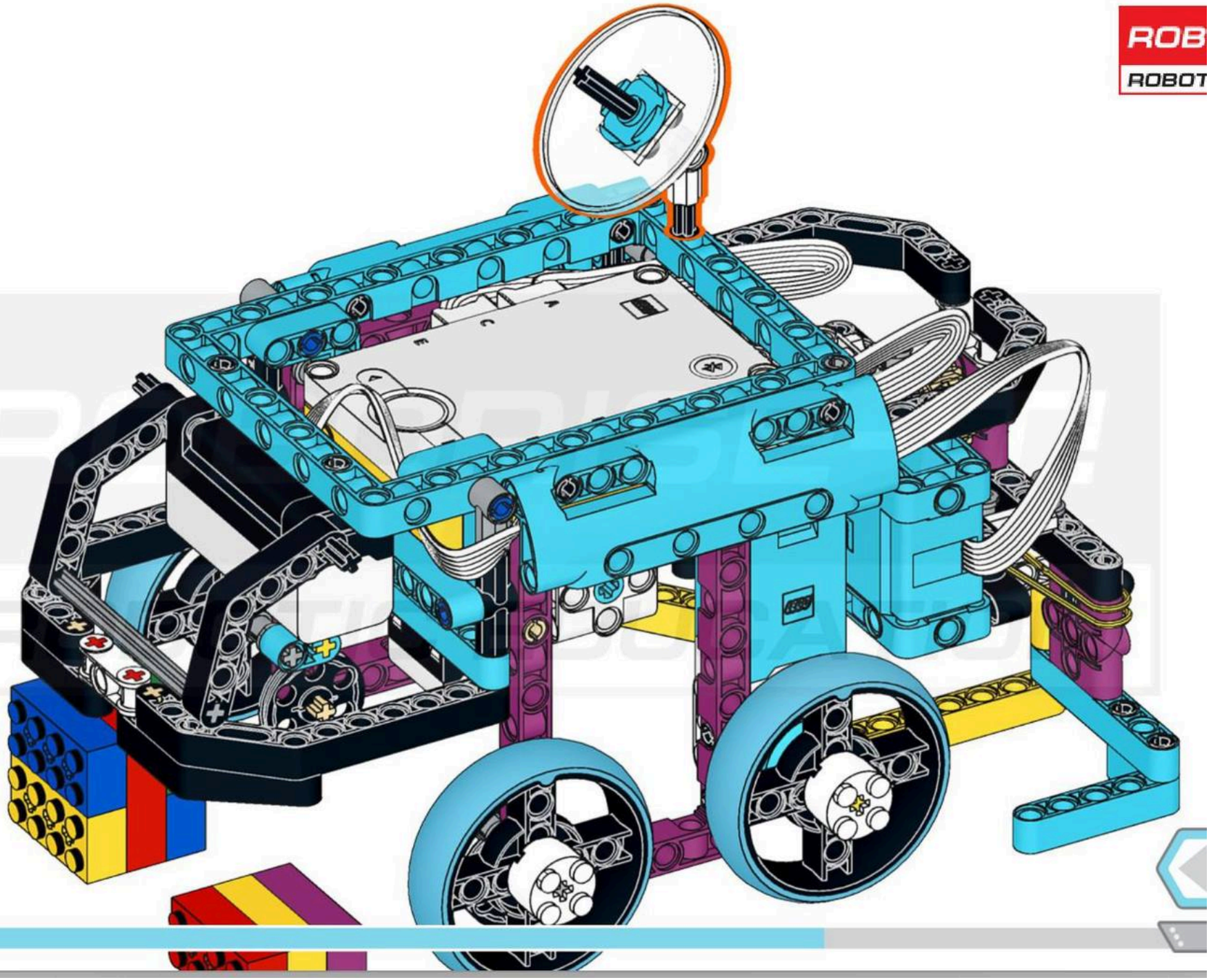


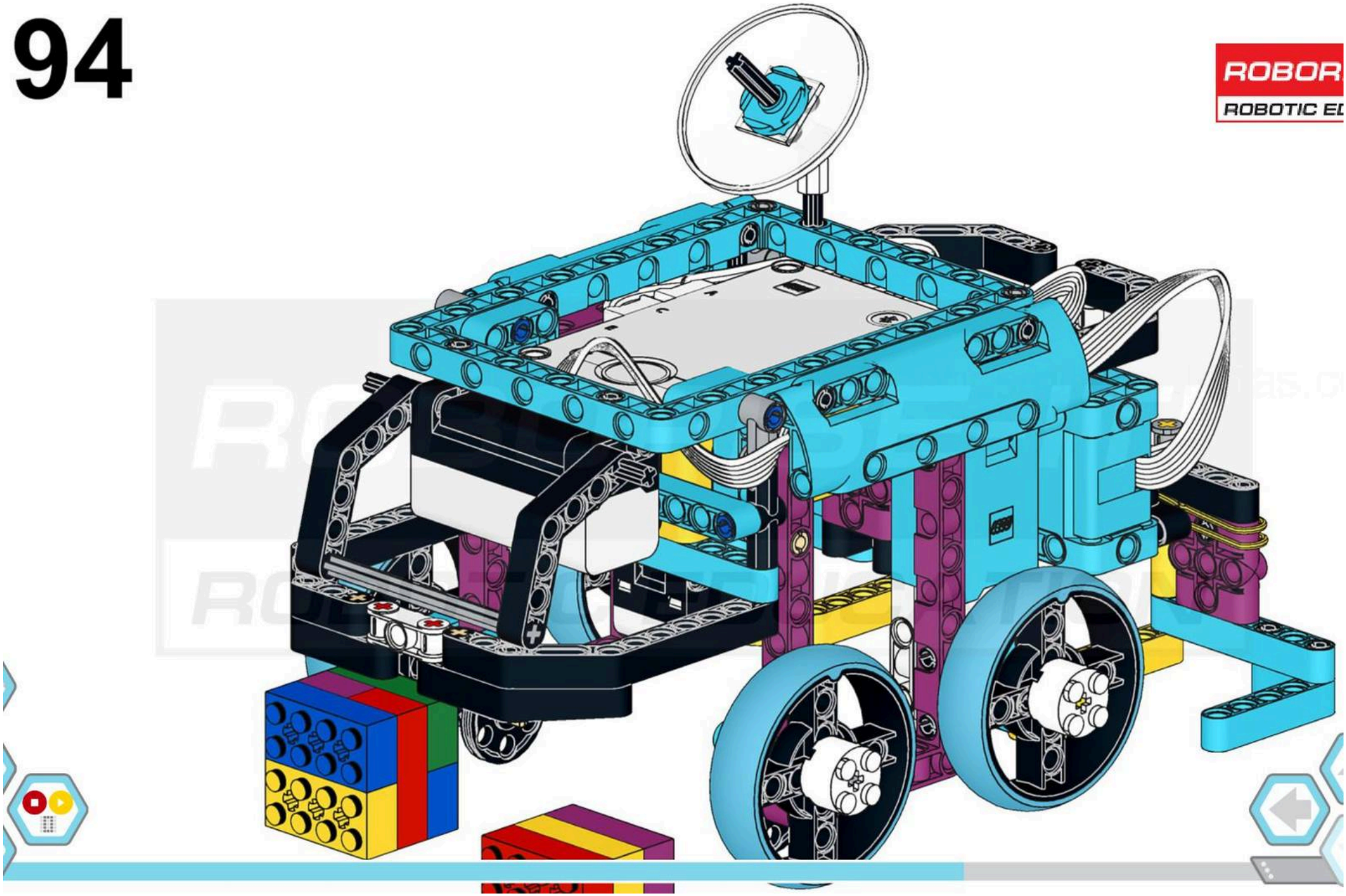
92



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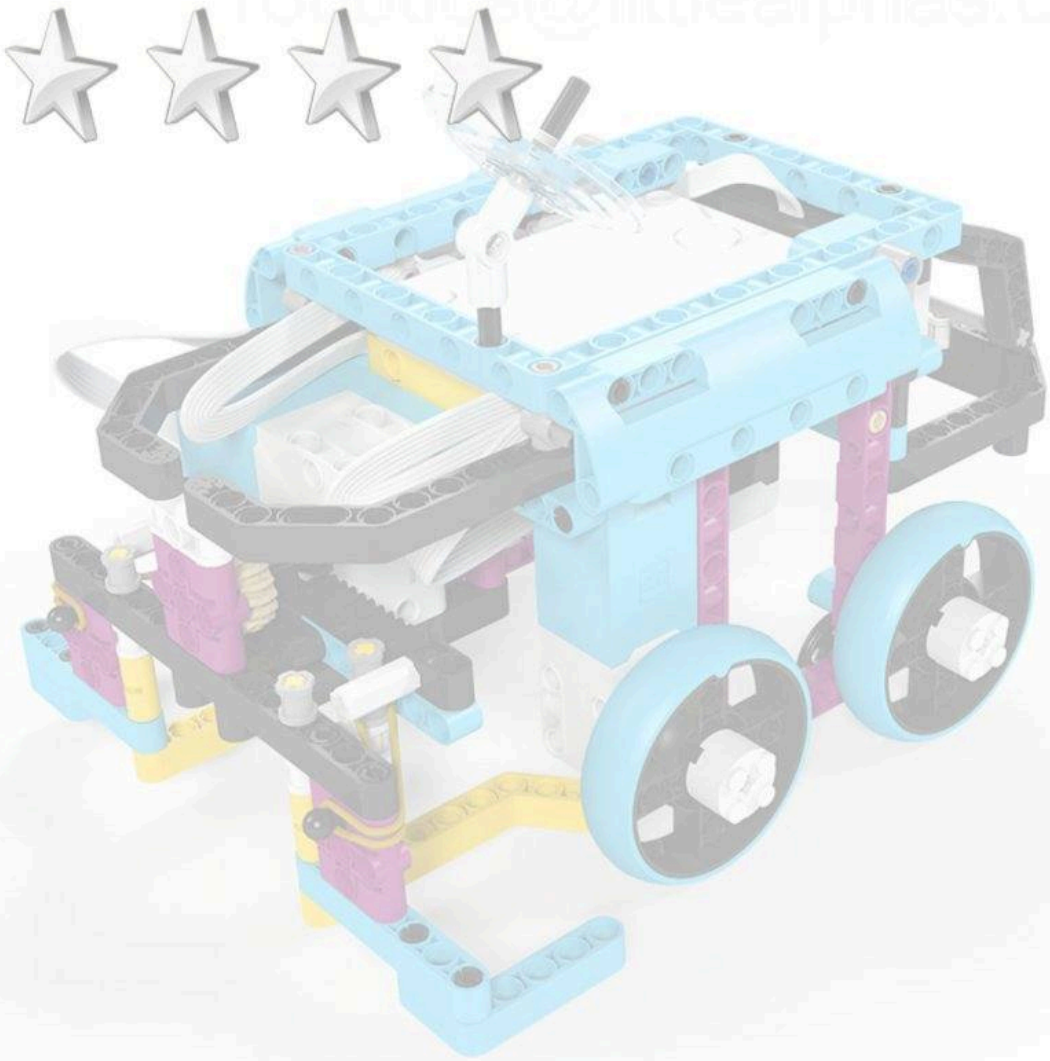
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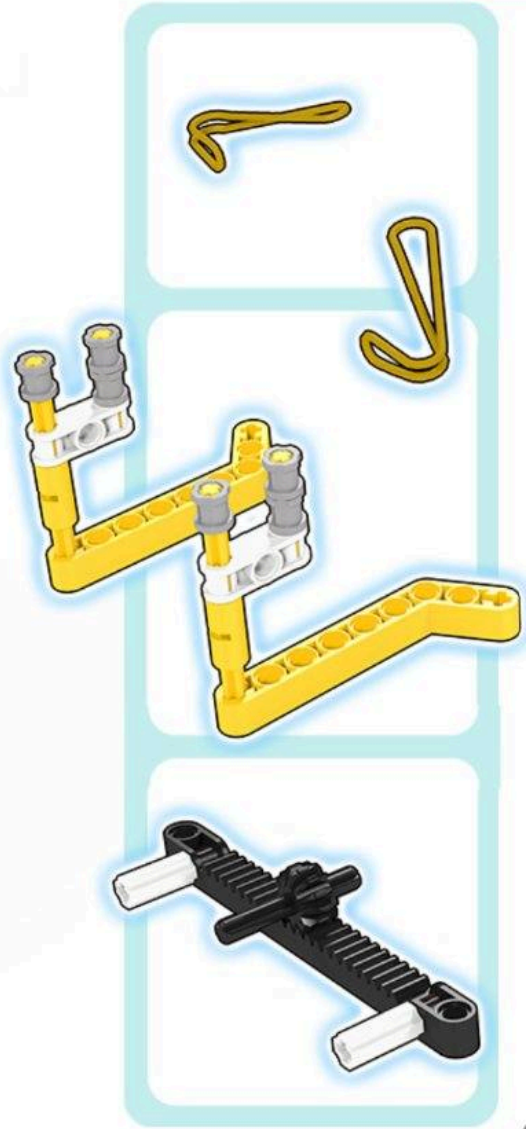




Task

Place the parts of the robot in the correct places





Task



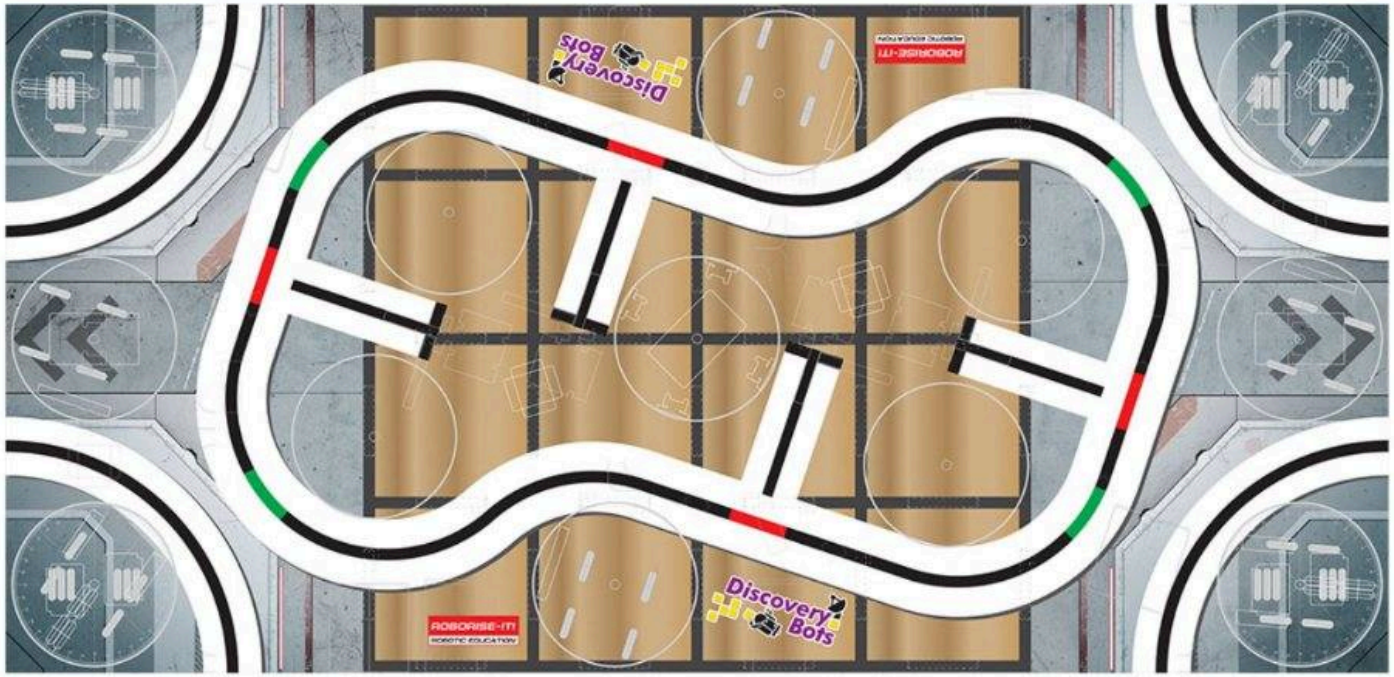
After the meteorite falls, we will use robots to collect samples. At the same time, it is possible to program the collection of samples of any size, or to collect only large or only small fragments of meteorites.



Task field

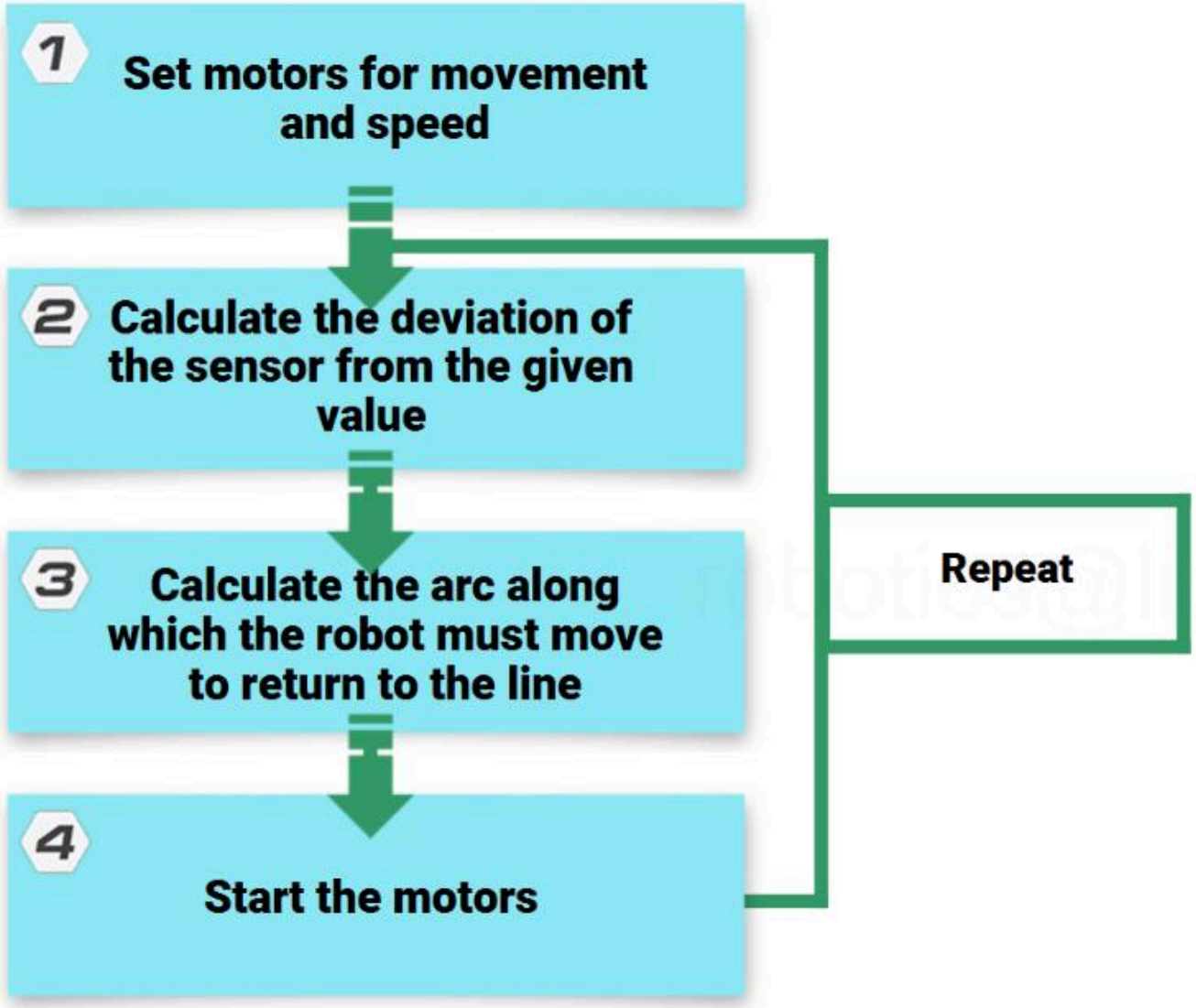


Take a look at the field assigned to the tasks in this course. It has markings that will help you correctly place objects for tasks, as well as markings for the movement of the robot itself. At the same time, tasks can be performed by two, and sometimes four teams!

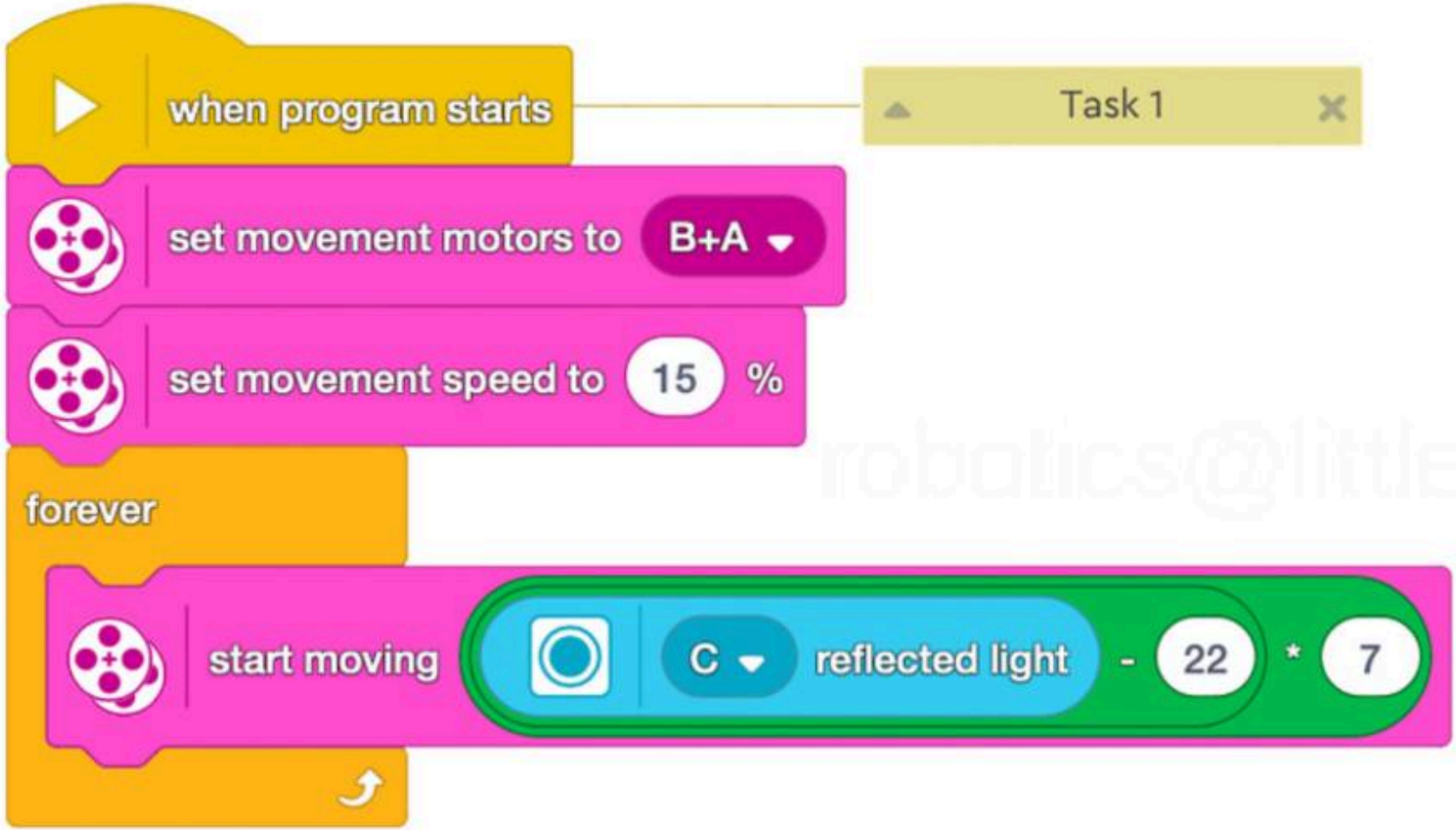


Please note that if the field is printed in such a way that the robot sensor does not correctly detect the colors, you can cover the corresponding areas with self-adhesive films "Oracal"

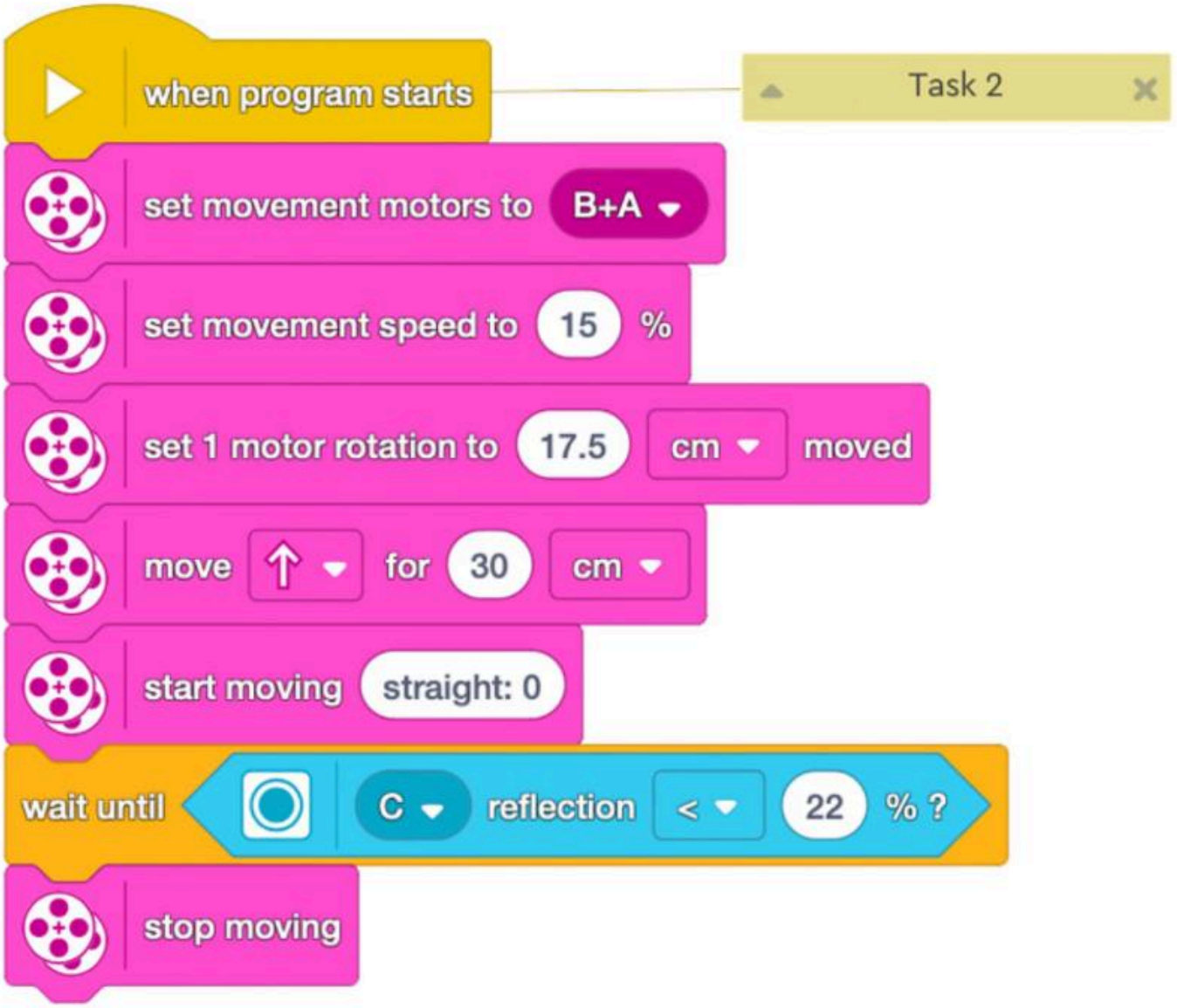
The robot movement algorithm along the marking line using a proportional regulator is as follows:



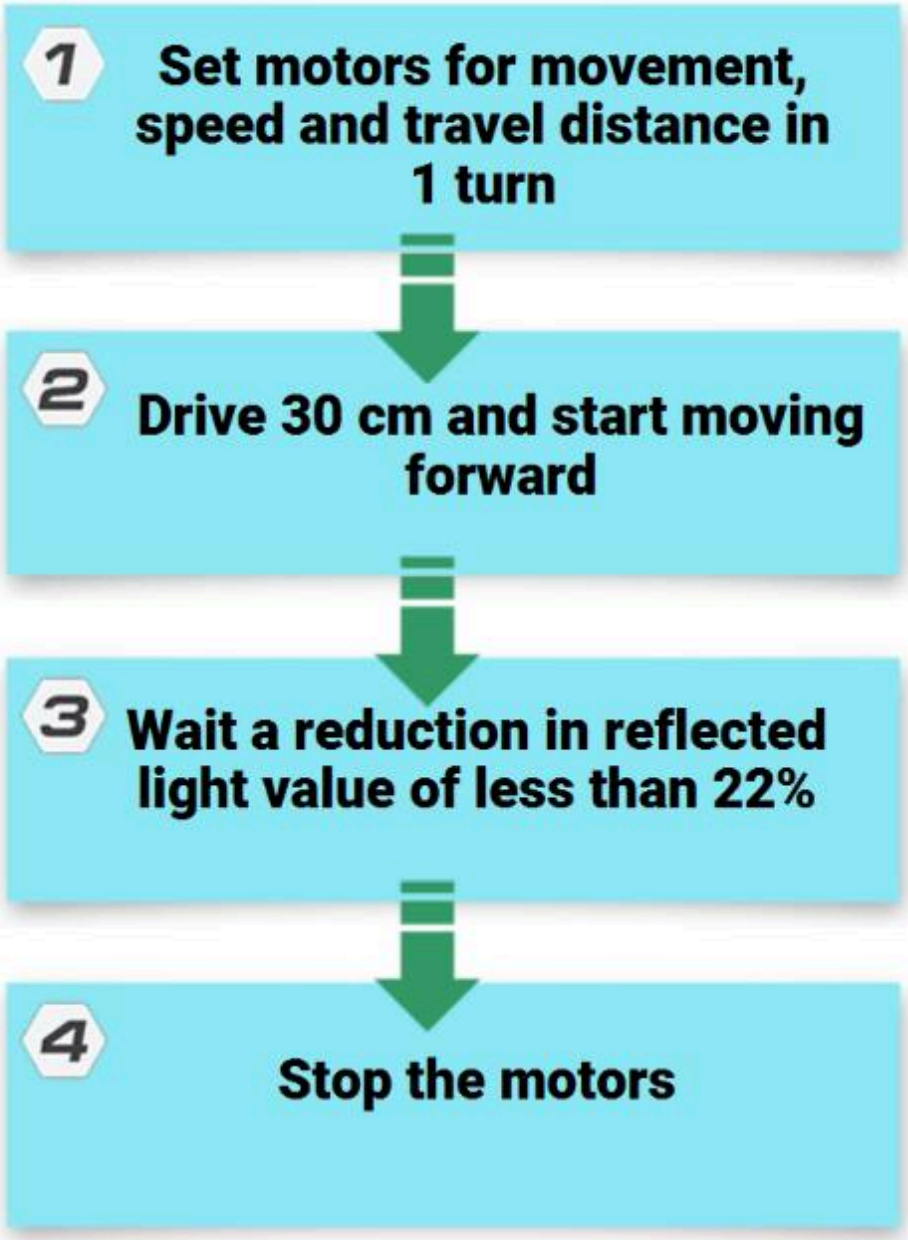
The program may look like this:



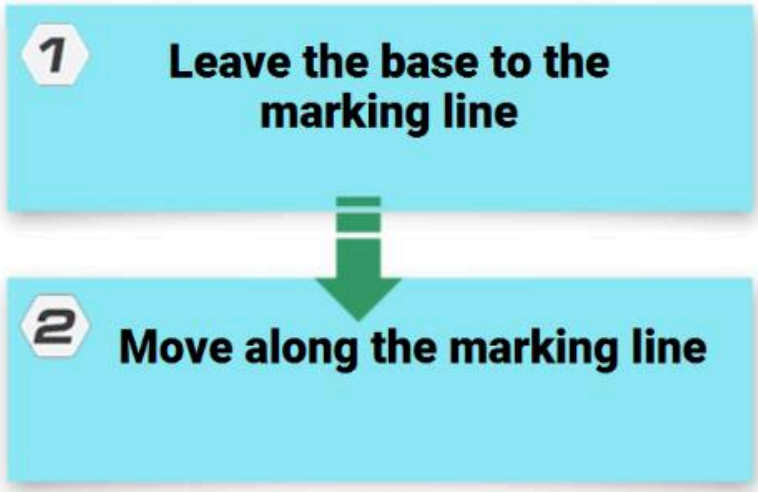
The program may look like this. Please note that journey times may differ when starting from another base!



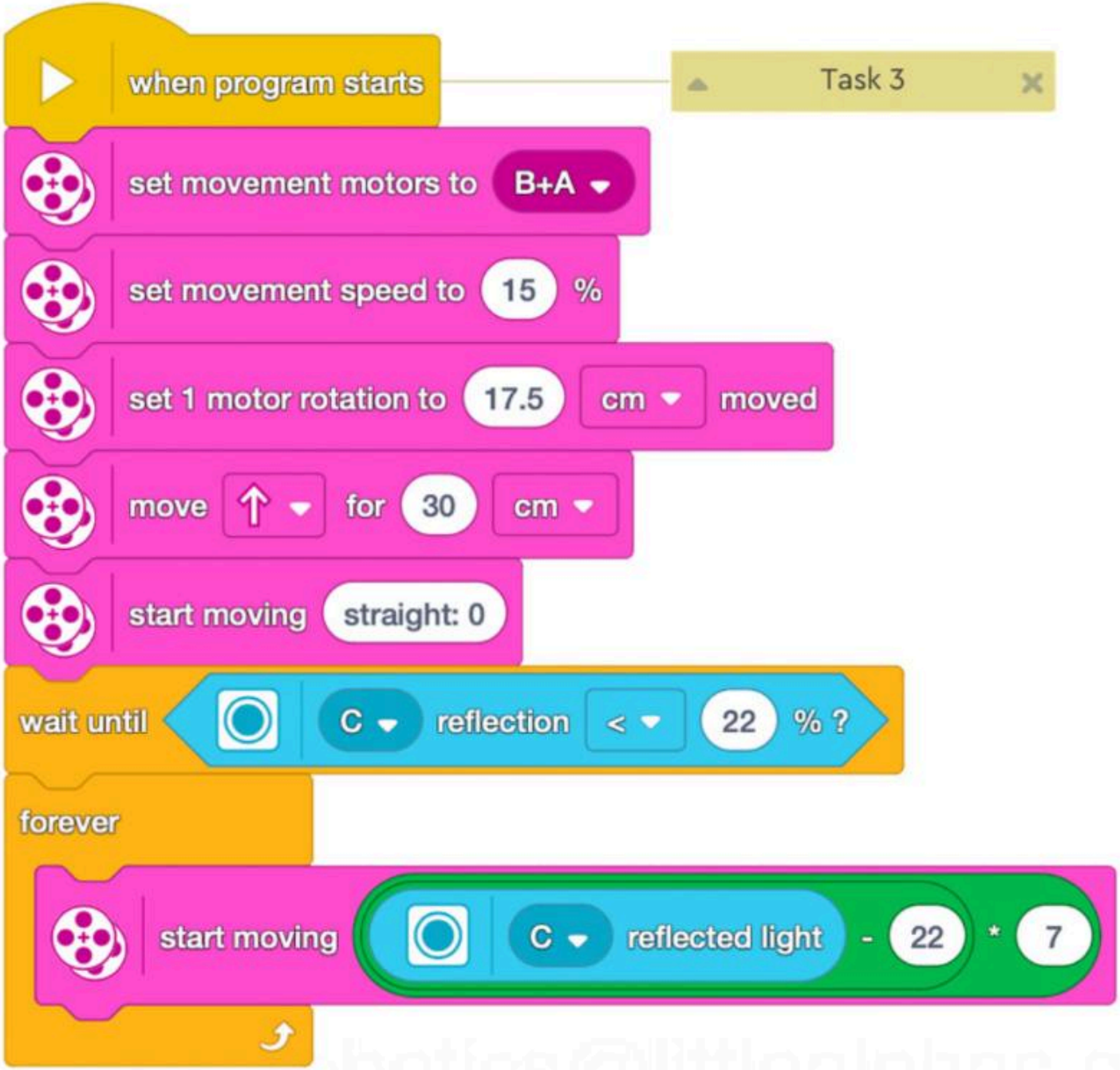
The algorithm for leaving the marking line can look like this:



To perform the assigned task, the program must work according to the following algorithm:

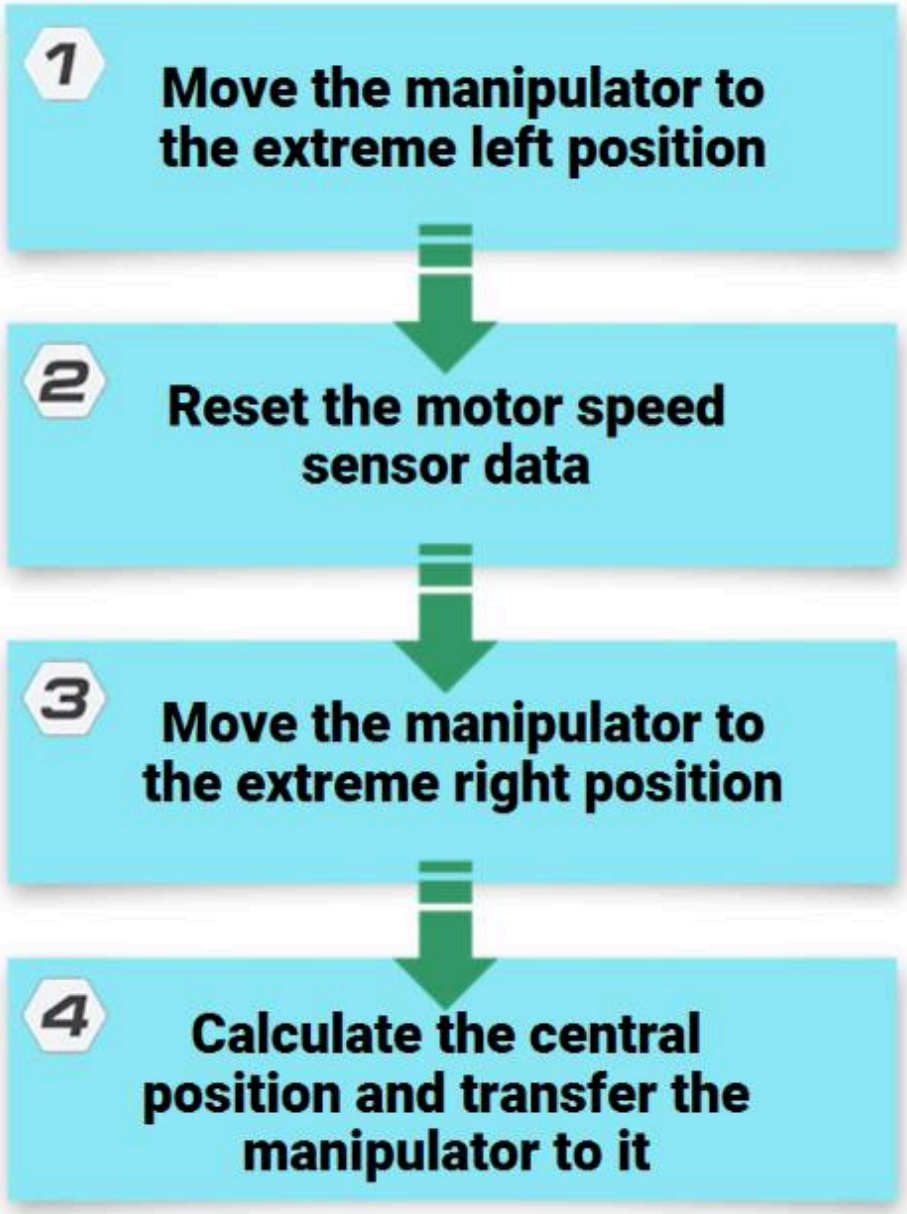


The combined program looks like this:

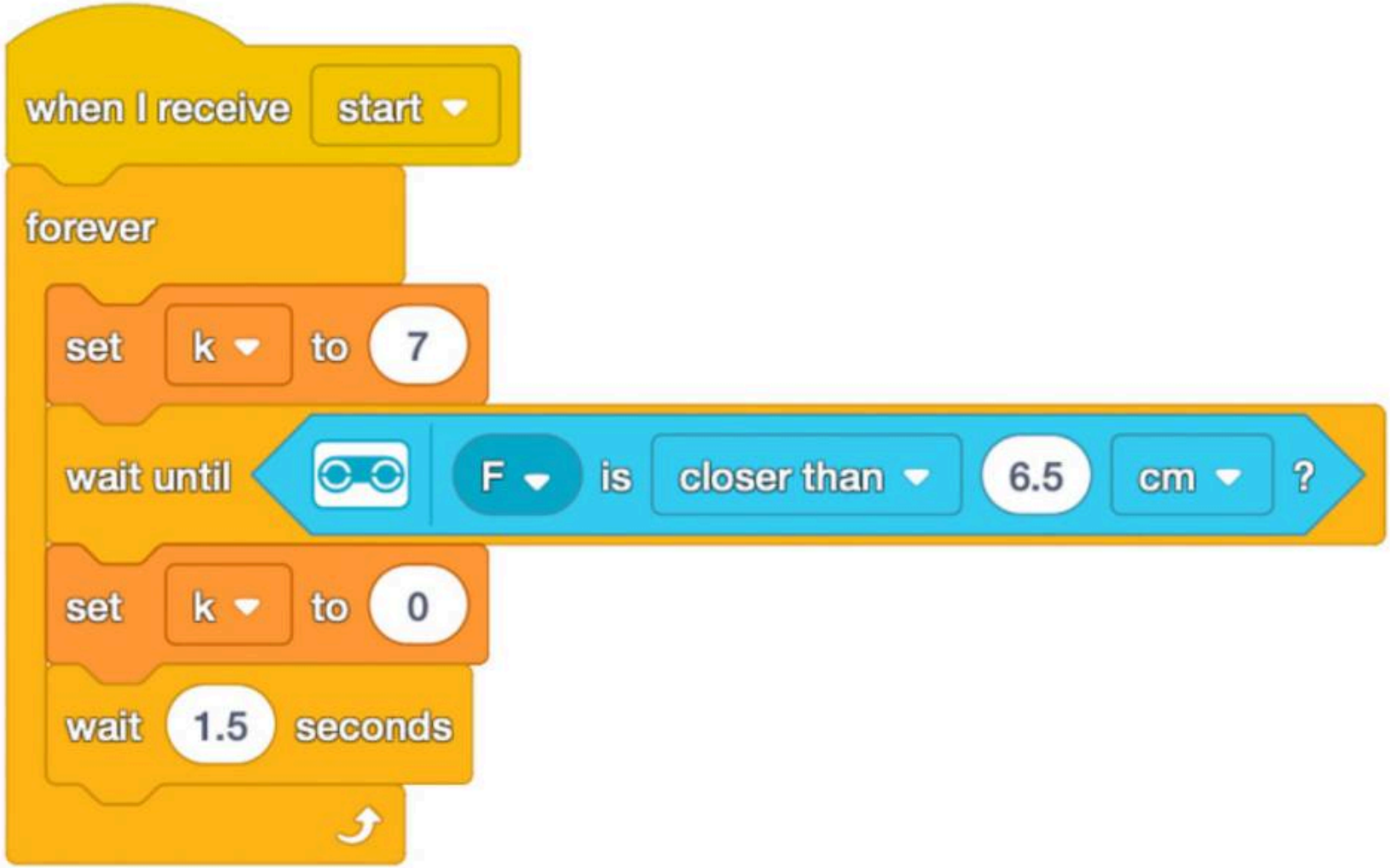


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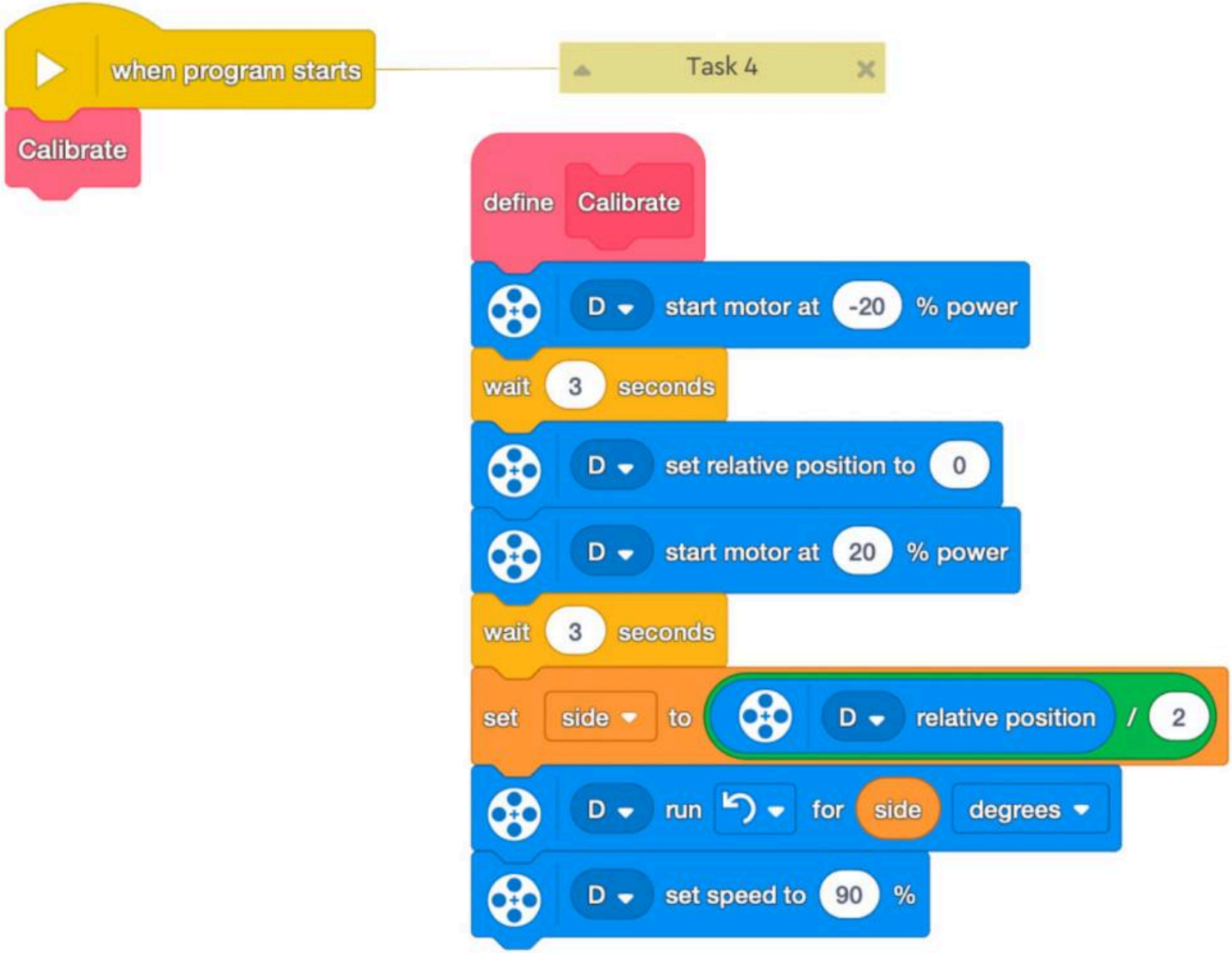
The calibration algorithm can be as follows:



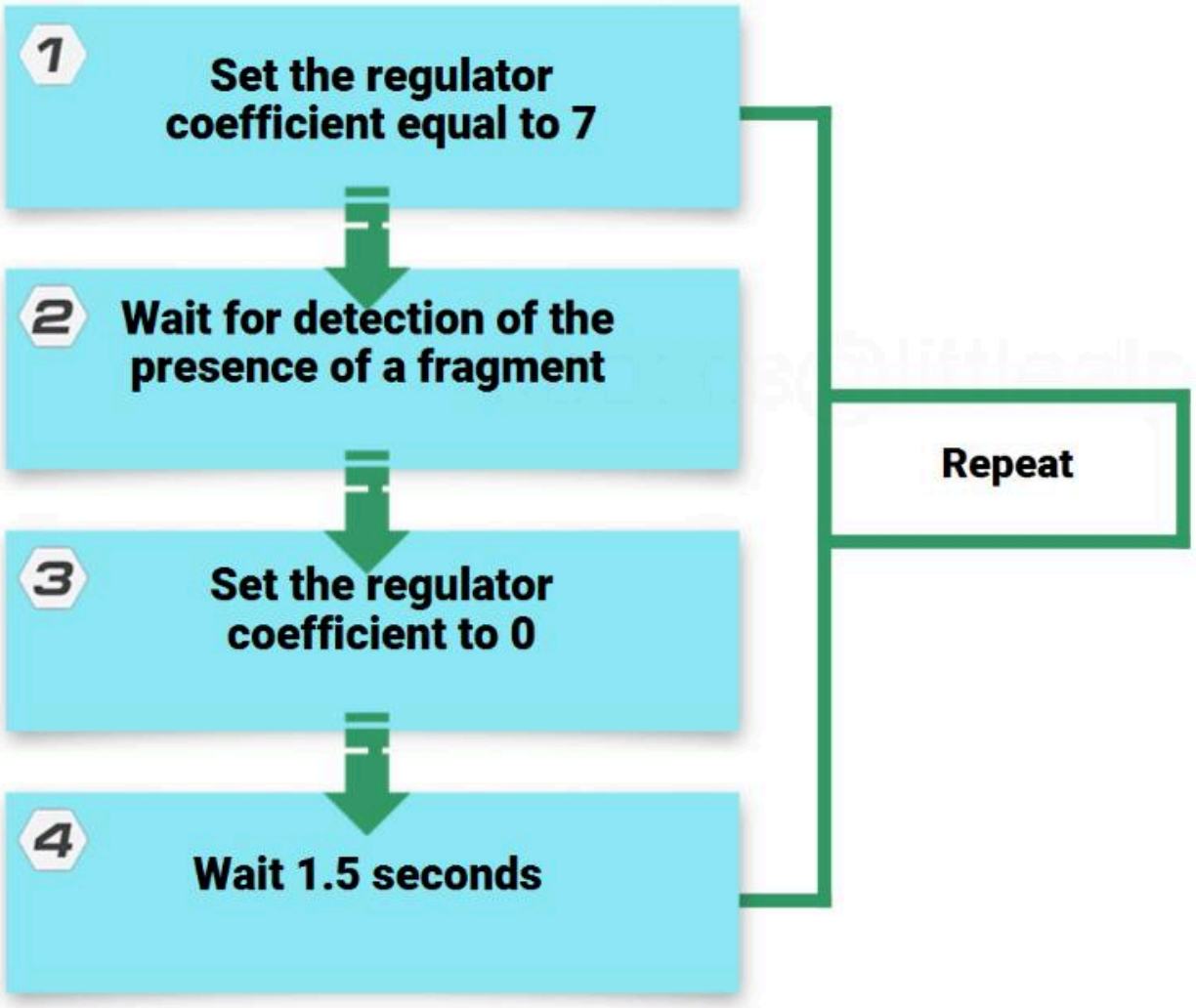
According to the following program, the robot will drive along the line with a coefficient of 7, and when detecting a fragment, it will shake hands for 1.5 seconds straight (the coefficient is equal to 0).



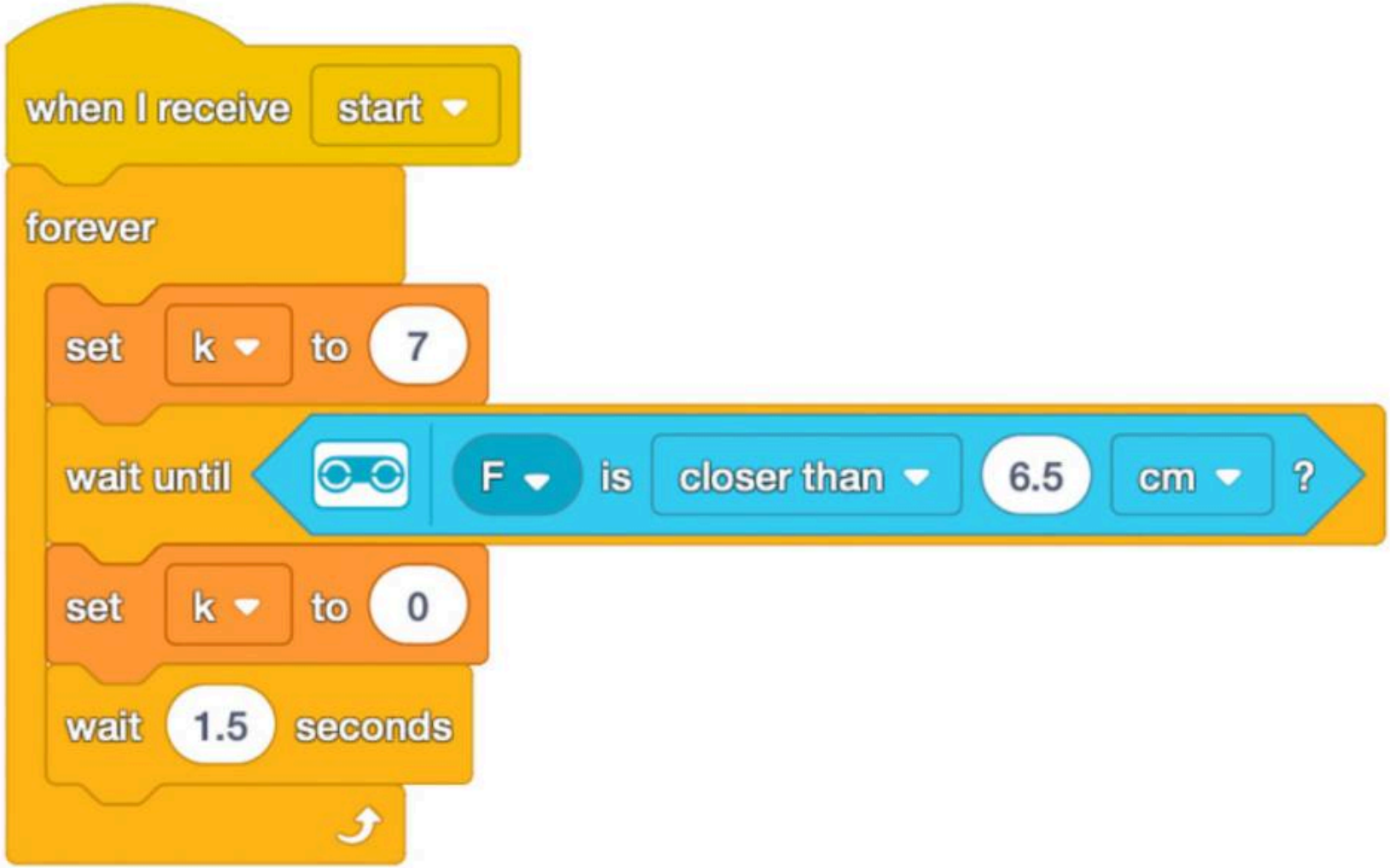
Create a 'My block' for convenient use of the calibration program:



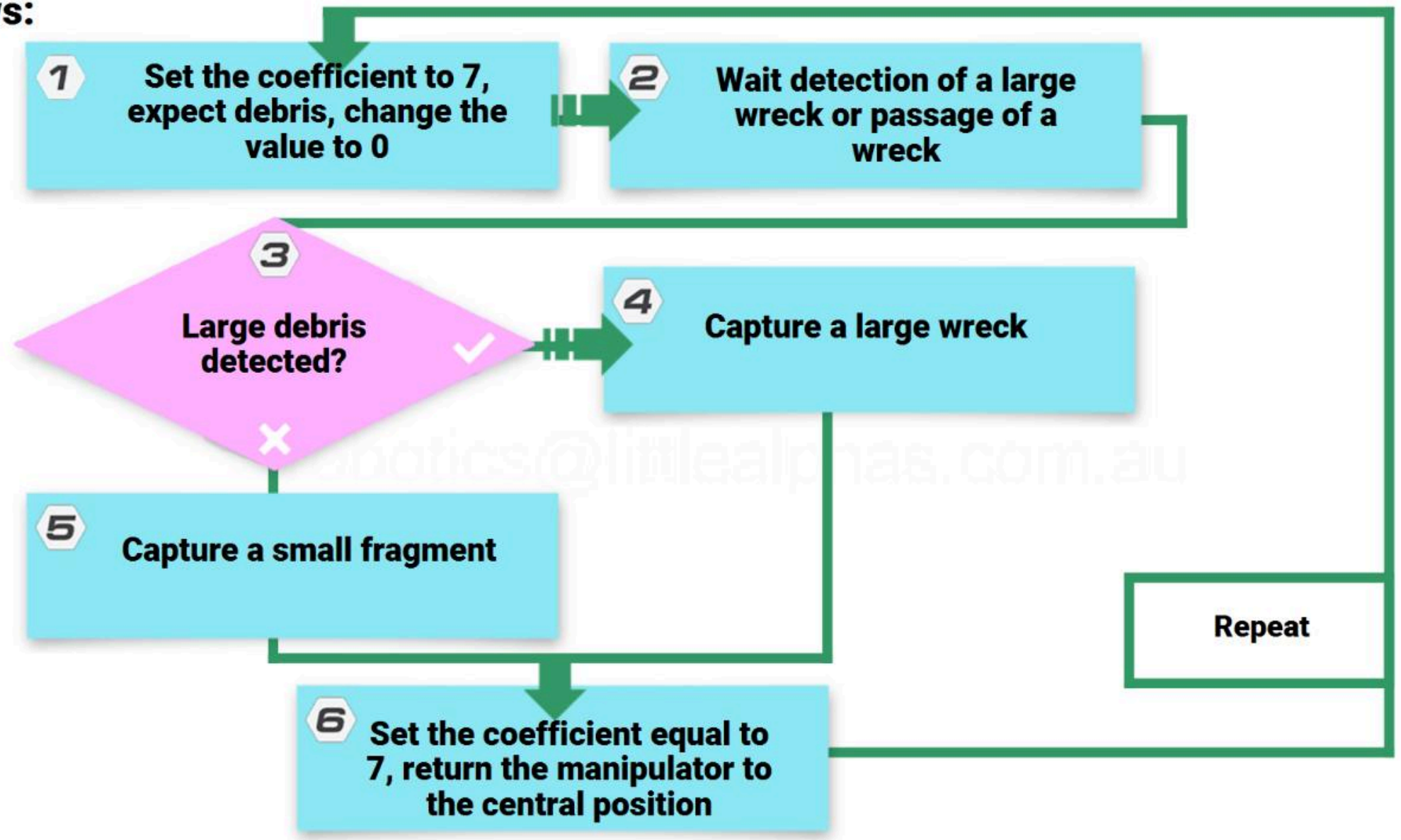
The algorithm for responding to the presence of debris on the trajectory can be as follows:



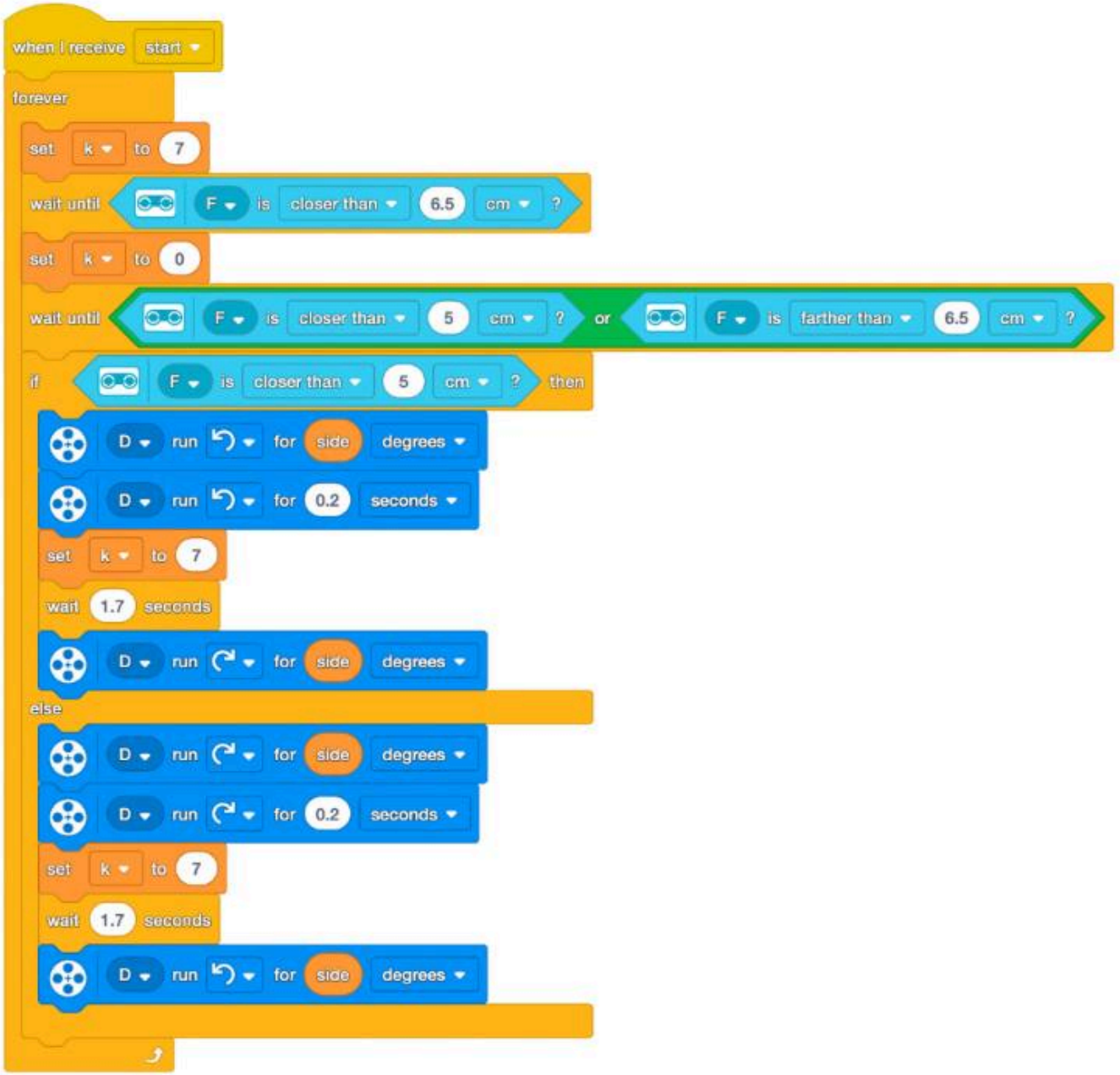
According to the following program, the robot will drive along the line with a coefficient of 7, and when detecting a fragment, it will shake hands for 1.5 seconds straight (the coefficient is equal to 0).



The algorithm for responding to the presence of debris on the trajectory can be as follows:

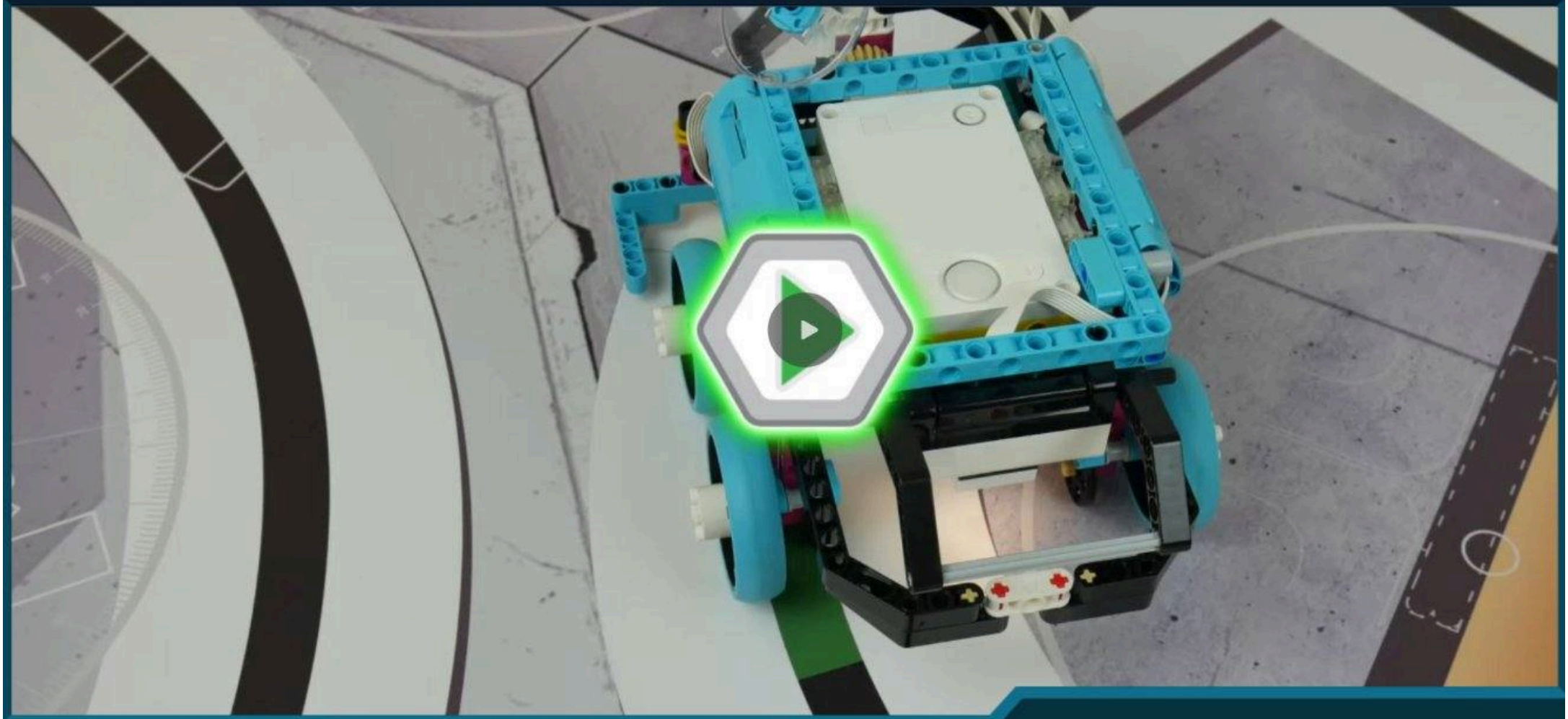


The updated program may look like this:

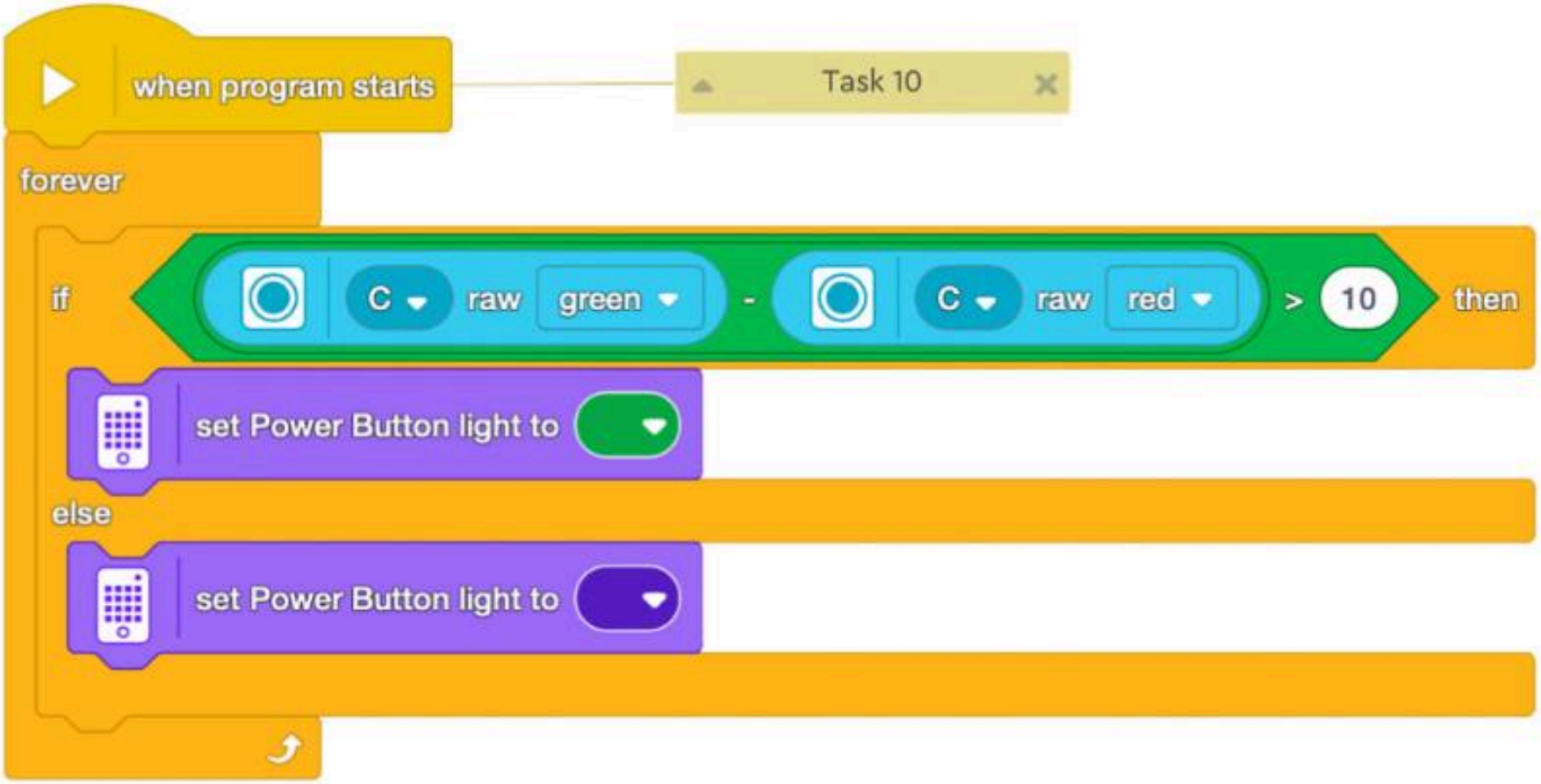


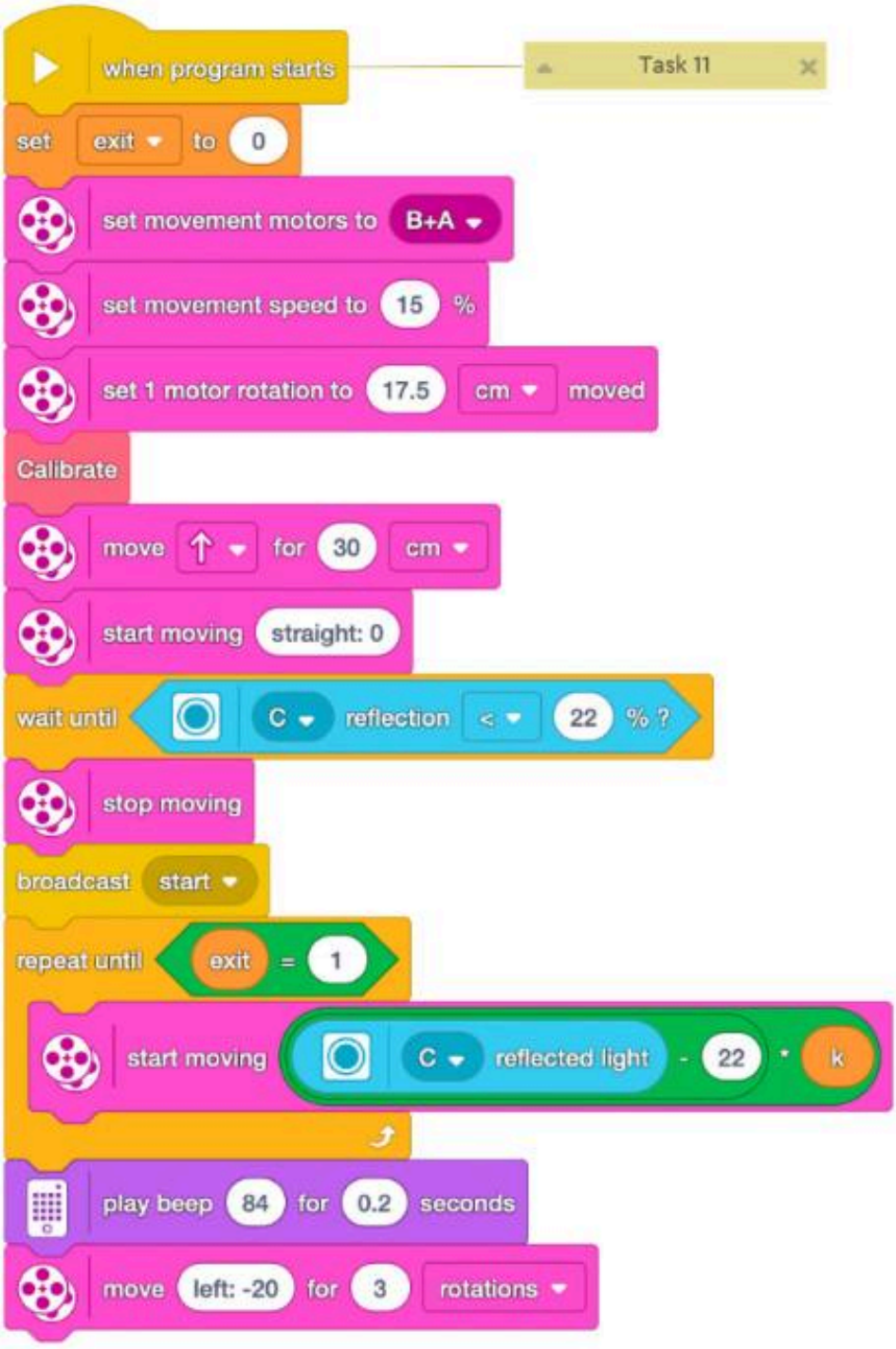


To complete the movement along the marking line, the robot must be able to count the number of green zones it passes. Test the following program to see if the robot 'sees' the green area and adjust it as needed



The test program may look like this:




when program starts
Task 11
set exit to 0
set movement motors to B+A
set movement speed to 15 %
set 1 motor rotation to 17.5 cm moved
Calibrate
move for 30 cm
start moving straight: 0
wait until C reflection < 22 % ?
stop moving
broadcast start
repeat until exit = 1
start moving C reflected light - 22 * k
play beep 84 for 0.2 seconds
move left: -20 for 3 rotations

Discuss!

- What event happened near your base on Mars?
- How do meteorites differ from asteroids and meteors?
- What types of meteorites do you know?
- What celestial bodies have you observed yourself?
- How did your robot distinguish the size of meteorites?
- What sensor is used to program the movement of the robot along the line?



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